Entry into Innovation Areas *

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August 2026

Abstract

This paper investigates the entry decisions of U.S. electronics firms across innovation areas. I apply topic modeling to over 630,000 patents to endogenously define innovation areas and merge those areas with market-implied patent valuations. Firms occupy several areas at once, and the number of firms differs widely across areas. An entry game, estimated by moment inequalities, finds that additional rivals reduce expected profit and that firms gain from occupying more areas at once. In the case study of semiconductor, a payment based on a quarter of predicted value shifts 0.1–0.5% into those areas and leads up to 0.8% of firms to add an extra area. Meanwhile, a 25% reduction in the occupancy cost shifts up to 7% into those areas and increases up to 10% of firms to add an extra area.

Keywords: Entry; Innovation; Patents; Technological diversification. **JEL:** L25, O31, O34

*I would like to thank my supervisors, Sabien Dobbelaere and José-Luis Moraga-González, for their constant guidance and support. I subscribe to Werning's Dogma 26: generative AI tools were used in a supporting role for exposition, proofreading, and consistency checks of the arguments, calculations, code, and references. I take full responsibility for the content and the correctness of this paper. All remaining errors are on my own.

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1 Introduction

Firms operating in the technology sector face a strategic problem: deciding which specific innovation and technological areas to enter and invest their resources. This paper builds an empirical model to untangle the economic determinants that govern these multi-area entry decisions. Using a panel of publicly listed firms in the United States electronics industry from 1990 to 2019, I quantify the congestion and yearly occupancy cost, measured against the gain from occupying more areas at once.

To analyze these dynamics, I construct a dataset that merges financial information from publicly listed electronics firms with rich, unstructured patent texts. I apply Latent Dirichlet Allocation (LDA) to the text of over 630,000 patent titles and abstracts. Coupled with iterative grid search optimization, this approach allows the data to endogenously define twenty distinct innovation areas based on semantic word co-occurrences, topic uniqueness, and semantic stability. A patent can sit in several of those areas with explicit weights; then, that weights is mapped as the choice set of technological areas to the entry game. I then integrate market-implied patent valuations from Kogan et al. (2017) to assign an expected monetary value to these technological areas.

Using the innovation areas derived from LDA, I document stylized facts regarding the dynamics of the U.S. electronics sector. First, I find that multi-area bundling is a dominant feature of the industry. Innovation rarely occurs in isolated silos; the average patent spans 3.58 technological areas, and the median firm-year occupies five areas, and that firms compete in several distinct areas simultaneously. These empirical patterns suggest that entry is a complex portfolio optimization problem subject to economies or diseconomies of scope.

Second, the number of active competitors is heterogeneous across innovation areas. Areas such as "Integrated Circuit Design, Verification, and Testing" and "Audio and Speech-Signal Processing" are consistently occupied by many players. In contrast, hardware domains such as "Display Panel Tech (LCD/OLED)" are occupied by few firms. Over the same years, the mix of occupied areas moved. Competitor shares fell in "Thin-Film Materials, Deposition and Chemistry" and "Rotational Electromechanics and Magnetic Actuation," and rose in "Wireless Base-Station and Mobile Communications" and "Networking Protocols and Data Infrastructure." Patent value moved with that mix, toward "Mobile UI and Content Services," "Wireless Base-Station and Mobile Communications," and "Power-Supply and Power-Management Electronics."

To rationalize these facts, I build a static, complete-information, simultaneous-move entry game. In this model, a firm chooses a bundle of innovation areas to maximize its total profit in a given year. The profit derived from occupying any single area depends on several components: the expected valuation of the patents within that area, a congestion effect driven by the number of active rival firms, a per-area cost paid in each year the firm occupies the area, and an area-specific shock. Because a firm typically occupies multiple areas simultaneously, its total profit is also subject to an organizational portfolio term. This portfolio term depends on the total count of occupied areas,

capturing either the synergistic economies of scope or the managerial coordination costs of maintaining a broad innovation portfolio. The model estimates answer a contemporaneous portfolio question[1]: which areas the firm occupies in a given year, and how congestion in that year shifts expected profit.

There are several challenges in estimating the model. First, because firms choose a bundle of areas, the strategy space explodes combinatorially. Second, strategic entry games generate multiple equilibria, which forces standard point identification methods to rely on arbitrary equilibrium selection rules. Finally, point identifying models with expected valuations requires the researcher to exactly specify the firm's unobserved internal information set, risking severe model misspecification. To overcome these issues, I abandon point identification and adopt a partial identification approach utilizing moment inequalities.

I construct these inequalities from the necessary conditions for a Nash equilibrium, using revealed preference arguments. Specifically, I evaluate the conditions where a firm unilaterally alters its observed portfolio by adding or dropping a subset of innovation areas, and by swapping one entered area for a non-entered area. By analyzing these specific deviations, the estimation isolates the marginal incentives that drive entry. In order to mitigate the measurement error issue in patent value, I project realized valuations onto several lagged firm covariates, and evaluate the inequalities at those fitted values.

First, I find that additional rivals reduce expected profit. Second, firms gain extra expected profit from occupying more areas at once: occupying an extra area raises expected profit by more when the firm already occupies other areas. I also find that treating extra areas as diminishing synergies is inconsistent with the observed portfolios in the data. Third, firms still leave many areas unentered. I find that the occupancy cost of one more area, counted in the gain from one pair of areas, can be zero and can be as large as about twelve. Take a firm that already occupies five areas and adds a sixth, then the gain from the five new pairs is five. Suppose the occupancy cost equals the gain from three pairs, then this occupancy cost is allowed. The gain exceeds the occupancy cost by two, and those two are a gain. Suppose now the occupancy cost equals the gain from twelve pairs, then this occupancy cost is also feasible. The gain is five and the occupancy cost is twelve, so subtracting the gain from the occupancy cost leaves a cost of seven.

In a numerical simulation, I try to mimic the areas covered by the U.S. CHIPS manufacturing incentive and the EU CHIPS Act. In particular, the coverage of those policies are memory, lithography and fabrication, packaging, and the transistor. A payment equal to one quarter of the predicted valuations shifts 0.1–0.5% into those areas and leads up to 0.8% of firms to add an extra area. Meanwhile, a 25% reduction in the occupancy cost shifts up to 7% into those areas and increases up to 10% of firms to add an extra area.

Related Literature.—This paper contributes to several strands of literature. First, it relates to the

[1] Appendix B.3 discusses what a dynamic treatment of first entry would require.

use of text to define markets and product spaces. Gentzkow et al. (2019) survey the methods, and Ash and Hansen (2023) review later algorithms. I build on Hoberg and Phillips (2016) and Eisfeld (2025) for continuous market definitions from filings of the listed companies and descriptions of products from startups in the software industry. For patent text, the closest work is topic modeling of recombination and diversification (Kaplan and Vakili, 2015; Teodoridis et al., 2022), breakthrough measurement from the text itself (Kelly et al., 2021), similarity and novelty from claims (Arts et al., 2018, 2021), joint patent-job-earnings text (Kalyani et al., 2025), transformer embeddings (Hain et al., 2022), and text measures of corporate innovation from analyst reports (Bellstam et al., 2021). I translate those textual outputs into state variables for the entry model, and I value each area with the grant-window returns of Kogan et al. (2017).

Second, I estimate an entry game with moment inequalities. This study is related to Bresnahan and Reiss (1991) where they relate the number of firms in a market to the profit of being there. That relation is similar to the congestion term here: profit in an innovation area depends on how many rivals occupy it. Furthermore, in this paper, the firm also chooses its set of areas for the year. Related literature also discuss that several equilibria can fit the same market, so the inequalities have to hold in all of them (Tamer, 2003; Ciliberto and Tamer, 2009). Pakes (2010) and Pakes et al. (2015) derive those inequalities using a revealed preference argument and demonstrate several applications. I use them to check the observed set against adding a few areas, dropping a few, and swapping one area for another. The information structure follows Dickstein and Morales (2018) and Morales et al. (2019): the firm chooses before the area shock is realized. This paper is also related to other literature such as, Eizenberg (2014) where he studies which configurations firms offer in home computers, and Wollmann (2018) where he studies which truck models a manufacturer offers. I use the test statistic following the methods developed by Chernozhukov et al. (2019).

Finally, the findings speak to the limits of the firm and to technology spillovers. Jaffe (1986) and Bloom et al. (2013) measure spillovers in technology space. Aghion et al. (2005) document an inverted-U between competition and innovation, and Hall and Ziedonis (2001) document the semiconductor patenting surge. I find that additional rivals reduce profit. On scope, related work includes pharmaceutical research (Henderson and Cockburn, 1996), product-line dynamics (Klette and Kortum, 2004), recombination (Weitzman, 1998; Fleming, 2001), relatedness and diversity (Breschi et al., 2003; Miller, 2006), and multi-location research (Chikis et al., 2025). The bounds recovered from the model suggest that synergies rise with portfolio size. Rotemberg and Saloner (1994), Seru (2014), and Rawley and Simcoe (2010) stress agency and overload costs. The semiconductor case study also touches on the current debate on industrial policy (Juhász et al., 2024; Goldberg et al., 2024; Barwick et al., 2025; Bloom et al., 2019). In the case study, the rival count in each area is the sum of the other firms' entry, updated until that sum agrees with the portfolios.

2 Data, Methodology, and Stylized Facts

2.1 Data

I construct a firm-level panel covering the U.S. electronics industry from 1990 to 2019. Firm's financial and organizationaal information, such as total assets and R&D expenses, are sourced from the Compustat dataset. I constrain the sample to publicly listed firms categorized under the two-digit Standard Industrial Classification (SIC) code 36 (Electronics and Electrical Components). To ensure the consistency of the data, I require consecutive yearly observations and impute missing R&D and asset values with zero. To proxy for firm size, I use total assets, aligning with standard empirical finance conventions (Seru, 2014). This procedure yields a final sample of 16,653 firm-year observations distributed across 1,373 unique firms. Of these, 755 firms have at least one patent.

To capture the economic value and nature of these firms' innovations, I integrate the patent dataset from Kogan et al. (2017), which provides market-implied patent valuations (derived from stock price movements subsequent to patent issuances), forward citation counts, and primary patent classifications. I link this data to the Compustat utilizing a correspondence table, i.e., from permco/permno to gvkey, provided by Wharton Research Data Services (WRDS)[2]. The data is the intersection of Compustat electronics firms and the Kogan et al. (2017) valuations. Those valuations are only available for a listed firm with a traded return around the patent grant date, so startups and privately held innovators are outside the panel by construction. By focusing on two-digit SIC 36, it also naturally omits firms in another industry code who have electronics patents, and similarly non-electronics firms that enter these topics. Of the 1,373 sample firms, 755 have at least one patent.

Because topic modeling requires rich textual inputs, I augment the Kogan et al. (2017) data by collecting unstructured text, specifically patent titles and abstracts, directly from Google Patents. My final textual corpus consists of 631,369 patents covering the electronics industry across the timeline[3]. Table 1 summarizes the firm characteristics I use.

2.2 Methodology: Topic Modeling

To parse the textual corpus of patents into economically meaningful dimensions, I employ Latent Dirichlet Allocation (LDA; Blei et al., 2003), treating each patent as a probabilistically weighted mixture of underlying "topics" or innovation areas. Conceptually, this mirrors the clustering methodologies frequently utilized in the literature to define product market based on business descriptions (e.g., Eisfeld, 2025), but it applies a sophisticated generative probabilistic model engineered for unstructured text.

[2]For an extensive discussion detailing the nuances of dynamically matching Compustat and USPTO entities, see Arora et al. (2021) and Dyevre and Seager (2023).

[3]A negligible fraction of patents (0.5%) were omitted from the final corpus due to missing records or irretrievable web elements.

Table 1: Summary statistics of firm characteristics

Variable	Statistic	Note
Number of firms (full sample)	1,373 (100%)	
Firms with at least one patent	755 (55%)	
R&D (millions USD, real 2012)	161.39	Average for patenting firms
Assets (millions USD, real 2012)	2663.77	Average for patenting firms
Patents (flow, annual)	54.16	Average annual patents
Patents (stock, cumulative)	4568.41	Average cumulative patents

Administrative patent classes such as the International Patent Classification and the Cooperative Patent Classification assign each patent based on the judgement of patent-examiner. Furthermore, the assigned-class for each patent have the properties of discrete, historically path dependent, and do not supply weights when the patent-examiner lists several classes for a patent. Meanwhile, LDA recovers a probability distribution over areas from the patent text, so a single invention can span several domains with explicit weights.[4] I select the number of topics and the Dirichlet priors by a five-fold iterative grid search that balances several metrics, e.g., perplexity, NPMI coherence, exclusivity, and decade-to-decade stability.[5] Appendix A.1 through Appendix A.5 detail the preprocessing, the interative grid search, and the metrics being used to select the hyperparameters

To assign labels, I extract the most heavily weighted terms for each area and iterate between two large language models (OpenAI's GPT-o3 and xAI's Grok 3 Think) until the proposed names converge. Table 2 lists the twenty labels, and top 25 words can be found in Appendix A.6.

Translating the raw LDA results into tractable inputs for the model requires aggregating from the patent level to the firm-year level. By dropping trivial topic exposures or outliers (i.e., weights below a certain threshold $\bar{w}$. See Appendix A.3 for further details), I construct the following key variables: (i) A monetary valuation of the patent and of the firm within each innovation area, A_{ipt} and v_{ijt}, along with the LDA weights, w_{pjt}, at the patent level. (ii) A binary indicator of firm i's entry into innovation area j at time t, d_{ijt}, which equals 1 if the firm has sufficient topic presence, and 0 otherwise. (iii) A congestion metric capturing the aggregate number of competing firms residing within area j, $N_{jt,-i} \equiv \sum_{i' \neq i} d_{i'jt}, \forall j, t$. (iv) The number of areas the firm occupies in each period, $k_{it} := |J_{it}| = \sum_j d_{ijt}$. (v) Predetermined firm covariates used as instruments: lagged R&D, RD_{it-1}, total assets, AT_{it-1}, and patent flow, PAT_{it-1}. These variables serve as the inputs for the model estimation introduced in the subsequent section.

[4] Appendix A.7 shows how many class codes examiners list on a grant in each decade, and which subclasses receive the most mass in each area (Tables 16 and 17). Certain important and crucial topics, such as, lithography, packaging, and transistor device physics all concentrate in subclass H01L. In the patent text those are different processes: etch and photoresist, die and bond pads, and gate and drain. A market defined as H01L would lump all of them together.

[5] From the optimization procedure, I find that twenty is the value at which those held-out metrics stop improving.

Table 2: Innovation areas

Topic	Label	Topic	Label
1	Semiconductor Memory Architecture	11	Power-Supply and Power-Management Electronics
2	Audio and Speech-Signal Processing	12	Networking Protocols and Data Infrastructure
3	Semiconductor Lithography and Fabrication Processes	13	Transistor Gate Structures and Semiconductor Device Physics
4	Integrated Circuit Design, Verification, and Testing	14	Imaging Sensors and Camera Pipelines
5	PCB and Electronic Hardware with Connectors	15	Processor Architecture, Caches, Instruction Execution
6	Video Recording, Encoding, and Media Processing	16	Display Panel Tech (LCD/OLED)
7	Semiconductor Packaging and Interconnect Bonding	17	Rotational Electromechanics and Magnetic Actuation
8	Thin-Film Materials, Deposition and Chemistry	18	Optics, Lasers, and Photonics
9	Wireless Base-Station and Mobile Communications	19	Analog and Differential Amplifier Circuits
10	Clock Generation, Timing Circuits, and Signal Processing	20	Mobile UI and Content Services

Notes: Labels are assigned from the twenty-topic LDA fit in Section 2.2. Top words for each area are in Appendix A.6.

2.3 Stylized Facts

I document two stylized facts: multi-area bundling and large differences in how many firms occupy each area. I then describe how competitor shares and patent value moved across decades.

Table 3: Patent-level summary statistics

Variable	Min	Q25	Q75	Max	Median	Mean	SD
w_{pjt} (LDA weight)	0	0	0.001	1	0	0.05	0.154
A_{ipt} (Patent value, million USD)	0	0.006	0.169	39.947	0.052	0.184	0.506
k_{ipt} (Number of areas per patent)	0	3	4	11	3	3.583	1.387

Stylized Facts 1 (On Multi-Area Bundling and Portfolio Scope) *Firms occupy several innovation areas in the same year. The average patent sits on 3.58 areas, and the median firm-year occupies five areas.*

Tables 3 and 4 demonstrate that multi-area bundling is a dominant feature of innovation in the electronics industry. At the patent level, the average patent spans 3.58 distinct innovation areas (k_{ipt}), indicating that a single invention relies on combinatorial knowledge. At the firm level, firms command broad innovation portfolios. The median firm-year occupies 5 distinct areas (k_{it}), with the top quartile participating in 10 or more areas simultaneously. These statistics show that entry is a joint portfolio decision, subject to economies or diseconomies of scope.

Stylized Facts 2 (On Competitive Congestion and Market Structure) *The number of firms differs widely across innovation areas. Some areas are occupied by many firms, and others by few, throughout*

Table 4: Firm-year summary statistics

Variable	Min	Q25	Q75	Max	Median	Mean	SD
d_{ijt} (Entry indicator)	0	0	1	1	0	0.269	0.443
$N_{jt,-i}$ (Competitor entry count)	20	83	141	218	105	110.255	39.913
k_{it} (Number of areas firm enters)	0	0	10	20	5	5.373	5.429
$\sum_p w_{pjt} A_{ipt}$ (Patent value sum, million USD)	0	0	0.721	112263.324	0	51.205	743.379

the sample.

Table 5: Composition of competitors by topic and decade

Topic	Label	1990s (%)	2000s (%)	2010s (%)	Mean (%)	SD
1	Semiconductor Memory Architecture	3.76	3.37	2.94	3.41	0.41
2	Audio and Speech-Signal Processing	6.97	7.19	7.1	7.08	0.23
3	Semiconductor Lithography and Fabrication Processes	4.39	4.33	4.27	4.34	0.32
4	Integrated Circuit Design, Verification, and Testing	7.49	7.68	7	7.44	0.41
5	PCB and Electronic Hardware with Connectors	6.97	6.47	6.73	6.72	0.4
6	Video Recording, Encoding, and Media Processing	3.7	3.73	3.35	3.62	0.31
7	Semiconductor Packaging and Interconnect Bonding	3.77	4.61	5.18	4.44	0.64
8	Thin-Film Materials, Deposition and Chemistry	6.31	5.14	4.66	5.45	0.79
9	Wireless Base-Station and Mobile Communications	3.62	4.35	4.74	4.18	0.53
10	Clock Generation, Timing Circuits, and Signal Processing	6.87	6.83	6.4	6.74	0.34
11	Power-Supply and Power-Management Electronics	5.97	6.14	6.27	6.11	0.28
12	Networking Protocols and Data Infrastructure	3.54	4.36	4.45	4.08	0.54
13	Transistor Gate Structures and Semiconductor Device Physics	3.6	3.78	3.72	3.7	0.2
14	Imaging Sensors and Camera Pipelines	3.12	3.3	4.22	3.46	0.54
15	Processor Architecture, Caches, Instruction Execution	5.1	5.47	5.53	5.35	0.33
16	Display Panel Tech (LCD/OLED)	1.86	1.81	1.92	1.85	0.19
17	Rotational Electromechanics and Magnetic Actuation	7.39	5.82	5.95	6.43	0.87
18	Optics, Lasers, and Photonics	3.84	3.78	3.91	3.84	0.22
19	Analog and Differential Amplifier Circuits	6.24	6.12	5.9	6.11	0.28
20	Mobile UI and Content Services	5.51	5.73	5.78	5.66	0.26
	Total	100.02	100.01	100.02	100.01	–

Table 5 and Table 6 reports the share of active firms and share of value in each area, respectively. Foundational areas, such as "Integrated Circuit Design, Verification, and Testing" (Topic 4) and "Audio and Speech-Signal Processing" (Topic 2), has around 7 percent share of the industry. "Display Panel Tech (LCD/OLED)" (Topic 16), however, remains sparse, under 2 percent throughout.

The results also show that legacy hardware such as "Rotational Electromechanics and Magnetic Actuation" (Topic 17) and "Thin-Film Materials, Deposition and Chemistry" (Topic 8) experienced a decline in share from the 1990s to the 2010s, while "Semiconductor Packaging and Interconnect Bonding" (Topic 7), "Wireless Base-Station and Mobile Communications" (Topic 9), and "Networking Protocols and Data Infrastructure" (Topic 12) experienced a gain. In the 1990s, "Rotational Electromechanics and Magnetic Actuation" (Topic 17) and "Thin-Film Materials, Deposition and Chemistry" (Topic 8) together accounted for more than 23 percent of patent value. By the 2010s their combined share had fallen, and value had moved toward "Mobile UI and Content

Table 6: Composition of patent valuation by topic and decade

Topic	Label	1990s (%)	2000s (%)	2010s (%)	Mean (%)	SD
1	Semiconductor Memory Architecture	2	2.12	1.62	1.94	5.15
2	Audio and Speech-Signal Processing	6.42	7.1	6.99	6.84	10.98
3	Semiconductor Lithography and Fabrication Processes	3.27	3.65	3.74	3.55	7.91
4	Integrated Circuit Design, Verification, and Testing	5.66	6.12	5.14	5.7	8.43
5	PCB and Electronic Hardware with Connectors	10.2	8.73	8.19	9.08	16.36
6	Video Recording, Encoding, and Media Processing	2.17	2.16	2	2.12	5.38
7	Semiconductor Packaging and Interconnect Bonding	2.42	4.4	4.96	3.88	9.98
8	Thin-Film Materials, Deposition and Chemistry	9.42	7.2	5.89	7.6	16.49
9	Wireless Base-Station and Mobile Communications	2.34	3.02	4.14	3.1	7.45
10	Clock Generation, Timing Circuits, and Signal Processing	7.76	7.85	6.79	7.53	11.91
11	Power-Supply and Power-Management Electronics	4.71	5.31	6.09	5.32	10.53
12	Networking Protocols and Data Infrastructure	3.4	5.2	5.19	4.58	10.87
13	Transistor Gate Structures and Semiconductor Device Physics	2.59	3.38	3.73	3.21	8.51
14	Imaging Sensors and Camera Pipelines	1.98	2.06	3.06	2.31	5.83
15	Processor Architecture, Caches, Instruction Execution	4.57	4.46	5	4.64	9.42
16	Display Panel Tech (LCD/OLED)	1.07	1.35	1.33	1.25	4.83
17	Rotational Electromechanics and Magnetic Actuation	14.14	8.85	8.39	10.52	18.68
18	Optics, Lasers, and Photonics	3.4	3.98	4.08	3.81	9.45
19	Analog and Differential Amplifier Circuits	7.19	7.38	6.22	7	12.44
20	Mobile UI and Content Services	5.28	5.68	7.45	6.03	12.5
	Total	99.99	100	100	100.01	–

Services" (Topic 20), "Wireless Base-Station and Mobile Communications" (Topic 9), and "Power-Supply and Power-Management Electronics" (Topic 11). Topic 20's share rose from 5.28 percent in the 1990s to 7.45 percent in the 2010s. Appendix A.8 reports the corresponding average count and dollar levels.

3 Model and Implementation

To explain the bundling and congestion patterns in Section 2.3, I develop a simple entry model following Pakes (2010), Pakes et al. (2015), Dickstein and Morales (2018), and Morales et al. (2019). For empirical implementation, I adopt the moment inequalities approach of Canay and Shaikh (2017), Chernozhukov et al. (2019), and Canay et al. (2023).

3.1 Information Structure and The Game

I model the firm's decision to enter innovation areas as a static, simultaneous-move, aggregative game under symmetric uncertainty, or complete information over observable state variables. When deciding to enter area j in period t, firm i uses its *ex-ante* information set $\mathcal{F}_{ijt}$ to evaluate expected profits. That set contains observable states, including the firm's own lagged R&D stock, lagged total assets, and its past patenting in specific areas, and characteristics of rivals, which the econometrician observes only in part. Further, I assume that unobservable area-specific market shocks u_{ijt} is realized only after entry, such that $E[u_{ijt} \mid \mathcal{F}_{ijt}] = 0$. Firms therefore play a simultaneous-move Nash equilibrium over expected payoffs.

This information structure follows Dickstein and Morales (2018) and Morales et al. (2019): firms do not select on shocks they have not yet seen, which yields me linear moment inequalities. On the static game, this setup answers how a firm composes its contemporaneous portfolio and how rival counts in the same period shift expected profit.[6]

3.2 Firm Valuation and Expected Profit

Each patent p filed by firm i in period t has a dollar value A_{ipt}. Using topic modeling from the prior section, weights w_{pjt} (where $0 \leq w_{pjt} \leq 1$ and $\sum_j w_{pjt} = 1$) allocate each patent across innovation areas. As such, I can define the area-specific value as $A_{ipjt} = w_{pjt} A_{ipt}$ [7].

I define the *realized ex-post* firm valuation as follows

$$v_{ijt} = b\left(\sum_{p \in \mathcal{P}_{it}} w_{pjt} A_{ipt}\right) + e_{ijt},$$

where $\mathcal{P}_{it}$ is firm i's patent set in period t, and $b(\cdot)$ is a monotonically increasing transformation function. This function maps the raw aggregated patent values into a tractable scale, which manages the high skewness of valuation data. The term e_{ijt} captures valuation measurement error, absorbing the noise inherent to LDA topic assignments and the stock market's reaction in the event studies of Kogan et al. (2017). Firms do not observe the exact realized error e_{ijt} prior to entry, meaning their ex-ante expectation is strictly deterministic, i.e., $E[e_{ijt} \mid \mathcal{F}_{ijt}] = 0$. Then, the expected valuation is simply $\check{v}_{ijt} \equiv E[v_{ijt} \mid \mathcal{F}_{ijt}]$.

The true realized profit for firm i of entry into area j in period t can then be written as

$$\pi_{ijt} = v_{ijt} + \alpha h(N_{jt,-i}) + \beta f(k_{it}) - \kappa + u_{ijt}$$

where both the second and third terms reflect competitive and organizational factors, respectively. By substituting the expected valuation, $v_{ijt} = \check{v}_{ijt} + e_{ijt}$, into the true profit function, the estimating equation of the profit can be written as

$$\pi_{ijt} = \check{v}_{ijt} + \alpha h(N_{jt,-i}) + \beta f(k_{it}) - \kappa + \epsilon_{ijt}$$

where $N_{jt,-i}$ is the number of competitors (excluding the firm itself) in area j, and $k_{it} \equiv |J|_{it}$ is the total number of innovation areas firm i is active in during period t. $\check{v}_{ijt} \equiv E\left[b\left(\sum_p w_{pjt} A_{ipt}\right) \mid \mathcal{F}_{ijt}\right]$ is the expected firm valuation, and $h(\cdot)$ and $f(\cdot)$ denote functional forms for both rival counts and portfolio size, respectively. As for ϵ_{ijt}, the unobserved error term in the estimating equation, $\epsilon_{ijt} \equiv$

[6] Two features of the data support treating rival counts as given. The number of rivals in an area is large, with a mean of 110 and a minimum of 20, so one firm's entry moves its rivals' payoffs little. Rival counts are also stable from year to year, with an autocorrelation of 0.97, so the count a firm expects is close to the count it observes. Appendix A.10 reports how persistent occupancy is in the panel.

[7] The construction of A_{ipt} is in Appendix B.1. The construction of the LDA weights w_{pjt} is in Appendix A.2.

$e_{ijt} + u_{ijt}$, can be thought of as a *composite ex-post forecasting error*. It absorbs both the valuation measurement noise (e_{ijt}) and the market shocks (u_{ijt}). Because both components are mean-zero conditional on the firm's information set, the conditional expected value of ϵ_{ijt} is zero, i.e., $E[\epsilon_{ijt} \mid \mathcal{F}_{ijt}] = 0$.

Parameter α captures congestion effects: positive values indicate spillovers (e.g., net-agglomeration effect), while negative values reflect crowding out. Parameter β measures the impact of breadth of innovation portfolio, with positive values suggesting benefits from diversification (e.g., economies of scope) and negative values indicating costs of overextension (e.g., diluted focus). As for parameter κ, it is a per-area fixed cost paid for each occupied area in year t. Consequently, a firm that drops an area in that year stops paying it. Let $\theta = (\alpha, \beta, \kappa)$ be the vector of parameters of interest.

Let d_{ijt} be an indicator whether firm i enters into area j at time t. The firm chooses a bundle of areas to maximize expected total profit:

Result 1 (Firm's Expected Total Profit) *Because $E[\epsilon_{ijt} \mid \mathcal{F}_{ijt}] = 0$, the expected total profit simplifies to its deterministic components:*

$$\begin{aligned}\mathbb{E}[\Pi_{it} \mid \mathcal{F}_{ijt}] &= \sum_{j=1}^{J} d_{ijt}\mathbb{E}[\pi_{ijt} \mid \mathcal{F}_{ijt}] = \sum_{j=1}^{J} d_{ijt}\left(\check{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa\right) + \beta f(k_{it}) \sum_{j=1}^{J} d_{ijt} \\ &= \sum_{j \in S}\left(\check{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa\right) + \beta g(k_{it})\end{aligned}$$

where $S = \{j \mid d_{ijt} = 1\}$, $k_{it} = |J|_{it}$, and $g(k_{it}) \equiv k_{it} f(k_{it})$. Result 1 highlights the importance of the bundling problem: the presence of the portfolio term $\beta g(k_{it})$ creates interdependencies across choices[8].

Firm i selects the bundle $\mathbf{d}_{it} = (d_{ijt})_j \in \{0,1\}^J$ to maximize expected profit $\mathbb{E}[\Pi_{it} \mid \mathcal{F}_{ijt}]$ in a simultaneous-move game. Solving for the full equilibrium fixed point over a strategy space of 2^J bundles is computationally intractable due to the curse of dimensionality. However, in any Nash equilibrium, the firm's observed bundle must be its optimal best response, taking the equilibrium actions of rival firms ($N_{jt,-i}$) as given.

Because the portfolio cost depends only on the *count* of entered areas (k_{it}) and not their specific identities, the firm's combinatorial optimization problem simplifies, conditional on these rival actions. This yields a characterization of the firm's optimal bundle:

[8] A standard independent single-agent discrete choice framework is ill-posed in this setting because the portfolio term $\beta g(k_{it})$ renders the entry decisions non-separable; the marginal payoff of entering area j explicitly depends on the endogenous decision to enter area m. Consequently, the firm is not making J independent choices, but rather selecting one bundle out of 2^J possible combinations. With $J = 20$ innovation areas, evaluating standard choice probabilities over 2^{20} (more than one million) alternative bundles per firm-year is computationally intractable, necessitating the moment inequalities approach.

Result 2 (Characterization of the Optimal Bundle) *The expected profitability index for area j is:*

$$\zeta^e_{ijt} = \check{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa$$

For a fixed portfolio size k_{it}, the firm maximizes profit by choosing the k_{it} areas with the largest ζ^e_{ijt}. That is, the areas are sorted such that: $\zeta^e_{(1)} \geq \zeta^e_{(2)} \geq \cdots \geq \zeta^e_{(J)}$. Then, the optimized expected profit for a portfolio of size k is:

$$U^e_{k_{it}} = \sum_{m=1}^{k_{it}} \zeta^e_{(m)} + \beta g(k_{it})$$

An optimal size is any $k^*_{it} \in \arg\max_{k_{it}=0,1,\ldots,J} U^e_{k_{it}}$. Any optimal bundle is a set of k^*_{it} areas with the largest ζ^e_{ijt}, allowing ties at the cutoff, and $d_{ijt} = 1$ if and only if j belongs to that bundle.

Proof. See Appendix B.2. ■

Assuming that the portfolio cost $\beta g(k_{it})$ depends only on the count k_{it}, it simplifies the firm's optimization problem: the firm ranks areas by ζ and selects a top-k^*_{it} set. However, this simplification does not solve the problem faced by the econometrician. The decision to enter area j is still not separable from area m[9]. Furthermore, although Result 2 provides a clean characterization of the problem, the idea of point identification of θ remains challenging due to multiple equilibria and combinatorial explosion. Which configuration materializes depends on an unobserved equilibrium-selection rule (as shown by Tamer, 2003; Ciliberto and Tamer, 2009).

Instead, I pursue partial identification following Pakes (2010) and Pakes et al. (2015). This replaces unattainable moment equalities with inequalities satisfied by every equilibrium. Based on Result 2, any deviation from the observed choice S^* must yield a lower (or equal) expected profit. To operationalize this notion, I consider 'step and swap deviations': (i) entering M additional areas, (ii) exiting M current areas, or (iii) swapping the entry portfolio while keeping k_{it} constant.

Let $Z_{ijt} \in \mathbb{R}^n$ be a vector of instruments observable to both the econometrician and the firm, and that this instrument is assumed to be a subset of $\mathcal{F}_{ijt}$. Let $l : \mathbb{R}^n \to \mathbb{R}_+$ map Z_{ijt} to a non-negative scalar. Following Canay et al. (2023), these instruments are weights that partition the sample and tighten the inequalities. Note that these instruments do not instrument rival counts $N_{jt,-i}$ or portfolio size k_{it} in the two-stage least squares sense. As for the common area-time attractiveness, it is absorbed in the first-stage projection of expected value.

First, I will derive the moment functions for adding M areas.

Result 3 (Moment Functions for Adding M Areas (Non-Entrant Deviation)) *Let $\mathcal{M}$ be an exogenously drawn subset of M areas ($M = |\mathcal{M}|$). I consider the event where the firm chose to enter none of*

[9] That is, te marginal cost of entering j depends on k_{it}, which is endogenously determined by the entry decisions for all other areas. Because the choices are jointly determined, those decisions cannot be estimated as J independent single-choice models (e.g., in a typical logit). The outcome is a single chosen bundle S^*.

the areas in $\mathcal{M}$, indicated by $D^0_{\mathcal{M},it} \equiv \prod_{j\in\mathcal{M}}(1-d_{ijt}) = 1$. The difference in expected profit between the deviation (adding $\mathcal{M}$) and the observed choice is, $\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}]$:

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] = \sum_{j\in\mathcal{M}} \left(\breve{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa\right) + \beta\Delta g^+_M(k_{it}) = \eta^+_{i,\mathcal{M},t}$$

where $\Delta g^+_M(k_{it}) = g(k_{it}+M) - g(k_{it})$. If the firm chose not to enter any area in $\mathcal{M}$, the expected deviation profit must be non-positive: $\eta^+_{i,\mathcal{M},t} \leq 0$. Multiplying this by the choice indicator and the non-negative instrument yields the unconditional moment function as follows:

$$\mathbf{M}_1(Z_{i,\mathcal{M},t};\theta) = E\left[D^0_{\mathcal{M},it} \cdot \eta^+_{i,\mathcal{M},t} \cdot l(Z_{i,\mathcal{M},t})\right] \leq 0 \tag{1}$$

Proof. See Appendix B.2. ■

Now, I will derive the moment functions for dropping M areas.

Result 4 (Moment Functions for Dropping M Areas (Entrant Deviation)) *Let $\mathcal{M}$ be an exogenously drawn subset of M areas ($M = |\mathcal{M}|$). I consider the event that the firm entered every area in $\mathcal{M}$, indicated by $D^1_{\mathcal{M},it} \equiv \prod_{j\in\mathcal{M}} d_{ijt} = 1$. The expected profit difference of dropping $\mathcal{M}$ is:*

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] = -\sum_{j\in\mathcal{M}} \left(\breve{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa\right) - \beta\Delta g^-_M(k_{it}) = -\eta^-_{i,\mathcal{M},t}$$

where $\Delta g^-_M(k_{it}) = g(k_{it}) - g(k_{it}-M)$. If the firm chose to enter all areas in $\mathcal{M}$, dropping them cannot yield strictly higher expected profit. Thus, $-\eta^-_{i,\mathcal{M},t} \leq 0$. I can then define the unconditional moment function as follows:

$$\mathbf{M}_2(Z_{i,\mathcal{M},t};\theta) = E\left[D^1_{\mathcal{M},it} \cdot (-\eta^-_{i,\mathcal{M},t}) \cdot l(Z_{i,\mathcal{M},t})\right] \leq 0 \tag{2}$$

Proof. See Appendix B.2. ■

These M-step deviations $\mathbf{M}_1$ and $\mathbf{M}_2$ moments are all valid and necessary conditions, yet they do not fully exploit the sorting characterization outlined in Result 2. To avoid endogenous selection bias, I define swap moments over exogenously chosen pairwise deviations. Because the portfolio term depends solely on the count k_{it}, any profitable swap of an entered area for a non-entered area, while keeping k_{it} fixed, is ruled out. I can then derive this 'swap' moment function as follows.

Result 5 (Moment Functions for Pairwise Swap Deviation) *Let (j,m) be an ordered pair where firm i chose to enter j but not m: $D^{swap}_{j,m,it} \equiv d_{ijt}(1-d_{imt}) = 1$. The expected profit of deviating by substituting m for j leaves the portfolio size constant ($\Delta g = 0$):*

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] = \left(\breve{v}_{imt} + \alpha h(N_{mt,-i})\right) - \left(\breve{v}_{ijt} + \alpha h(N_{jt,-i})\right) = \mu_{ijmt}(\theta)$$

The optimality condition requires that the expected swap profit is non-positive: $\mu_{ijmt}(\theta) \leq 0$. Then, the unconditional linear moment function can be written as follows:

$$\mathbf{M}_3(Z_{ijmt};\theta) = E\left[D^{swap}_{j,m,it} \cdot \mu_{ijmt}(\theta) \cdot l(Z_{ijmt})\right] \leq 0 \tag{3}$$

Proof. See Appendix B.2. ■

The swap moment, $\mathbf{M}_3$, isolates the congestion effect α. Now consider a firm that occupies area j and does not occupy area m. A swap asks whether that firm would be better off occupying m instead of j. Since the firm still occupies the same number of areas, the gain from occupying more areas at once does not change, and the firm still pays occupancy cost for the same number of areas. However, what can still differ is expected valuations and how crowded each area is. Suppose the two areas look similar in expected valuations. If the firm occupies the less crowded of the two, then the ranking identifies a congestion effect.

3.3 Empirical Implementation

I define entry as $d_{ijt} = \mathbf{1}\{\frac{1}{|\mathcal{P}_{it}|}\sum_{p\in\mathcal{P}_{it}} w_{pjt} \geq \bar{w}\}$ using the weights from the LDA model with a threshold $\bar{w}$. The reason to choose the threshold is that weights below this threshold are the long tail of the LDA model weights[10]. As for constructing $\check{v}_{ijt}$, I apply the inverse hyperbolic sine to realized v_{ijt} and project onto lagged R&D stocks, lagged assets, and lagged patent flow, together with firm and year effects, in a separate regression for each area, so that the fitted value carries firm-by-area and year-by-area effects. As such, the fitted value is the proxy that enters the estimating profit equation.

On the congestion effect, I define the congestion function as $h(N) = \text{asinh}(N)$, which is steep in thin markets and flatter once an area is crowded. As for the functional form for the portfolio term, I define it as $g(k) = \binom{k}{2}$, which counts pairs among entered areas and delivers the sorting in Result 2. I use pair and triplet add and drop deviations, which vary how many areas enter the portfolio term. Add and drop change how many areas the firm occupies. Under assumed pairwise interactions for the portfolio term, those comparisons can only discipline the occupancy-to-scope ratio $\rho = \kappa/\beta$[11]. Furthermore, I can only identify the ratio, I have to assume that β cannot be zero under the pairwise interactions. As such, I can only identify $\tilde{\theta} = (\alpha, \rho)$ instead of $\theta = (\alpha, \beta, \kappa)$.

To operationalize the model estimation, I first rank candidate add and drop subsets from predicted values $\check{v}_{ijt}$ before looking at realized d_{ijt}. For drop moments I keep the subsets with the lowest

[10]The brief explanation of the threshold $\bar{w}$ is the following: at the patent level, keep only the positive shares, i.e., $w_{pjt} > 0$. Let f be the density of $\ln(w_{pjt})$ on that set, and let $m_L < m_R$ be its two modes. The minimum point on the log scale then can be defined as $x^* = \arg\min_{x\in(m_L,m_R)} f(x)$. The threshold in the entry is simply $\bar{w} = \exp(x^*)$. I find that $\bar{w} = \exp(x^*) = 0.0094 \approx 1\%$. Further details and motivation can be found in Appendix A.3.

[11]See Appendix B.4 for further discussion on the identification of ρ

average predicted value, where the decision to stay is most informative about the cost. Similarly, for add moments I keep those with the highest average predicted value, where the decision not to enter is most informative. For swaps I rank the 380 ordered pairs by divergence in $\check{v}$ and draw from the top, the bottom, and the middle of that ranking to cover a representative sample of the ranked pairs. The reported sets use samples of 20 or 40 per deviation type, and the specification with all three instruments uses the largest of those samples. The test statistic is the max-type studentized sample moment of Chernozhukov et al. (2019), with their two-step self-normalized critical value at significance level 0.01. As for the model estimation, I first draw parameter vectors at random inside a bounded region and keep those for which the test statistic does not exceed the critical value. If the best draw still exceeds it, I minimize that excess and use the resulting vector as a starting value. The main reason to conduct this first step is that the criterion from the moment inequalities is noncovex, flat in many places, and non-differentiable. From those starting values I search the same region by simulated annealing a la Grieco (2014) and keep the vectors that satisfy the criterion, and I further profile each parameter in $\tilde{\theta} = (\alpha, \rho)$ by its lowest and highest value subject to that same restriction. Appendix B.3 details the search over a bounded parameter region and the profiled bounds.

4 Results

I first present the first-stage projection that replaces noisy realized patent values with the component that is predictable from several lagged covariates. I then show a reduced-form linear probability model of entry. Finally, I present the estimated identified sets from the moment inequalities.

4.1 First-stage Projection: Extracting Expected Valuations

The raw aggregated patent value, $v_{ijt} = b\left(\sum_{p\in\mathcal{P}_{it}} w_{pjt} A_{ipt}\right)$, is an *ex-post* realization inherently plagued by measurement errors. These errors originate from three primary sources: (i) Latent Dirichlet Allocation (LDA) misclassification, which introduces random probabilistic noise into the weights w_{pjt}; (ii) stochastic noise in market-implied patent valuations, A_{ipt}, derived from Kogan et al. (2017); and (iii) the mechanical aggregation frictions of mapping patent-level text distributions to the firm-year-topic level. If v_{ijt} is directly inserted into the entry model, classical measurement error would attenuate the valuation term. The first-stage projection keeps the component of that realized value that is predictable from lagged R&D, assets, and patent flow, together with firm and year-by-area effects. To form a proxy for expected private value, I project the observed v_{ijt} onto predetermined the instruments $Z_{ijt} = (\mathrm{RD}_{it-1}, \mathrm{AT}_{it-1}, \mathrm{PAT}_{it-1})$ and firm and year-by-area effects. This exercise yields a fitted value $\check{v}_{ijt}$. Here, the assumption being made is that $E[\epsilon_{ijt} \mid Z_{ijt}] = 0$, i.e., the composite forecast error is mean independent of the instruments used as weights.

As discussed in the previous section, the variables Z_{ijt} are discretized into binary indicators to

manage the extreme skewness of the data. Specifically, past R&D stocks (RD_{it-1}), past total assets (AT_{it-1}), and past patent flow (PAT_{it-1}) take a value of 1 for firm-year observations in the upper tier (medium-to-high levels). The lower tier (zero or bottom-quartile) serves as the baseline[12].

Table 7: First Stage: Expected Profit Projections

Dependent Variable:	$v_{ijt} = b(\sum_p w_{pjt} A_{ipt})$		
Model:	(1)	(2)	(3)
Variables			
Constant	-0.464***		
	(0.006)		
RD_{it-1}	0.320***	0.072***	0.072***
	(0.022)	(0.021)	(0.021)
AT_{it-1}	0.206***	0.109***	0.109***
	(0.021)	(0.021)	(0.021)
PAT_{it-1}	1.90***	0.837***	0.837***
	(0.130)	(0.072)	(0.072)
Fixed-effects			
Firm		Yes	Yes
Year		Yes	
Area		Yes	
Year × Area			Yes
Fit statistics			
Observations	225,220	225,220	225,220
R^2	0.502	0.709	0.710
Within R^2		0.063	0.063

Clustered (Firm) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is realized patent value after asinh and a global standardization.

Table 7 reports the first-stage projection. Focusing on the specification with firm and year-by-area fixed effects (Column 3), these covariates exhibit significant predictive relationships. Because the dependent variable v_{ijt} is transformed via an inverse hyperbolic sine and then standardized, the estimated coefficients translate to standard deviation units. Moving from a zero or near-negligible patenting history to a medium-to-high past patent flow (PAT_{it-1}) shifts the firm's expected area valuation upward by a 0.837 standard deviation. Similarly, total assets (AT_{it-1}) and R&D stocks (RD_{it-1}) provide significant premiums of 0.109 and 0.072 standard deviations, respectively. Across the twenty areas, the mean absolute deviation of the fitted valuation is clustered around 0.34. Comparison of predictive performance across specifications can be found in Appendix A.9.

[12]As discussed in the implementation of the instrument Z_{ijt}, transforming these covariates into non-negative binary scalars allows them to function efficiently as weights $l(\cdot)$.

4.2 Reduced-Form Evidence on Firm Entry

Before implementing the moment inequalities, it is useful to assess firm's entry likelihood via standard tools. Table 8 estimates a reduced-form linear probability model on entry by regressing the entry indicator d_{ijt} on the fitted-value of valuation valuation $\check{v}_{ijt}$, competitive congestion $\text{asinh}(N_{jt,-i})$, and the portfolio scale variable $(k_{it}-1)/2$. Focusing on the specification with fixed effects (Column 2), I find the following results: First, valuations matter, i.e., higher expected profits increase the likelihood of entry. Second, the coefficient on $\text{asinh}(N_{jt,-i})$ is positive and remains that way after expected valuation and portfolio size enter the same regression.

Table 8: Entry, expected valuation, rival counts, and portfolio size

Dependent Variable:	d	
Model:	(1)	(2)
Variables		
Constant	-0.8239***	
	(0.0334)	
$\tilde{v}_{ijt} = b(\hat{v}_{ijt})$	7.43×10^{-6}**	0.0003***
	(3.72×10^{-6})	(3.9×10^{-5})
$\text{asinh}(N_{jt,-i})$	0.1643***	0.0799***
	(0.0063)	(0.0140)
$(k_{it}-1)/2$	0.0996***	0.0995***
	(0.0002)	(0.0001)
Fixed-effects		
Firm		Yes
Year		Yes
Area		Yes
Fit statistics		
Observations	225,220	225,220
R^2	0.3974	0.4075

Clustered (Firm) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

The same OLS specification also generates a positive coefficient on the portfolio measure (0.0995). However, recall that the regressor is $(k_{it}-1)/2$, so a coefficient near 0.1 is the adding-up identity of the occupancy rate. Furthermore, unobserved firm capability can also raise both k_{it} and the probability of entering any given area, which is a reason the linear probability model cannot solely identify and isolate scope.

4.3 Model Estimates

I partially identify congestion, α, and the occupancy-to-scope ratio, ρ, with the add, drop, and swap comparisons ($\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3$) derived in the previous section. Table 9 reports 99% confidence sets under the two-step self-normalized procedure of Chernozhukov et al. (2019). The procedure constructs a confidence set using a max-type studentized statistic and a two-step critical value that recenters slack moments. The table shows three cases: (i) no instruments with $|M| = 20$, (ii) lagged patent flow as an instrument with $|M| = 20$, and (iii) all three instruments with $|M| = 40$. Appendix C provides the complete results from using no instruments, a single instrument, and all three instruments.

Table 9: Estimated parameter bounds

Setup	$\lvert M\rvert$	α	$\rho = \kappa/\beta$
No IV	20	$[-1.1084, -0.1111]$	—
PAT_{it-1}	20	$[-1.1106, -0.0966]$	$[0, 12.6250]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	40	$[-1.1160, -0.1847]$	$[0, 12.3750]$

Notes: The table reports 99 percent confidence sets for congestion α and occupancy-to-scope ratio ρ under pairwise interactions among entered areas, add and drop deviations of size two and three, and swap moments. A dash means that ratio is not identified. I further profile each parameter and find that I reject the possibility of $\alpha \geq 0$ and $\rho < 0$, which also implies that $\beta < 0$ is not possible. Table 24 reports the full instrument and subset-size grid. Table 25 repeats that grid with add and drop deviations of size two only. Inference uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

At the specification that uses all three instruments and the largest subset sample, congestion α lies in $[-1.1160, -0.1847]$. Meanwhile, the occupancy-to-scope ratio ρ lies in $[0, 12.3750]$. First, take a look at the result without instrument, each inequality is then simply a single average over all firm-years. In that average, congestion α is already negative, and the occupancy-to-scope ratio ρ is not identified. Moving on to using lagged patent flow as a single instrument already provides a non-negativeset for ρ. Finally, using all three instruments and the largest subset sample, the set is bounded above by 12.375, a tighter bound than the one using only the lagged patent as an instrument. Furthermore, from further profiling of the parameters, the 99% sets in Table 9 do reject the possibility that $\alpha \geq 0$ and $\rho < 0$, which implies that $\beta < 0$ is not possible. I also explore alternative portfolio functions, and the results of such exercises are reported in Table 10.

Competitive congestion. The baseline identified set for congestion α is $[-1.1160, -0.1847]$. The economic interpretation is that firms require compensation to enter a more crowded area. An incremental unit of inverse-hyperbolic-sine rival counts lowers the expected profit by between 0.18 and 1.12 valuation units. Since rival counts enter through the inverse hyperbolic sine, this penalty is steep in thin markets (i.e., markets with few or no competitors) and flattens once an area is saturated (i.e., markets with many competitors). That is, moving from two rivals to three changes the congestion index by about $asinh(3) - asinh(2) \approx 0.37$. At the same time, adding the 51st rival to a market of 50 changes it by about $asinh(51) - asinh(50) \approx 0.02$. As expected, the result from the moment inequalities put congestion in negative, as opposed to the positive coefficient from

the reduced-form linear probability model. As usual, since I am using an aggregative rival counts, I can only interpret α as a net effect of business stealing, litigation, and any residual unobserved area attractiveness that the first-stage projection did not absorb.

Occupancy cost relative to scope. Under pairwise interactions among entered areas, add and drop comparisons recover $\rho = \kappa/\beta$. At all three instruments and the largest subset sample, that set is $[0, 12.3750]$. Take a firm that already occupies five areas (A, B, C, D, E), and is deciding whether to occupy a sixth area, F. Pairwise interactions mean F is linked with each of A, B, C, D, and E. The firm therefore gains five new pairs. The extra scope benefit of F is β times five. If $\rho = 5$, the fee equals five pairs' worth of scope. The firm already occupies five areas, so the five new pairs from F match the fee. If $\rho = 3$, the fee equals three pairs' worth of scope, and the five new pairs more than match the fee. If $\rho = 12$, the fee equals twelve pairs' worth of scope, and the five new pairs fall short of the fee. In this case, subtracting the gain from the occupancy cost leaves a cost of seven.

Since the 99% set is $[0, 12.3750]$, values such as $\rho = 0$, $\rho = 3$, $\rho = 5$, and $\rho = 12$ remain inside the set. A value such as $\rho = 20$ does not. From further profiling of the parameters, I also find that the data allow the yearly cost κ to be zero, and they allow it to be as large as about twelve pairs' worth. The intuition of this result is that the data reject a fee so large that a firm would need more than about twelve areas already before one more area's new pairs matched that fee. Take the example of the median firm that occupies five areas. Because the set includes both $\rho < 5$ and $\rho > 5$, that median firm may already have enough pairs to match the fee, or it may not. To conclude, rival congestion and the ratio of occupancy-to-scope bound together rationalize both the observed multi-area bundling and the fact that even large electronics firms leave many topics unentered[13].

Alternative portfolio functions. Table 10 reports congestion and the occupancy object under alternative portfolio functions, at all three instruments and the largest subset sample. When the portfolio function is linear in the number of occupied areas, take again as an example the firm that already occupies five areas (A, B, C, D, E), and is deciding whether to occupy a sixth area, F. Again, the extra scope benefit of F is β. It is also the case that the extra occupancy cost of F is κ. Similar to the pairwise interactions case, add and drop moment comparisons recover $\delta = \beta - \kappa$. Suppose now that $\delta = 2$, it means that the extra scope exceeds the fee by 2. Similarly, if $\delta = 3$, then extra scope exceeds the fee by 3. From the table, the 99% set is $[1.6250, 3.3750]$. That is, values such as $\delta = 2$ and $\delta = 3$ are feasible since they remain inside the set. However, values such as $\delta = 1$ or $\delta = 4$ do not. From the table, congestion α on this row lies in $[-1.1160, -0.8500]$.

Let's now move to the case of multiplying the portfolio size by its inverse hyperbolic sine, $kasinh(k)$, where this remains feasible. First, I find that the congestion α lies in $[-1.1160, -0.2000]$, and the occupancy-to-scope ratio ρ lies in $[0, 4.1250]$. Again, take the example of the firm that already occupies five areas (A, B, C, D, E), and is deciding whether to occupy a sixth area, F. In this case, the

[13]The same pattern holds when I consider restricting $M = 2$ (Table 25 in Appendix C).

Table 10: Identified sets under alternative portfolio functions

$g(k)$	$\lvert M\rvert$	α	ρ or δ
$g(k) = \binom{k}{2}$	40	$[-1.1160, -0.1847]$	$[0, 12.3750]$
$g(k) = k$	40	$[-1.1160, -0.8500]$	$[1.6250, 3.3750]$
$g(k) = \text{asinh}(k)$	40	—	—
$g(k) = k\,\text{asinh}(k)$	40	$[-1.1160, -0.2000]$	$[0, 4.1250]$
$g(k) = \sqrt{k}$	40	—	—
$g(k) = \ln(1+k)$	40	—	—

Notes: Each row uses all three instruments and the largest subset sample ($|M| = 40$). Each row uses all three instruments, lagged R&D, assets, and patent flow, and add and drop deviations of size two and three together with the swap moments. On the pairwise rows $g(k) = \binom{k}{2}$ and on $g(k) = k\,\text{asinh}(k)$, add and drop moments depend on occupancy cost and scope through their ratio $\rho = \kappa/\beta$. On the linear row $g(k) = k$, one more area always adds the same scope benefit, so add and drop moments restrict only $\delta = \beta - \kappa$. On $g(k) = \text{asinh}(k)$, $g(k) = \sqrt{k}$, and $g(k) = \ln(1+k)$, the inequalities found no jointly feasible point, so both columns are empty and neither ρ nor δ is reported. Those empty rows are why I do not use a concave portfolio function as the main specification. A dash in the last column means that object is not identified. I further profile each parameter and find that I reject the possibility of $\alpha \geq 0$ and $\rho < 0$, which also implies that $\beta < 0$ is not possible. Inference for the last column uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

extra scope benefit of F is $\beta\big(6\text{asinh}(6) - 5\text{asinh}(5)\big) \approx 3.39\beta$. If $\rho = 2$, then the fee equals 2, and the extra scope benefit of 3.39 more than matches the fee. If $\rho = 4.125$, then the fee equals 4.125, and the extra scope benefit falls short of the fee. Finally, when extra areas enter as diminishing synergies, through the inverse hyperbolic sine, the square root, or the logarithm of portfolio size, the 99% set is empty.

Fixed and sunk costs. The occupancy cost above is a fee paid in every year the firm occupies an area. The same comparisons can also allow a separate fee paid only in the year the firm enters, i.e., a sunk cost. To measure that sunk cost separately from the fixed cost, I extend the add, drop, and swap comparisons by one period, borrowing the idea outlined by Morales et al. (2019). The deviation changes the portfolio only in one year and then follows the path the firm actually chose, so continuation values cancel except for the sunk cost that this year's choice brings forward or postpones[14]. Table 23 in Appendix B.5 shows the results of this exercise for both fixed-cost ratio ρ_f and the sunk-cost ratio ρ_s. Both ratios measure the fee against the gain from occupying more areas at once. Each row holds congestion α at one value from Table 9. Take for example the case at the midpoint of that set, when firms absent from the sample next year are dropped (Panel A), the accepted fixed-cost ratio runs from 0 to 9.2 and the accepted sunk-cost ratio from 0 to 7.4. From this exercise, a zero sunk cost is still accepted. Suppose the sunk cost is zero, then the firm only pays the fixed cost, as in the static model. At the weaker end of congestion $\alpha = -0.1847$, the fixed-cost ratio runs from 3.5 to 9.4 and the sunk-cost ratio from 0 to 6.8, still the zero sunk cost is feasible. In this case, when an extra rival reduces expected profit only a little, the results call for a larger fixed cost. Treating firms that are absent next year as having left (Panel B) yields similar interpretation. These exercise provide lessons on how far off the above static results when we

[14] Appendix B.5 derives the step by step derivation of my exercise here.

decompose ρ into the sunk-cost ratio ρ_s and the fixed-cost ratio ρ_f.

Alternative entry thresholds. In the baseline results, I define entry whenever a firm enters an area when, on average, its patents in that year exceeds the threshold $\bar{w} \approx 1\%$ of their text there (see Appendix A.3). For sensitivity analysis, I re-estimate using the same specification of the third row of Table 9 at $\tilde{w} = 0.5\%$ and $\tilde{w} = 2\%$. These two points are not the minimum point in the density distribution of LDA weights, but represent the beginning points of the left and right modes, respectively (see Figure 1). Table 26 in Appendix C.2 reports the results. At $\tilde{w} = 0.5\%$, an extra rival still negatively affects expected profit, and from further profiling of α I reject the possibility that $\alpha \geq 0$. However, I find that the occupancy-to-scope ratio ρ is not identified at $\tilde{w} = 0.5\%$. Among the chief reason of unidentified ρ at $\tilde{w} = 0.5\%$ is that 'unimportant' patent texts or texts that do not cleanly define an area are now being included. At $\tilde{w} = 2\%$, I observe that this cutoff lies above the fourth-largest share and below the third (see Figure 2). Therefore, based on this cutoff, the average firm then occupies only 4.34 areas in a year, against 5.37 at $\bar{w} \approx 1\%$. Furthermore, for this specification, I find no combination of congestion and occupancy-to-scope ratio that satisfies the inequalities, so the set is empty.

5 Case Study: Semiconductor

In these numerical exercises, I try to mimic a targeted subsidy for certain areas as outlined in the U.S. CHIPS incentive[15] and the EU Chips Act[16]. I map the areas listed in these regulations to memory, lithography and fabrication, packaging, and the transistor. There are two numerical exercises: (i) a subsidy equal to one quarter of predicted value in these four areas, and (ii) a subsidy that lowers the occupancy-to-scope ratio ρ by 25% in those areas. For each of these two exercises, firms can either keep the number of areas in their portfolios ("fixed areas") or add an extra area ("extra areas")[17]. For all the exercises, I use the result in the third row of Table 9, and the rival count in each area is the sum of the other firms' entry, updated until that sum agrees with the portfolios. The details of my numerical exercises, along with additional exercises on halving rival congestion (Table 29), applying a uniform subsidy (Table 30), and the simulation where I hold the rival counts (see Tables 27 and 28), can be found in Appendix D.

[15]The CHIPS Act of 2022, Public Law 117–167, Division A, 136 Stat. 1366 (August 9, 2022), funds assistance under 15 U.S.C. § 4652 for facilities and equipment in the United States for the fabrication, assembly, testing, advanced packaging, production, or research and development of semiconductors, of materials used to manufacture semiconductors, or of semiconductor manufacturing equipment. Public Law 119–21, 139 Stat. 72 (July 4, 2025), § 70308, raises the advanced manufacturing investment credit in Internal Revenue Code § 48D from 25 percent to 35 percent for property placed in service after December 31, 2025. That credit remains the credit for a facility whose primary purpose is manufacturing semiconductors or semiconductor manufacturing equipment.

[16]Regulation (EU) 2023/1781 of the European Parliament and of the Council of 13 September 2023, OJ L 229, 18.9.2023, pp. 1–53, Article 2.

[17]For the "extra areas" exercises, I use several values of b. Since the estimates of the model can only recover α and ρ, the level of $\beta = b$ is a scale that has to be chosen. A higher b raises the per-area occupancy cost $\kappa = b\rho$, and it also changes which pairs (α, ρ) the inequalities still accept.

5.1 Case I: a 25% subsidy based on predicted value

For this exercise, I add a quarter of predicted value, $0.25\check{v}$, for each of the four semiconductor areas. In the first column, firms can only reshuffle the areas within the portfolio they actually occupy ("fixed-size"). I find that the four semiconductor areas all gain share. Memory gains between 0.16% and 0.46%, packaging between 0.26% and 0.54%, lithography between 0.24% and 0.53%, and the transistor between 0.14% and 0.37%. Meanwhile, every other area loses share. Video moves between a loss of 0.06% and a gain of 0.01%. For example, imaging falls by 0.01% to 0.08%, optics and photonics by 0.02% to 0.07%, and analog circuits by 0.10% to 0.24%.

As for the remaining columns, the "extra-area" columns indicate that firms are allowed to add empty areas to their portfolios. I again find that the four semiconductor areas all gain share, and that the gain is smaller at a higher value of b. At $b = 0.25$, memory gains between 0.04% and 0.06%, the transistor gains between 0.05% and 0.06%, packaging gains between 0.04% and 0.05%, and lithography gains between 0.03% and 0.04%. At $b = 1$, memory gains between 0.02% and 0.03%, the transistor gains 0.02%, packaging gains between 0.00% and 0.01%, and lithography gains 0.01%. At $b = 4$, each of the four areas moves by at most 0.01%. I also find that integrated circuit design, audio, and clock circuits lose their shares.

5.2 Case II: a 25% occupancy-to-scope ratio reduction in semiconductor areas

For this exercise, I lower the occupancy-to-scope ratio by 25% in each of the four semiconductor areas. In the fixed-area columns, firms can only reshuffle the areas within the portfolio they actually occupy. I find that the four semiconductor areas gain share, and that the gain is larger at a higher value of b. At $b = 0.25$, the gains run from 0% to 1.06%. At $b = 1$, lithography gains between 5.06% and 5.86%, packaging between 5.07% and 5.80%, the transistor between 5.15% and 5.98%, and memory between 5.18% and 6.10%. At $b = 4$, memory gains between 6.36% and 7.11%, the transistor between 6.30% and 6.89%, lithography between 6.20% and 6.46%, and packaging between 6.06% and 6.28%. Meanwhile, every other area loses share at $b = 1$ and at $b = 4$. For example, at $b = 1$, display falls by 1.36% to 1.47%, imaging by 1.34% to 1.50%, and integrated circuit design by 1.38% to 1.69%.

As for the remaining columns, the extra-area columns indicate that firms are allowed to add empty areas to their portfolios. I again find that the four semiconductor areas gain share, and that the gain is larger at a higher value of b. At $b = 0.25$, each of the four areas moves by at most 0.03%. At $b = 1$, memory gains between 0.26% and 0.35%, the transistor between 0.21% and 0.28%, lithography between 0.15% and 0.21%, and packaging between 0.11% and 0.15%. At $b = 4$, memory gains between 0.48% and 0.61%, the transistor between 0.41% and 0.52%, lithography between 0.32% and 0.41%, and packaging between 0.23% and 0.31%. I also find that integrated circuit design, audio, and clock circuits lose their shares. At $b = 4$, integrated circuit design falls by 0.52% to 0.60%, audio by 0.46% to 0.54%, and clock circuits by 0.38% to 0.45%.

Table 11: Case I: a 25% subsidy based on predicted value

	Fixed size	Extra areas		
Area		$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.16, 0.46]	[0.04, 0.06]	[0.02, 0.03]	[−0.00, 0.01]
Audio and Speech-Signal Processing	[−0.09, −0.06]	[−0.03, −0.03]	[−0.02, −0.02]	[−0.02, −0.00]
Semiconductor Lithography and Fabrication Processes	[0.24, 0.53]	[0.03, 0.04]	[0.01, 0.01]	[−0.00, 0.01]
Integrated Circuit Design, Verification, and Testing	[−0.18, −0.08]	[−0.03, −0.02]	[−0.03, −0.02]	[−0.02, −0.00]
PCB and Electronic Hardware with Connectors	[−0.20, −0.09]	[−0.02, −0.02]	[−0.02, −0.01]	[−0.01, 0.00]
Video Recording, Encoding, and Media Processing	[−0.06, 0.01]	[0.00, 0.01]	[0.00, 0.02]	[0.00, 0.01]
Semiconductor Packaging and Interconnect Bonding	[0.26, 0.54]	[0.04, 0.05]	[0.00, 0.01]	[0.00, 0.01]
Thin-Film Materials, Deposition and Chemistry	[−0.15, −0.05]	[−0.01, −0.00]	[−0.01, 0.00]	[−0.00, 0.00]
Wireless Base-Station and Mobile Communications	[−0.10, −0.02]	[−0.01, −0.00]	[0.00, 0.01]	[0.00, 0.01]
Clock Generation, Timing Circuits, and Signal Processing	[−0.13, −0.04]	[−0.03, −0.03]	[−0.02, −0.01]	[−0.01, −0.00]
Power-Supply and Power-Management Electronics	[−0.13, −0.06]	[−0.03, −0.02]	[−0.02, −0.01]	[−0.01, −0.00]
Networking Protocols and Data Infrastructure	[−0.08, −0.01]	[−0.01, 0.00]	[0.00, 0.01]	[0.00, 0.01]
Transistor Gate Structures and Semiconductor Device Physics	[0.14, 0.37]	[0.05, 0.06]	[0.02, 0.02]	[−0.00, 0.01]
Imaging Sensors and Camera Pipelines	[−0.08, −0.01]	[0.01, 0.02]	[0.01, 0.02]	[−0.00, 0.01]
Processor Architecture, Caches, Instruction Execution	[−0.11, −0.03]	[−0.02, −0.01]	[−0.01, −0.00]	[−0.01, 0.00]
Display Panel Tech (LCD/OLED)	[−0.07, −0.03]	[0.01, 0.02]	[0.03, 0.03]	[0.00, 0.02]
Rotational Electromechanics and Magnetic Actuation	[−0.15, −0.09]	[−0.02, −0.01]	[−0.01, −0.01]	[−0.01, 0.00]
Optics, Lasers, and Photonics	[−0.07, −0.02]	[−0.01, −0.01]	[0.01, 0.01]	[0.00, 0.01]
Analog and Differential Amplifier Circuits	[−0.24, −0.10]	[−0.03, −0.02]	[−0.02, −0.02]	[−0.01, −0.00]
Mobile UI and Content Services	[−0.11, −0.03]	[−0.02, −0.02]	[−0.01, −0.00]	[−0.01, −0.00]
Firm-years adding an area	0	[0.76, 0.83]	[0.34, 0.43]	[0.07, 0.19]

Notes: Each area row is the change, in percentage points, in that area's share of occupied slots. The fixed-size column holds the number of areas at the observed count. The extra-area columns start from the observed portfolio and keep occupied areas occupied. The last row is the change in the share of firm-years that add at least one area. Rival counts are those implied by the portfolios.

Table 12: Case II: a 25% occupancy-to-scope ratio reduction

	Fixed size			Extra areas		
Area	$b = 0.25$	$b = 1$	$b = 4$	$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.00, 1.00]	[5.18, 6.10]	[6.36, 7.11]	[−0.00, 0.03]	[0.26, 0.35]	[0.48, 0.61]
Audio and Speech-Signal Processing	[−0.20, 0.00]	[−1.51, −1.28]	[−1.73, −1.57]	[−0.02, 0.00]	[−0.29, −0.24]	[−0.54, −0.46]
Semiconductor Lithography and Fabrication Processes	[0.00, 1.04]	[5.06, 5.86]	[6.20, 6.46]	[0.00, 0.02]	[0.15, 0.21]	[0.32, 0.41]
Integrated Circuit Design, Verification, and Testing	[−0.23, 0.00]	[−1.69, −1.38]	[−2.18, −1.73]	[−0.01, 0.00]	[−0.32, −0.25]	[−0.60, −0.52]
PCB and Electronic Hardware with Connectors	[−0.20, 0.00]	[−1.53, −1.30]	[−1.83, −1.62]	[−0.01, 0.00]	[−0.23, −0.17]	[−0.35, −0.30]
Video Recording, Encoding, and Media Processing	[−0.34, 0.00]	[−1.47, −1.30]	[−1.55, −1.47]	[−0.00, 0.00]	[0.11, 0.14]	[0.18, 0.21]
Semiconductor Packaging and Interconnect Bonding	[0.00, 1.02]	[5.07, 5.80]	[6.06, 6.28]	[0.00, 0.03]	[0.11, 0.15]	[0.23, 0.31]
Thin-Film Materials, Deposition and Chemistry	[−0.23, 0.00]	[−1.33, −1.16]	[−1.56, −1.42]	[−0.00, 0.00]	[−0.02, 0.01]	[−0.02, −0.00]
Wireless Base-Station and Mobile Communications	[−0.32, 0.00]	[−1.55, −1.36]	[−1.66, −1.59]	[−0.00, 0.00]	[0.02, 0.05]	[0.04, 0.08]
Clock Generation, Timing Circuits, and Signal Processing	[−0.18, 0.00]	[−1.38, −1.15]	[−1.63, −1.48]	[−0.02, 0.00]	[−0.24, −0.19]	[−0.45, −0.38]
Power-Supply and Power-Management Electronics	[−0.23, 0.00]	[−1.58, −1.30]	[−2.08, −1.67]	[−0.01, 0.00]	[−0.20, −0.17]	[−0.34, −0.30]
Networking Protocols and Data Infrastructure	[−0.30, 0.00]	[−1.42, −1.28]	[−1.52, −1.48]	[−0.00, 0.00]	[0.05, 0.08]	[0.09, 0.11]
Transistor Gate Structures and Semiconductor Device Physics	[0.00, 1.06]	[5.15, 5.98]	[6.30, 6.89]	[0.00, 0.03]	[0.21, 0.28]	[0.41, 0.52]
Imaging Sensors and Camera Pipelines	[−0.35, 0.00]	[−1.50, −1.34]	[−1.58, −1.49]	[−0.00, 0.01]	[0.13, 0.15]	[0.19, 0.24]
Processor Architecture, Caches, Instruction Execution	[−0.23, 0.00]	[−1.41, −1.21]	[−1.60, −1.49]	[−0.01, 0.00]	[−0.07, −0.06]	[−0.16, −0.11]
Display Panel Tech (LCD/OLED)	[−0.30, 0.00]	[−1.47, −1.36]	[−1.53, −1.34]	[−0.00, 0.01]	[0.29, 0.35]	[0.52, 0.60]
Rotational Electromechanics and Magnetic Actuation	[−0.23, 0.00]	[−1.50, −1.26]	[−1.83, −1.61]	[−0.01, 0.00]	[−0.15, −0.11]	[−0.25, −0.21]
Optics, Lasers, and Photonics	[−0.35, 0.00]	[−1.46, −1.31]	[−1.53, −1.39]	[−0.01, 0.00]	[0.05, 0.07]	[0.17, 0.20]
Analog and Differential Amplifier Circuits	[−0.20, 0.00]	[−1.47, −1.20]	[−1.78, −1.53]	[−0.02, 0.00]	[−0.17, −0.14]	[−0.27, −0.23]
Mobile UI and Content Services	[−0.25, 0.00]	[−1.47, −1.27]	[−1.71, −1.54]	[−0.01, 0.00]	[−0.13, −0.10]	[−0.26, −0.20]
Firm-years adding an area	0	0	0	[0.00, 0.34]	[4.10, 5.51]	[8.84, 10.43]

Notes: Each area row is the change, in percentage points, in that area's share of occupied slots. The fixed-size column holds the number of areas at the observed count. The extra-area columns start from the observed portfolio and keep occupied areas occupied. The last row is the change in the share of firm-years that add at least one area. Rival counts are those implied by the portfolios.

6 Conclusion

This paper develops a data-driven framework to analyze the strategic entry of U.S. electronics firms into multiple innovation areas. I define those areas from patent text, so that a single invention can sit in several domains with explicit weights. Applying Latent Dirichlet Allocation (LDA) to a large corpus of titles and abstracts yields twenty innovation areas. The resulting areas display two stylized facts: a multi-bundle innovation in the electronics industry, and the heterogeneous congestion effect. To identify the mechanisms behind these patterns, I construct an entry model and estimate the parameters by partial identification, using moment inequalities to handle bundling, equilibrium multiplicity, and unobserved firm heterogeneity.

A reduced-form linear probability model of whether a firm occupies an area reveals three patterns: (i) a higher predicted patent valuation raises the probability of entry, (ii) a larger number of rivals also raises that probability, and (iii) there is a positive relationship between portfolio size and the probability of entry. However, crowded areas could be crowded because they are attractive, so this positive relation mixes a congestion effect with the unobserved attractiveness of those areas. Similarly, the coefficient on portfolio size is about one tenth. With twenty areas, that is the rate implied by counting a firm as present in each area already included in its portfolio. A more capable firm also occupies more areas and enters any given area more often, so the reduced-form result does not isolate a gain from scope.

The model estimates separate the above concerns. Taking predicted patent value as the unit of profit, I find that additional rivals reduce expected profit. The yearly occupancy cost, measured against the gain from occupying more areas at once, can be zero, and it can be large. Negative values of that comparison are rejected. The penalty from rivals is steep when an area has few rivals and flattens once the area is crowded. Firms also gain from occupying more areas at once, through the pairwise interactions among the areas they enter. Treating extra areas as diminishing synergies is inconsistent with the data.

In a numerical simulation, I try to mimic the areas covered by the U.S. CHIPS manufacturing incentive and the EU CHIPS Act. In particular, the coverage of those policies are memory, lithography and fabrication, packaging, and the transistor. A payment equal to one quarter of the predicted valuations shifts 0.1–0.5% into those areas and leads up to 0.8% of firms to add an extra area. Meanwhile, a 25% reduction in the occupancy cost shifts up to 7% into those areas and increases up to 10% of firms to add an extra area.

The paper has three limitations. First, the model is a static game. It answers how a firm composes its portfolio in a given year and how rival counts in that year shift expected profit. The occupancy cost recovered from those comparisons is a fixed cost, paid in each year of occupancy. The game abstracts from learning, option value, preemption, and sunk costs of R&D. If firms occupy an area in part because it makes later expansion cheaper, that occupancy cost is net of that benefit and understates the per-year occupancy cost. If entry is sunk and waiting is valuable, the same

comparisons overstate that cost. The same revealed-preference argument identifies a sunk cost of entering an area. I compare the observed path with a counterfactual that differs in only one year and then follows the firm's actual subsequent choices. Appendix B.5 carries out that comparison. It recovers how large the fixed cost and the sunk cost are relative to the gain from occupying more areas at once. Second, the sample consists of listed U.S. electronics firms from 1990 to 2019. The findings and the counterfactuals are a case study of that industry. Startups and privately held innovators are outside the sample, because market-implied patent values require a traded return. Third, the local cost and congestion simulation exercises start from observed portfolios, and the rival count in each area is the sum of the other firms' entry, updated until that sum agrees with the portfolios. Appendix D reports the same exercises with rival counts held at their observed values.

References

P. Aghion, N. Bloom, R. Blundell, R. Griffith, and P. Howitt. Competition and innovation: An inverted-U relationship. *The Quarterly Journal of Economics*, 120(2):701–728, 2005.

A. Arora, S. Belenzon, and L. Sheer. Matching patents to compustat firms, 1980–2015: Dynamic reassignment, name changes, and ownership structures. *Research Policy*, 2021.

S. Arts, B. Cassiman, and J. C. Gomez. Text matching to measure patent similarity. *Strategic Management Journal*, 39(1):62–84, 2018.

S. Arts, J. Hou, and J. C. Gomez. Natural language processing to identify the creation and impact of new technologies in patent text: Code, data, and new measures. *Research Policy*, 2021.

E. Ash and S. Hansen. Text algorithms in economics. *Annual Review of Economics*, 15:659–688, 2023.

P. J. Barwick, M. Kalouptsidi, and N. B. Zahur. Industrial policy implementation: Empirical evidence from China's shipbuilding industry. *The Review of Economic Studies*, 92(6):3611–3648, 2025.

G. Bellstam, S. Bhagat, and J. A. Cookson. A text-based analysis of corporate innovation. *Management Science*, 2021.

J. Bischof and E. M. Airoldi. Summarizing topical content with word frequency and exclusivity. In *Proceedings of the 29th International Conference on Machine Learning (ICML)*, pages 201–208, Edinburgh, Scotland, 2012. Omnipress. URL `https://icml.cc/Conferences/2012/papers/113.pdf`.

D. M. Blei and J. D. Lafferty. Dynamic topic models. In *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, pages 113–120. ACM, 2006. doi: 10.1145/1143844.1143859.

D. M. Blei, A. Y. Ng, and M. I. Jordan. Latent dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022, 2003.

N. Bloom, M. Schankerman, and J. Van Reenen. Identifying technology spillovers and product market rivalry. *Econometrica*, 2013.

N. Bloom, J. Van Reenen, and H. Williams. A toolkit of policies to promote innovation. *Journal of Economic Perspectives*, 33(3):163–184, 2019.

S. Breschi, F. Lissoni, and F. Malerba. Knowledge-relatedness in firm technological diversification. *Research Policy*, 32(1):69–87, 2003.

T. F. Bresnahan and P. C. Reiss. Entry and competition in concentrated markets. *Journal of Political Economy*, 99(5):977–1009, 1991.

I. A. Canay and A. M. Shaikh. Practical and theoretical advances in inference for partially identified models. In B. Honoré, A. Pakes, M. Piazzesi, and L. Samuelson, editors, *Advances in Economics and Econometrics: Eleventh World Congress*, Econometric Society Monographs, pages 271–306. Cambridge University Press, Cambridge, 2017.

I. A. Canay, G. Illanes, and A. Velez. A user's guide for inference in models defined by moment inequalities. *Journal of Econometrics*, 257:105558, 2023. doi: 10.1016/j.jeconom.2023.105558.

J. Chang, S. Gerrish, C. Wang, J. L. Boyd-Graber, and D. M. Blei. Reading tea leaves: How humans interpret topic models. In *Advances in Neural Information Processing Systems 22 (NIPS 2009)*, pages 288–296. MIT Press, 2009. URL `https://papers.nips.cc/paper/3700-reading-tea-leaves-how-humans-interpret-topic-models`.

J. Chen, K. Li, J. Zhu, and W. Chen. Warplda: a cache efficient o(1) algorithm for latent dirichlet allocation. *Proceedings of the VLDB Endowment*, 9(10):744–755, 2016. doi: 10.14778/2977797.2977801.

V. Chernozhukov, D. Chetverikov, and K. Kato. Inference on causal and structural parameters using many moment inequalities. *The Review of Economic Studies*, 2019.

C. A. Chikis, B. Kleinman, and M. Prato. The geography of innovative firms. NBER Working Paper 34010, National Bureau of Economic Research, 2025.

F. Ciliberto and E. Tamer. Market structure and multiple equilibria in airline markets. *Econometrica*, 77(6):1791–1828, 2009. doi: 10.3982/ECTA5368.

M. J. Dickstein and E. Morales. What do exporters know? *The Quarterly Journal of Economics*, 2018.

A. Dyevre and O. Seager. 70 years of patents matched to compustat firms: Methodology and insights about firm heterogeneity. *Working Paper*, 2023.

L. Eisfeld. Entry and acquisitions in software markets. *Working Paper*, 2025.

A. Eizenberg. Upstream innovation and product variety in the u.s. home pc market. *The Review of Economic Studies*, 2014.

L. Fleming. Recombinant uncertainty in technological search. *Management Science*, 47(1):117–132, 2001.

M. Gentzkow, B. Kelly, and M. Taddy. Text as data. *Journal of Economic Literature*, 57(3):535–574, 2019.

P. K. Goldberg, R. Juhász, N. J. Lane, G. Lo Forte, and J. Thurk. Industrial policy in the global semiconductor sector. NBER Working Paper 32651, National Bureau of Economic Research, 2024.

P. L. E. Grieco. Discrete games with flexible information structures: An application to local grocery markets. *The RAND Journal of Economics*, 45(2):303–340, 2014. doi: 10.1111/1756-2171.12052.

D. S. Hain, R. Jurowetzki, T. Buchmann, and P. Wolf. A text-embedding-based approach to measuring patent-to-patent technological similarity. *Technological Forecasting and Social Change*, 177: 121559, 2022.

B. H. Hall and R. H. Ziedonis. The patent paradox revisited: An empirical study of patenting in the U.S. semiconductor industry, 1979–1995. *The RAND Journal of Economics*, 32(1):101–128, 2001. URL `https://www.jstor.org/stable/2696400`.

R. Henderson and I. Cockburn. Scale, scope, and spillovers: The determinants of research productivity in drug discovery. *The RAND Journal of Economics*, 27(1):32–59, 1996. URL `https://www.jstor.org/stable/2555791`.

G. Hoberg and G. Phillips. Text-based network industries and endogenous product differentiation. *Journal of Political Economy*, 2016.

A. B. Jaffe. Technological opportunity and spillovers of R&D: Evidence from firms' patents, profits, and market value. *American Economic Review*, 76(5):984–1001, 1986.

R. Juhász, N. Lane, and D. Rodrik. The new economics of industrial policy. *Annual Review of Economics*, 16:213–242, 2024.

A. Kalyani, N. Bloom, M. Carvalho, T. A. Hassan, J. Lerner, and A. Tahoun. The diffusion of new technologies. *The Quarterly Journal of Economics*, 140(2):1299–1365, 2025.

S. Kaplan and K. Vakili. The double-edged sword of recombination in breakthrough innovation. *Strategic Management Journal*, 2015.

B. Kelly, D. Papanikolaou, A. Seru, and M. Taddy. Measuring technological innovation over the long run. *American Economic Review: Insights*, 3(3):303–320, 2021. doi: 10.1257/aeri.20190499.

T. J. Klette and S. Kortum. Innovating firms and aggregate innovation. *Journal of Political Economy*, 112(5):986–1018, 2004.

L. Kogan, D. Papanikolaou, A. Seru, and N. Stoffman. Technological innovation, resource allocation, and growth. *The Quarterly Journal of Economics*, 132(2):665–712, 2017. doi: 10.1093/qje/qjw040.

J. Lu, M. Henchion, and B. Mac Namee. Extending jensen shannon divergence to compare multiple corpora. In *Proceedings of the 25th Irish Conference on Artificial Intelligence and Cognitive Science (AICS)*, pages 76–88, Dublin, Ireland, 2017. URL `https://ceur-ws.org/Vol-2086/AICS2017_paper_20.pdf`.

D. J. Miller. Technological diversity, related diversification, and firm performance. *Strategic Man-*

agement Journal, 27(7):601–619, 2006.

D. Mimno, H. M. Wallach, E. M. Talley, M. Leenders, and A. McCallum. Optimizing semantic coherence in topic models. In *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 262–272, Edinburgh, Scotland, UK, July 2011. Association for Computational Linguistics. URL `https://aclanthology.org/D11-1024`.

E. Morales, G. Sheu, and A. Zahler. Extended gravity. *The Review of Economic Studies*, 2019.

A. Niekler and P. Jähnichen. Matching results of latent dirichlet allocation for text. In *Proceedings of the 11th International Conference on Cognitive Modeling (ICCM 2012)*, pages 317–322, Berlin, 2012. Universitätsverlag der TU Berlin.

A. Pakes. Alternative models for moment inequalities. *Econometrica*, 2010.

A. Pakes, J. Porter, K. Ho, and J. Ishii. Moment inequalities and their application. *Econometrica*, 83 (1):315–334, 2015. doi: 10.3982/ECTA6865.

E. Rawley and T. S. Simcoe. Diversification, diseconomies of scope, and vertical contracting: Evidence from the taxicab industry. *Management Science*, 56(9):1534–1550, 2010. doi: 10.1287/mnsc.1100.1207.

J. J. Rotemberg and G. Saloner. Benefits of narrow business strategies. *The American Economic Review*, 1994.

A. Seru. Firm boundaries matter: Evidence from conglomerates and r&d activity. *Journal of Financial Economics*, 2014.

E. Tamer. Incomplete simultaneous discrete response model with multiple equilibria. *The Review of Economic Studies*, 2003.

Y. W. Teh, M. I. Jordan, M. J. Beal, and D. M. Blei. Sharing clusters among related groups: Hierarchical dirichlet processes. In *Advances in Neural Information Processing Systems 17 (NIPS 2004)*, pages 1385–1392. MIT Press, 2004.

F. Teodoridis, J. Lu, and J. Furman. Mapping the knowledge space: Exploiting unassisted machine learning tools. *Working Paper*, 2022.

X. Wang and A. McCallum. Topics over time: A non-markov continuous-time model of topical trends. In *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 424–433, Philadelphia, PA, USA, 2006. ACM. URL `https://dl.acm.org/doi/10.1145/1150402.1150450`.

M. L. Weitzman. Recombinant growth. *The Quarterly Journal of Economics*, 113(2):331–360, 1998.

T. G. Wollmann. Trucks without bailouts: Equilibrium product characteristics for commercial vehicles. *American Economic Review*, 108(6):1364–1406, 2018.

A Data, Methodology, and Stylized Facts

A.1 Latent Dirichlet Allocation (LDA)

Before performing LDA, I first combine the title of the patent and its abstract[18]. Then, I perform preprocessing on the combined text[19]. I proceed to tokenize the combined text, create a set of 'vocabs' consisting of unigrams (a single word) and bigrams (a combination of two words) that will be considered (V), and perform vocab-pruning[20]. This process leaves me with 7,588 vocabs V with other information such as how many times the term appears in the whole corpus (term-frequency) and how many documents have the term (document-frequency)[21]. From here, I construct the document-term matrix of dimension $P \times V$, where P is the number of patents, that will be used as an input for fitting the LDA.

I apply LDA to model electronics patents as mixtures of latent topics, such as semiconductors or wireless communication, each characterized by a distribution over words. Sampling recovers two outputs: a document-topic distribution that assigns topic proportions to each patent, and a topic-word distribution that links words to topics. Besides the document-term matrix, LDA takes three hyperparameters: the number of topics K; the Dirichlet prior τ on patent-topic proportions, which governs how many topics a patent is allowed to cover; and the Dirichlet prior ω on topic-word distributions, which governs how concentrated each topic's vocabulary is. In the notation of Blei et al. (2003), τ is their document-topic prior and ω is their topic-word prior.

A Dirichlet prior on the simplex produces sparse mixtures when its parameters are below one and more even mixtures when they exceed one. Each patent p draws topic shares, and each topic j draws word probabilities,

$$w_{p\cdot t} \sim \text{Dirichlet}(\tau), \qquad \psi_j \sim \text{Dirichlet}(\omega).$$

Words are then drawn by first sampling a topic and then sampling a word from that topic. Posterior means have the usual Dirichlet-multinomial form: topic shares mix the prior τ with token counts, and word probabilities mix the prior ω with word counts. I recover those posteriors with the WarpLDA sampler of Chen et al. (2016), which assigns tokens by Metropolis-Hastings. The same Dirichlet-multinomial structure underlies hierarchical, correlated, and dynamic topic models (Teh et al., 2004; Blei and Lafferty, 2006). Those extensions introduce additional hyperparameters that I do not estimate here.

Among the chief challenges of LDA is to identify the optimal hyperparameters, in particular, the number of topics K. One might also argue that different initial priors of τ and ω can yield different results. I try to mitigate these concerns by computing several measures, summarized in Table 13, that guide the selection of the optimal hyperparameters. The first one is 'perplexity,' which

[18] The results of my topic modeling are robust if I include title, abstract, and claims, at the cost of higher RAM usage and a significantly longer computation time. Including description is not feasible on my personal machine; that is, the data is about 33.6 GB, and it is difficult to perform topic modeling given the data size and the machine limitation.

[19] First, I translate the combined text into lower case, remove any punctuation and stopwords from the combined text, and I remove all standalone numbers (one or more digits) or letters from the combined text. Finally, I lemmatize the combined text. Example of sentence "Activity 1: I bought 2 coffee makers and the machine is amazing!". After the preprocessing, it becomes "activity buy coffee maker machine amazing".

[20] I set the following restriction: 1. Both unigrams and bigrams must appear at a maximum of 10% of my patent sample. This reduces the redundant words, such as 'invent', being considered in the final set of vocabs V, 2. Both unigrams and bigrams must appear at least 0.1% of my patent sample. The idea of this restriction is to limit overtly technical words, such as 'Fe_3O_4' or 'SiC', being considered in vocabs V.

[21] Using title, abstract, and claims, I have 12,788 vocabs.

measures how well the model predicts a held-out test set. It is conceptually similar to the inverse likelihood: a lower perplexity score implies that the model generalizes better to unseen data. However, perplexity does not necessarily align with human interpretability, especially in large corpora like patents (see Chang et al. (2009) for detailed discussion on the measurement). For this measure, I will be looking for an 'elbow', an inflection point where the decrease in perplexity is marginal compared to the increasing number of topics being considered.

Table 13: LDA Evaluation Metrics

Measures	**Purpose**	**Goal**	**Example**
Perplexity	Out-of-sample predictive fit	Lower	"Elbow"
Coherence	Within-topic semantic tightness	Higher	Mean NPMI: $0.2 - 0.3$
Exclusivity	Distinctness between topics	Higher	≥ 0.45
JSD	How much each topic's meaning drifts	Lower	≤ 0.3
Drifted Share	Fraction of topics whose top-10 words change substantially	Lower	max 40% topics drifted

To complement perplexity, I consider coherence, which evaluates how frequently top words within each topic co-occur across the documents. Coherence serves as a proxy for interpretability: higher coherence suggests that the top words in a topic form a semantically meaningful group, increasing confidence that the topic reflects a real innovation cluster (e.g., "chip," "circuit," "transistor" likely point to semiconductors). In particular, I will be using the NPMI-based coherence in evaluating the topic quality (see Mimno et al. (2011) for detailed discussion on the measurement). I optimize the topic model's hyperparameters to achieve a mean NPMI between 0.2 and 0.3, while also ensuring that the minimum NPMI remains around 0.1, to preserve the interpretability of even the weakest topics.

In addition to coherence, I include exclusivity, which assesses how uniquely a word belongs to a single topic, i.e., topic distinctiveness. A high exclusivity score indicates that top words of a topic do not overlap significantly with other topics, thus ensuring distinct innovation areas. While coherence favors commonality among words, exclusivity ensures separation between topics, both of which are desirable but often trade off (see Bischof and Airoldi (2012) for more discussions).

Beyond internal topic quality metrics, I also assess the semantic stability of topics over time. To evaluate this, I split the corpus into three time waves (e.g., 1990-1999, 2000-2009, and 2010–2019), which generally capture the standard narrative of technological evolution in the electronics industry, and compute the Jensen-Shannon divergence (JSD) between the topic-word distributions estimated in each wave. The JSD is a symmetric measure of how dissimilar the vocabulary of each topic is across time periods. In my implementation, JSD can take values in the range of zero and one. A low JSD indicates that topics retain similar meanings over time, which is desirable for maintaining a consistent mapping between topics and innovation areas. In particular, I am looking for the mean JSD below 0.3 when considering the optimal hyperparameters[22].

[22]See for example Lu et al. (2017). Their treatment of the extended JSD measure is applied to compare vast Twitter

In addition, I define a measure called drifted share, which calculates the proportion of topics whose top keywords change substantially between time waves. A low drifted share implies that most topics remain stable in meaning, reinforcing their interpretation as persistent innovation domains. For example, there should be a maximum of 40% topics with share drifted more than 30% across period, when evaluating the optimal hyperparameter. Together, JSD and drifted share help identify topic models that are not only coherent and exclusive, but also temporally stable, which is essential for defining innovation areas and for the model implementation used in the next part of this paper[23]. One could also think of this as a way to mitigate the concerns raised in the dynamic, correlated topic model, as discussed by Blei and Lafferty (2006).

To find the optimal hyperparameters, I employ the following procedure (See Algorithm 1 in Appendix A.4, and Appendix A.5 for the fine-tuning result). First, I employ a 5-fold cross validation. That is, for each year, I randomly divide the patents into five parts. I will then fit the LDA for four parts and use the estimated model for the other part when evaluating and optimizing the hyperparameters. I repeat this exercise five times, each with different folds as training and validation sets. During this cross validation, I begin the iterative search[24]. In the first step, I grid search for the number of optimal topics K, while keeping other hyperparameters (τ and ω) fixed. Based on the evaluation metrics described above, I identify the optimal range for K. Next, I search over values of τ, again holding the other parameters fixed, and select its optimal range using the same set of metrics, then obtain the optimal τ and its bounds. I then repeat the process for ω. This procedure is iterated until further refinements to the hyperparameters do not yield significant improvements in the evaluation metrics.

As discussed earlier, the outputs of the LDA model are the document-topic and topic-word distributions. For each topic, I extract the top 25 most probable words and assign an interpretable label to the topic. To assist with labeling, I utilize two large language models (OpenAI's ChatGPT o3 and xAI's Grok 3 + Think) to propose candidate labels based on the top words associated with each topic. I then iterate between the two models until their suggested labels converge or are manually reconciled, using the agreed-upon label as the final topic label.

A.2 From topic proportions to area weights

The document-topic posterior derived above is the source of the patent-area weights used in Section 3. For a patent p filed by firm i in period t, the Dirichlet-multinomial posterior mean is a distribution over the J areas,

$$w_{pjt} \;=\; \frac{n_{pj} + \tau}{\sum_{j'=1}^{J} (n_{pj'} + \tau)},$$

where n_{pj} is the number of tokens in patent p assigned to area j, τ is the Dirichlet prior on patent-topic proportions, and $J = 20$ in the fitted model. By construction of the Dirichlet,

$$w_{pjt} \geq 0 \qquad \text{and} \qquad \sum_{j=1}^{J} w_{pjt} = 1.$$

(now X) corpora across different time periods.

[23] I employ two procedures to match the distance matrix: greedy matching and Hungarian matching. Greedy matching does not require a 1-to-1 match between topics in one period to another, while Hungarian matching will enforce a 1-to-1 match. See Wang and McCallum (2006), Niekler and Jähnichen (2012), and Lu et al. (2017) for discussions.

[24] This algorithm is conceptually similar to the Expectation-Maximization (EM) algorithm in statistics, or to coordinate ascent and iterative grid search methods used in machine learning.

A single patent can therefore assign positive weight to several areas at once, with the weights giving the patent's proportionate exposure to each area.

When I move from the patent to the firm-year objects used in estimation, I set any weight below a threshold $\bar{w}$ to zero, so that a patent does not contribute to areas where its estimated presence is negligible[25]. The remaining fitted weights are left as they are. The area-specific dollar value of the patent is then the product

$$A_{ipjt} \;=\; w_{pjt}\, A_{ipt},$$

where A_{ipt} is the market-implied patent value constructed in Appendix B.1. Summing over the firm's patents in the period gives the realized area valuation that enters the first-stage projection,

$$\sum_{p \in \mathcal{P}_{it}} A_{ipjt} \;=\; \sum_{p \in \mathcal{P}_{it}} w_{pjt}\, A_{ipt}.$$

The same weights define the firm-year entry indicator in Section 3: firm i is defined as entering area j in period t when the average patent weight in that area is at least higher or equal to the threshold $\bar{w}$[26],

$$d_{ijt} \;\equiv\; \mathbf{1}\left\{ \frac{1}{|\mathcal{P}_{it}|} \sum_{p \in \mathcal{P}_{it}} w_{pjt} \geq \bar{w} \right\}, \qquad \bar{w} = 0.01.$$

A.3 The entry threshold

The cutoff $\bar{w}$ is constructed from the patent-level weights w_{pjt}. First, I find that many of those weights are exact zeros, i.e., the fitted model assigns none of that patent's text to the area. On the strictly positive weights, $w_{pjt} > 0$, I take the natural logarithm $\ln(w_{pjt})$. Let f be the density of this logarithm, and let $m_L < m_R$ be its two modes, the tops of the two concentrations in Figure 1. A point x on the horizontal axis of that figure is a possible value of the logarithm, and $f(x)$ is the height of the density at x. The low point between the modes is the logarithm at which positive weights are least common,

$$x^* \;=\; \arg \min_{x \in (m_L, m_R)} f(x),$$

and the share at that point is e^{x^*}. In these weights e^{x^*} is about $0.0094 \approx 0.01$ or $\bar{w} = 1\%$. I then define entry as the proportion of firm-year averages that are above this cutoff,

$$d_{ijt} \;=\; \mathbf{1}\left\{ \frac{1}{|\mathcal{P}_{it}|} \sum_{p \in \mathcal{P}_{it}} w_{pjt} \geq \bar{w} \right\}.$$

Figure 1 shows the density f. I keep every patent-area weight that is strictly positive, replace it by its natural logarithm, and group those logarithms into bins of equal width. The height of a bin is the fraction of positive weights whose logarithm falls in the bin. The lines denote the shares of 0.5%, 1%, 2%, and 5%. The solid red line is $\ln(0.01)$, i.e., where we find the lowest point between the two modes.

Reading Figure 1 from left to right, a point further to the right is a larger share of the patent's text, and the height of the curve is how often that share appears. The left hill is the pile of small positive shares. Once a patent has placed most of its text in one or two areas, a little weight is often left

[25] See Appendix A.3 for further details.

[26] See Appendix A.3 for further details.

over on other areas. Those leftover shares are common, and they sit well below 1 percent. The right hill is the pile of shares that account for the patent itself, from several percent of the text up to nearly the whole patent. Shares near 1 percent are the least common size in between, so a cutoff there keeps the shares that make up the patent and sets aside the leftover tail. The line at one half of one percent is still on the left hill, among those leftovers. The line at 5 percent is already on the right hill, among the shares that describe what the patent is about.

I also provide Figure 2 where it sorts the twenty weights inside each patent from largest to smallest. At each rank I plot the median and the mean across patents. The right panel begins at the third-largest weight, so that the same four lines can be read against the smaller ranks. In the median patent the largest weight is about 61% of the text, the second about 23%, the third about 8%, and the fourth just about 0.9%. The 1% line falls at that fourth weight.

Figure 1: Density of the natural log of positive patent-area weights. The solid line is a share of 1 percent.

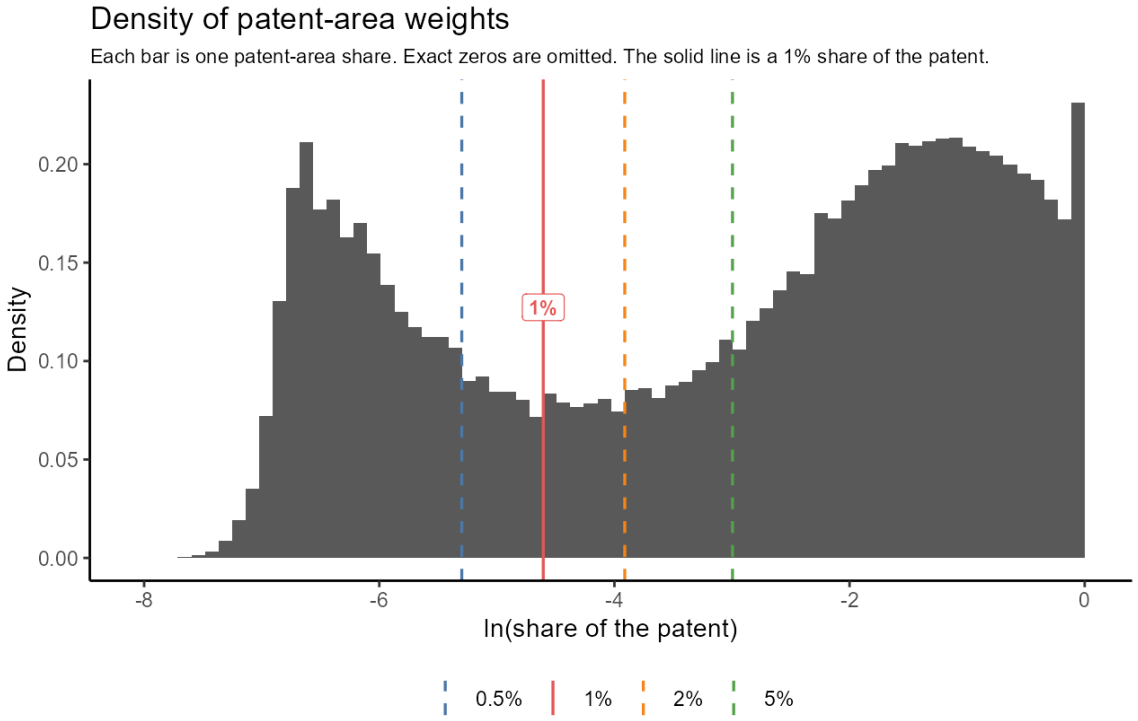
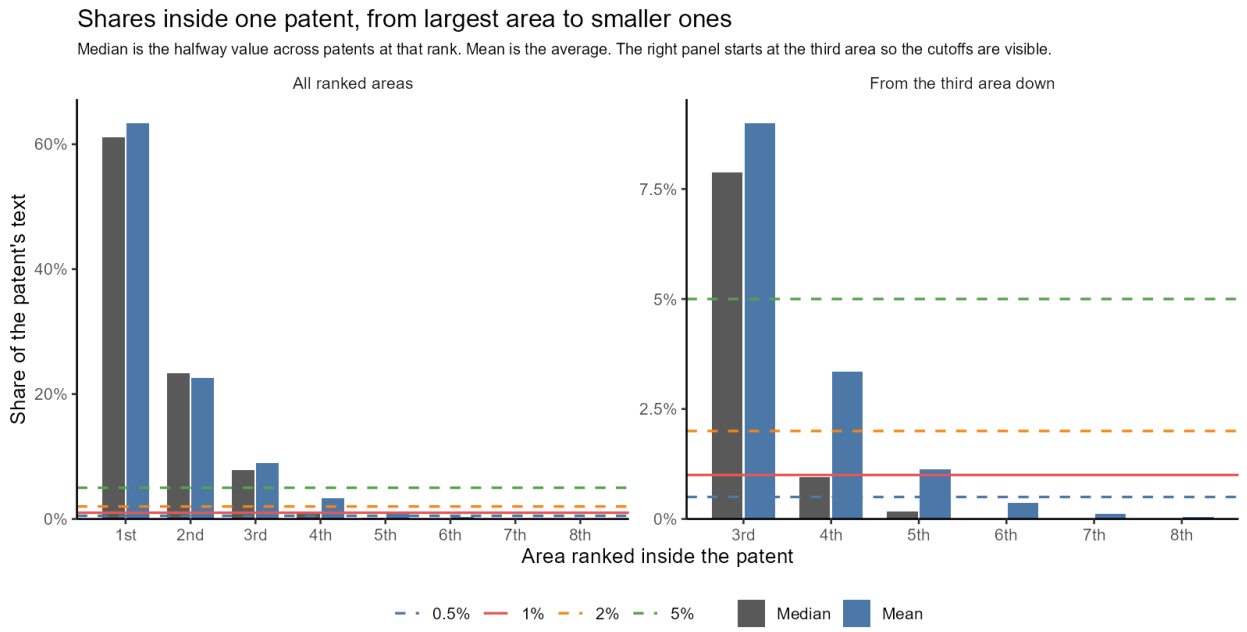


Figure 2: Patent-area weights ranked from largest to smallest within each patent. Gray bars are medians and blue bars are means.



A.4 LDA Algorithm

Algorithm 1 Heuristic Hyperparameter Tuning Procedure for LDA

Input: Document-term matrix (DTM), Term-Co-occurence matrix (TCM), initial values for (K, τ, ω)
Set seed for the random number generator, e.g., 1234
Split data using 5-fold cross validation
Fit LDA on the set of training data
Cross-validate on set of test data
repeat
    Step 1: Tune K ▷ Grid search over K, fix τ, ω
        Evaluate perplexity, coherence, exclusivity, JSD, drifted share
        Update $K \leftarrow K^*$
    Step 2: Tune τ ▷ Grid search over τ, fix K, ω
        Evaluate same metrics
        Update $\tau \leftarrow \tau^*$
    Step 3: Tune ω ▷ Grid search over ω, fix K, τ
        Evaluate same metrics
        Update $\omega \leftarrow \omega^*$
until No significant improvement in evaluation metrics
Output: Optimal (K^*, τ^*, ω^*)

Important notions:

- Document-term matrix (DTM) will be used as the main input for fitting LDA. Dimension:

$P \times V$, where P is the number of patents and V is the number of vocabs.

- Term Co-occurence matrix (TCM) is an important input to compute one of the evaluation metrics 'coherence' of a topic. Dimension $V \times V$. To construct TCM, I do the following:
  - First, convert DTM into a binary matrix B of shape $P \times V$.
  - Then, TCM is simply $B^T \cdot B$.

A.5 Fine-tuning LDA

As indicated in Algorithm 1, I first start with grid search of the number of topics K. I do this sequence in two parts; first, I sequence from 5 to 100, and qualitatively evaluate the metrics explained earlier in Section 2.2. Then, I sequence from 15 to 25. From here, I settle the number of topics at $J = 20$.

The second step is to perform a grid search for τ. First, I search for τ in this region, $\{0.01, 0.02, 0.05, 0.1, 0.5, 1\}$. Then, I sequence, with width of 0.01, from 0.01 to 0.15. For this step, I finally chose the hyperparameter τ to be at 0.05.

The final step is to perform a grid search for ω. I first search for ω in this region, (0.001, 0.002, 0.005, 0.007, 0.01, 0.02, 0.05, 0.07, 0.1, 0.2, 0.5, 0.7). Then, I sequence from 0.001 to 0.015 with a width of 0.001. From here, I chose the hyperparameter ω to be at 0.001.

To summarize the steps and the resulting optimal hyperparameters chosen in this paper, see Table 14 below.

Table 14: Summary of LDA finetuning

Hyperparameters	**Steps and Results**
Number of topics K	1. Grid search from 5 to 100, with width of 5 2. Grid search from 15 to 25, with width of 1 Optimal number of topics $J = 20$
Document-topic priors τ	1. Grid search in the region (0.01, 0.02, 0.05, 0.1, 0.5, 1) 2. Grid search from 0.01 to 0.15, with width of 0.01 Optimal $\tau^* = 0.05$
Topic-word priors ω	1. Grid search in the region (0.001, 0.002, 0.005, 0.007, 0.01, 0.02, 0.05, 0.07, 0.1, 0.2, 0.5, 0.7) 2. Grid search from 0.001 to 0.015, with width 0.001 Optimal $\omega^* = 0.001$

The coarse search over $K \in \{5, \ldots, 100\}$ is the plot that disciplines $J = 20$. Figure 3 shows that the decline in perplexity slows after $K = 20$, while mean NPMI sits in the target band of 0.2 to 0.3 and min NPMI is highest at $K = 20$. Figure 4 shows that mean exclusivity remains above 0.75 in that neighborhood. Figures 5 and 6 show decade-to-decade Jensen-Shannon divergence and drifted share. Mean JSD stays below 0.3 for K between 5 and 20 under Hungarian matching. I therefore set $J = 20$, $\tau = 0.05$, and $\omega = 0.001$. Finer grids around those values confirm the same choice.

Figure 3: Perplexity and coherence. Coarse grid of K from 5 to 100.

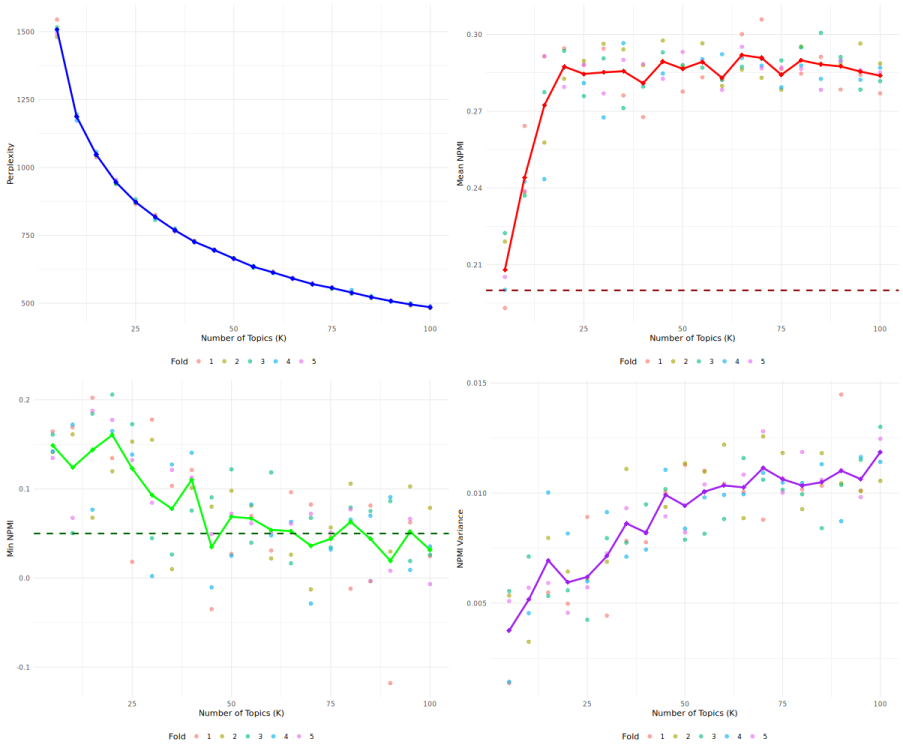


Figure 4: Perplexity and exclusivity. Coarse grid of K from 5 to 100.



Figure 5: Jensen-Shannon divergence and drifted share, 1990s to 2000s. Coarse grid of K from 5 to 100.

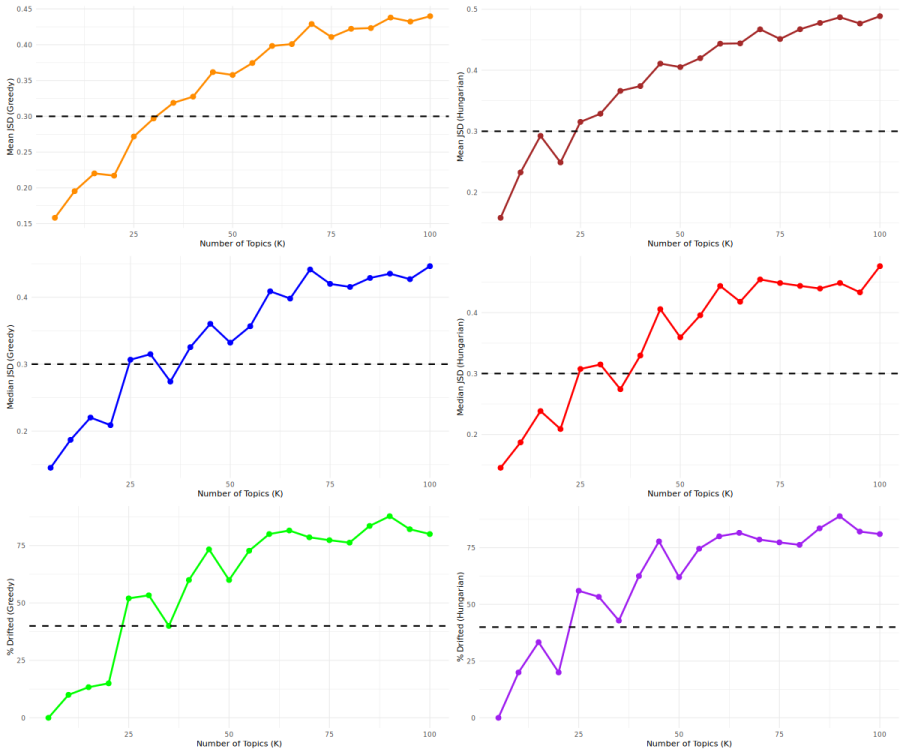
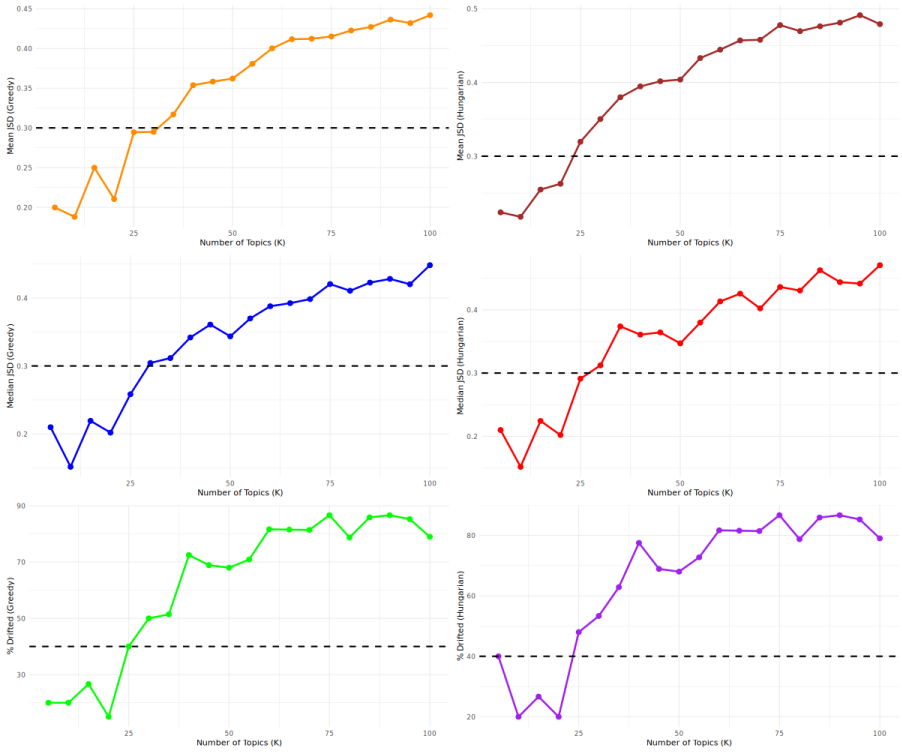


Figure 6: Jensen-Shannon divergence and drifted share, 2000s to 2010s. Coarse grid of K from 5 to 100.



A.6 Topic labels and top words

Table 15 reports the twenty labels and the highest-probability words. The main text lists labels only.

Table 15: Topics and Top Words

Topic	Label	Top Words
1	Semiconductor Memory Architecture	cell, memory_cell, line, row, array, bite_line, column, memory_device, memory_array, flash, word_line, nonvolatile, flash_memory, nonvolatile_memory, word, erase, cell_include, semiconductor_memory, bite, sram, memory_element, refresh, cell_array, transmission_line, dram
2	Audio and Speech-Signal Processing	value, estimate, symbol, calculate, speech, parameter, metric, measure, correlation, sequence, train, coefficient, sample, tone, set, noise, audio_signal, model, threshold, threshold_value, echo, audio, pilot, sound
3	Semiconductor Lithography and Fabrication Processes	dielectric, etch, dielectric_layer, mask, metal, pattern, silicon, metal_layer, deposit, layer_form, photoresist, material, barrier, nitride, expose, dielectric_material, first_dielectric, barrier_layer, mask_layer, oxide_layer, trench, remove, oxide, sacrificial
4	Integrated Circuit Design, Verification, and Testing	test, design, integrate, integrate_circuit, mask, logic, programmable, port, logic_device, ic, programmable_logic, layout, circuitry, circuit_design, output_port, pld, test_mode, method_test, device_test, test_signal, input_port, functional, test_circuit, architecture
5	PCB and Electronic Hardware with Connectors	connector, board, cable, circuit_board, house, module, end, print_circuit, assembly, electronic, conductor, mate, mount, antenna, 2_end, electronic_component, electrical_connector, first_end, socket, contact, print, receptacle, enclosure, component
6	Video Recording, Encoding, and Media Processing	record, video, stream, encode, reproduce, code, record_medium, block, picture, video_signal, decode, medium, frame, video_datum, reproduction, datum_stream, information_record, encoder, motion_vector, bit, compress, datum_record, datum_block, parity
7	Semiconductor Packaging and Interconnect Bonding	die, package, pad, bond, chip, conductive, solder, bump, first_surface, bond_pad, semiconductor_die, lead, semiconductor_package, semiconductor_chip, 2_surface, mems, microelectronic, ball, lead_frame, package_include, electrically, mold, interposer, leadframe
8	Thin-Film Materials, Deposition and Chemistry	composition, glass, temperature, particle, polymer, resin, gas, compound, film, coat, fuel, polish, ceramic, acid, mixture, reaction, electrolyte, carbon, catalyst, fuel_cell, melt, solution, hydrogen, powder
9	Wireless Base-Station and Mobile Communications	station, wireless, base_station, wireless_communication, transmission, ue, channel, radio, communication_system, uplink, user_equipment, downlink, mobile_station, equipment, resource, control_channel, equipment_ue, subframe, radio_communication, report, reception, wireless_device, communication_device
10	Clock Generation, Timing Circuits, and Signal Processing	clock, frequency, phase, clock_signal, delay, filter, analog, digital, oscillator, rf, output_signal, loop, input_signal, pulse, digital_signal, datum_signal, rf_signal, mixer, adc, pll, signal_2, signal_generate, gain, baseband
11	Power-Supply and Power-Management Electronics	power, power_supply, mode, switch, consumption, power_amplifier, supply, battery, power_source, low_power, transformer, ac, dc, power_consumption, power_converter, power_control, reduce_power, power_management, output_power, wireless_power, first_power, 2_power, primary, secondary
12	Networking Protocols and Data Infrastructure	network, packet, protocol, node, message, mode, link, network_node, communication_network, datum_packet, ip, service, traffic, network_device, access_point, subscriber, gateway, access_network, session, first_network, wireless_network, network_element, area_network, entity
13	Transistor Gate Structures and Semiconductor Device Physics	region, gate, fin, gate_electrode, drain, dope, sourcedrain, gate_structure, region_2, semiconductor_layer, conductivity_type, drain_region, source_drain, channel_region, region_form, conductivity, implant, region_first, epitaxial, sourcedrain_region, active_region, dopant, ptype, transistor
14	Imaging Sensors and Camera Pipelines	image, sensor, display, camera, image_datum, object, screen, information_process, image_process, capture, image_sensor, image_signal, process_apparatus, section, image_display, touch, color, display_unit, scene, capture_image, apparatus_image, image_capture, first_image, display_screen
15	Processor Architecture, Caches, Instruction Execution	processor, instruction, address, cache, bus, register, execution, command, store, host, memory_controller, interrupt, cpu, execute, memory_access, buff, thread, graphic, storage_device, controller, pipeline, entry, operand, transaction

Topic	Label	Top Words
16	Display Panel Tech (LCD/OLED)	electrode, liquid_crystal, panel, crystal, pixel, display_device, liquid, display, display_panel, first_electrode, 2_electrode, pixel_electrode, crystal_display, solidstate, organic, cpu, gate_line, organic_light, film_transistor, photoelectric, solidstate_image, electrode_2, lcd, plasma_display
17	Rotational Electromechanics and Magnetic Actuation	member, air, magnetic, head, shaft, rotate, roller, main_body, refrigerator, magnet, rotation, actuator, wash, motor, rotor, door, nozzle, cartridge, drum, inner, body, compressor, movable, laundry
18	Optics, Lasers, and Photonics	optical, light, lens, laser, fiber, beam, wavelength, light_source, optical_fiber, emit, laser_beam, waveguide, reflect, light_beam, optical_signal, refractive, incident, polarization, reflection, light_guide, illumination, grate, reflector, optical_system
19	Analog and Differential Amplifier Circuits	voltage, current, bias, differential, transistor, amplifier, resistor, reference_voltage, stage, current_source, input_terminal, output_terminal, voltage_level, first_voltage, supply_voltage, charge_pump, 2_voltage, terminal, output_node, driver, output_voltage, bias_voltage, voltage_first, voltage_2
20	Mobile UI and Content Services	user, key, content, server, user_interface, mobile_device, application, authentication, telephone, compute_device, item, computer, encrypt, mobile, security, client, encryption, location, database, computer_program, remote, phone, game, file

A.7 Administrative-class composition

Table 16 reports how many class codes examiners list on a grant in each decade. Examiners assign more codes to later patents.

Table 16: Listed Cooperative Patent Classification codes per grant

Decade	Grants	Mean listed codes	Median listed codes
1990s	109,872	5.1	3
2000s	250,567	6.1	4
2010s	270,930	8.0	6

Notes: Counts are class codes listed on grants in the electronics patent sample. A typical 1990s grant lists 5.1 codes; a typical 2010s grant lists 8.0. Examiners assign more codes to later patents. They do not assign shares across those codes.

Table 17 places each area's top words next to the Cooperative Patent Classification subclasses that receive the most of that area's mass. Lithography, packaging, and transistor device physics all concentrate in subclass H01L. In the patent text those are different processes: etch and photoresist, die and bond pads, and gate and drain. A market defined as H01L would merge them.

Table 17: Innovation areas, top words, and administrative classes

Topic	Area	Top words	CPC subclasses
1	Semiconductor Memory Architecture	cell, memory_cell, line, row, array, bite_line, column, memory_device	G11C (48.0%); H01L (21.8%); G06F (8.7%)
2	Audio and Speech-Signal Processing	value, estimate, symbol, calculate, speech, parameter, metric, measure	H04L (16.9%); G06F (10.7%); H04N (8.8%)
3	Semiconductor Lithography and Fabrication Processes	dielectric, etch, dielectric_layer, mask, metal, pattern, silicon, metal_layer	H01L (74.9%); G03F (4.2%); H05K (1.6%)
4	Integrated Circuit Design, Verification, and Testing	test, design, integrate, integrate_circuit, mask, logic, programmable, port	G06F (20.1%); H01L (14.5%); G01R (8.4%)

Topic	Area	Top words	CPC subclasses
5	PCB and Electronic Hardware with Connectors	connector, board, cable, circuit_board, house, module, end, print_circuit	H01R (15.6%); H01L (12.6%); H05K (9.5%)
6	Video Recording, Encoding, and Media Processing	record, video, stream, encode, reproduce, code, record_medium, block	H04N (33.7%); G11B (23.0%); H04L (9.3%)
7	Semiconductor Packaging and Interconnect Bonding	die, package, pad, bond, chip, conductive, solder, bump	H01L (70.5%); H05K (4.5%); G01R (1.9%)
8	Thin-Film Materials, Deposition and Chemistry	composition, glass, temperature, particle, polymer, resin, gas, compound	H01L (23.1%); H01M (5.9%); C08G (3.0%)
9	Wireless Base-Station and Mobile Communications	station, wireless, base_station, wireless_communication, transmission, ue, channel, radio	H04W (45.3%); H04L (24.5%); H04B (11.9%)
10	Clock Generation, Timing Circuits, and Signal Processing	clock, frequency, phase, clock_signal, delay, filter, analog, digital	H04L (10.3%); H04B (9.6%); H03L (7.4%)
11	Power-Supply and Power-Management Electronics	power, power_supply, mode, switch, consumption, power_amplifier, supply, battery	H02M (12.4%); G06F (10.7%); H02J (7.8%)
12	Networking Protocols and Data Infrastructure	network, packet, protocol, node, message, mode, link, network_node	H04L (37.8%); H04W (32.7%); G06F (6.3%)
13	Transistor Gate Structures and Semiconductor Device Physics	region, gate, fin, gate_electrode, drain, dope, sourcedrain, gate_structure	H01L (82.9%); G11C (2.3%); G06F (1.5%)
14	Imaging Sensors and Camera Pipelines	image, sensor, display, camera, image_datum, object, screen, information_process	H04N (26.5%); G06F (16.8%); G06T (9.4%)
15	Processor Architecture, Caches, Instruction Execution	processor, instruction, address, cache, bus, register, execution, command	G06F (55.1%); G11C (9.2%); H04L (8.0%)
16	Display Panel Tech (LCD/OLED)	electrode, liquid_crystal, panel, crystal, pixel, display_device, liquid, display	H01L (28.2%); G02F (21.6%); G09G (17.3%)
17	Rotational Electromechanics and Magnetic Actuation	member, air, magnetic, head, shaft, rotate, roller, main_body	G11B (8.1%); H01L (7.0%); D06F (4.6%)
18	Optics, Lasers, and Photonics	optical, light, lens, laser, fiber, beam, wavelength, light_source	G02B (23.6%); H01L (10.2%); G11B (9.9%)
19	Analog and Differential Amplifier Circuits	voltage, current, bias, differential, transistor, amplifier, resistor, reference_voltage	H03K (14.1%); G11C (12.0%); H03F (10.8%)
20	Mobile UI and Content Services	user, key, content, server, user_interface, mobile_device, application, authentication	G06F (23.8%); H04L (15.5%); H04W (11.0%)

Notes: The top words are the eight highest-probability tokens from the topic model. CPC mass is the topic weight of each patent, split equally across the class codes listed on that grant. Displayed ranks omit tagging section Y. Lithography, packaging, and transistor device physics all concentrate in subclass H01L. In the patent text those are different processes: etch and photoresist, die and bond pads, and gate and drain. A market defined as H01L would merge them.

A.8 Competitor counts and valuations by decade

Tables 18 and 19 report rival counts and patent values by topic and decade. The main text reports the corresponding share tables.

Table 18: Competitor counts by topic and decade

Topic	Label	1990s	2000s	2010s	Mean	SD
1	Semiconductor Memory Architecture	78.21	85.87	56.23	75.59	18.39
2	Audio and Speech-Signal Processing	144.49	182.38	135.87	156.65	31.26
3	Semiconductor Lithography and Fabrication Processes	90.13	110.33	81.42	95.58	19.25
4	Integrated Circuit Design, Verification, and Testing	155.05	194.85	134.22	164.87	35.76
5	PCB and Electronic Hardware with Connectors	142.59	164.38	127.97	147.16	23.77
6	Video Recording, Encoding, and Media Processing	76.85	94.41	63.95	80.25	17.79
7	Semiconductor Packaging and Interconnect Bonding	77.84	116.57	98.62	97.7	21.15
8	Thin-Film Materials, Deposition and Chemistry	128.76	130.93	89.09	119.62	24.89
9	Wireless Base-Station and Mobile Communications	75.97	110.2	89.96	92.42	20.77
10	Clock Generation, Timing Circuits, and Signal Processing	142.86	173.42	122.23	149.24	31.58
11	Power-Supply and Power-Management Electronics	123.27	155.58	119.92	134.65	25.09
12	Networking Protocols and Data Infrastructure	75.11	110.63	85	91.02	23.24
13	Transistor Gate Structures and Semiconductor Device Physics	74.05	95.88	71.04	81.55	16.07
14	Imaging Sensors and Camera Pipelines	64.76	83.04	80.26	75.56	13.67
15	Processor Architecture, Caches, Instruction Execution	106.24	139.03	105.6	118.48	25.26
16	Display Panel Tech (LCD/OLED)	38.63	45.8	36.6	40.83	8.18
17	Rotational Electromechanics and Magnetic Actuation	150.07	147.76	113.25	139.95	22.69
18	Optics, Lasers, and Photonics	78.65	96.04	74.21	84.11	14.26
19	Analog and Differential Amplifier Circuits	129.29	155.28	112.9	135.01	27.31
20	Mobile UI and Content Services	113.88	145.6	109.77	124.84	23.82

Rival counts are high in every area, and they are uneven. Integrated circuit design and audio average about 165 and 157 rivals. Display averages about 41, the thinnest area in the table. Memory averages about 76. Most areas are more crowded in the 2000s than in the 1990s, and less crowded in the 2010s than in the 2000s. Memory falls from about 86 rivals in the 2000s to about 56 in the 2010s. Integrated circuit design falls from about 195 to about 134 over the same decades. Thin-film materials fall from about 131 to about 89, and rotational electromechanics from about 148 to about 113. Wireless and networking recede from their 2000s peaks as well, and both remain above their 1990s counts.

These levels are the ones that enter $h(N) = \text{asinh}(N)$. At the display mean of about 41 rivals, one more rival changes the index by about $\text{asinh}(42) - \text{asinh}(41) \approx 0.024$. At the integrated-circuit-design mean of about 165, the same step changes the index by about $\text{asinh}(166) - \text{asinh}(165) \approx 0.006$. An extra rival therefore moves the congestion term by more in the thin areas of this table. The case study holds each firm-year at the count observed in that year, so the exercises inherit both the thick markets and the thin display market.

Table 19: Patent valuation by topic and decade

Topic	Label	1990s	2000s	2010s	Mean	SD
1	Semiconductor Memory Architecture	21.14	13.25	32.22	20.94	185.68
2	Audio and Speech-Signal Processing	59.98	67.45	164.38	89	946.12
3	Semiconductor Lithography and Fabrication Processes	37.22	27.85	57.59	38.79	334.05
4	Integrated Circuit Design, Verification, and Testing	48.72	44.95	99.48	60.03	511.19
5	PCB and Electronic Hardware with Connectors	33.69	36.43	116.4	55.48	702.7
6	Video Recording, Encoding, and Media Processing	21.46	19.92	43.85	26.5	263.56
7	Semiconductor Packaging and Interconnect Bonding	25.43	23.09	65.97	34.72	294.2
8	Thin-Film Materials, Deposition and Chemistry	57.21	53.77	63.61	57.52	563.75
9	Wireless Base-Station and Mobile Communications	32.71	37.62	159.88	66.48	1091.89
10	Clock Generation, Timing Circuits, and Signal Processing	56.24	46.29	85.24	59.75	421.75
11	Power-Supply and Power-Management Electronics	22.07	24.39	67.12	34.25	305.21
12	Networking Protocols and Data Infrastructure	32.29	39.73	90.41	49.69	499.55
13	Transistor Gate Structures and Semiconductor Device Physics	26.28	20.12	57.12	31.69	264.17
14	Imaging Sensors and Camera Pipelines	14.15	22.22	115.92	42.74	793.96
15	Processor Architecture, Caches, Instruction Execution	94.46	63.72	197.53	108.7	1345.49
16	Display Panel Tech (LCD/OLED)	4.44	6.15	27.71	10.93	185.31
17	Rotational Electromechanics and Magnetic Actuation	27.87	24.15	62.5	35.16	311.01
18	Optics, Lasers, and Photonics	17.74	18.64	39.9	23.64	265.96
19	Analog and Differential Amplifier Circuits	57.8	45.97	83	59.65	439.15
20	Mobile UI and Content Services	45.76	81.45	281.62	118.45	2037.15

A.9 First-stage fit by area

Table 20 reports, for each of the twenty areas, the standard deviation of realized v_{ijt} and of fitted $\check{v}_{ijt}$, together with mean absolute deviation. Fitted values shrink noise, and the fit is similar across topics.

A.10 Persistence of area occupancy

The static game treats each year's portfolio as a fresh choice. This subsection reports how much of a portfolio carries over from one year to the next, so that the reader can judge that abstraction. Table 21 summarizes transitions in the firm-area-year panel. An area that a firm occupied in the previous year is occupied again in 68.5 percent of cases, and an area the firm did not occupy is entered in 11.5 percent of cases. Occupancy therefore carries over from one year to the next, with frequent exits and returns. About one third of occupied areas are vacated the following year, the median occupancy spell lasts one year, and 61 percent of entries are returns to an area the firm had occupied earlier in the sample. Among three-year windows, entries that are abandoned the next year (10,330) outnumber entries that last at least two years (6,575). These spells and returns fit a yearly decision in which part of the portfolio carries over. Under irreversible entry with large sunk costs, spells would be long and returns rare.

Table 22 reports linear probability models of occupancy on last year's occupancy in the same area, the predicted valuation, the congestion term, and the number of other areas the firm occupied in the previous year. With year effects only, last year's occupancy raises the probability of occupancy this year by 57 percentage points. With firm-by-area and year effects, which absorb a firm's persistent advantage in an area, the coefficient falls to 9 percentage points, and adding area-by-year effects leaves it there. The remaining state dependence is what a sunk cost paid on entering

Table 20: Comparison of Predictive Performance Across Specifications

Topic	Label	$\text{sd}(v_{ijt})$	$\text{sd}(\hat{v}_{ijt}^{fe})$	$\text{sd}(\hat{v}_{ijt}^{pool})$	MAD^{fe}	MAD^{pool}
1	Semiconductor Memory Architecture	0.8525	0.8356	0.8279	0.2795	0.2757
2	Audio and Speech-Signal Processing	1.1342	0.8356	0.8442	0.3828	0.3820
3	Semiconductor Lithography and Fabrication Processes	0.9727	0.8356	0.8342	0.3160	0.3156
4	Integrated Circuit Design, Verification, and Testing	1.1036	0.8356	0.8396	0.3661	0.3663
5	PCB and Electronic Hardware with Connectors	1.0636	0.8356	0.8416	0.4055	0.4048
6	Video Recording, Encoding, and Media Processing	0.8756	0.8356	0.8283	0.2832	0.2788
7	Semiconductor Packaging and Interconnect Bonding	0.9823	0.8356	0.8425	0.3080	0.3070
8	Thin-Film Materials, Deposition and Chemistry	1.0148	0.8356	0.8290	0.3691	0.3729
9	Wireless Base-Station and Mobile Communications	0.9366	0.8356	0.8390	0.2875	0.2884
10	Clock Generation, Timing Circuits, and Signal Processing	1.1375	0.8356	0.8389	0.4072	0.4072
11	Power-Supply and Power-Management Electronics	1.0299	0.8356	0.8417	0.3325	0.3354
12	Networking Protocols and Data Infrastructure	0.9842	0.8356	0.8365	0.3221	0.3231
13	Transistor Gate Structures and Semiconductor Device Physics	0.9542	0.8356	0.8343	0.3013	0.2995
14	Imaging Sensors and Camera Pipelines	0.8659	0.8356	0.8366	0.2827	0.2807
15	Processor Architecture, Caches, Instruction Execution	1.0744	0.8356	0.8365	0.3245	0.3254
16	Display Panel Tech (LCD/OLED)	0.6922	0.8356	0.8274	0.3289	0.3229
17	Rotational Electromechanics and Magnetic Actuation	1.0386	0.8356	0.8342	0.4109	0.4102
18	Optics, Lasers, and Photonics	0.8809	0.8356	0.8339	0.3230	0.3215
19	Analog and Differential Amplifier Circuits	1.1271	0.8356	0.8386	0.3987	0.4002
20	Mobile UI and Content Services	1.0503	0.8356	0.8428	0.3337	0.3350
	Mean	0.9886	0.8356	0.8364	0.3382	0.3376

Notes: v_{ijt} is realized patent value after asinh and a global standardization (the same left-hand side as Table 7). $\hat{v}_{ijt}^{fe}$ and $\hat{v}_{ijt}^{pool}$ are fitted values from Table 7 Model 2 (firm, year, and area effects) and Model 3 (firm and year-by-area effects), respectively. MAD = Mean Absolute Deviation between actual and fitted values on this common scale.

an area, or a serially correlated unobserved shock, would have to explain. The predicted valuation $\check{v}_{ijt}$ already carries firm-by-area and year-by-area effects, because the first-stage projection is estimated separately for each area with firm and year effects.

The same table shows the size of the portfolio, alongside the area-by-area transitions. In 39.9 percent of firm-years the firm occupies no area. In 39.0 percent the number of occupied areas equals last year's number, and the correlation between this year's count and last year's count is 0.833. The list of areas inside that count still turns over. About one third of occupied areas are vacated within a year, the median spell lasts one year, and 61 percent of entries are returns. A portfolio can keep a similar size from year to year while the areas inside it change. Appendix B.5 asks how large a sunk cost of entering an area would have to be to account for the part of that pattern that survives firm-by-area effects.

The columns of Table 22 place that persistence next to the objects in the moment inequalities. Predicted valuation raises the probability of occupancy by 7.6 percentage points with year effects and by 10.3 percentage points once firm-by-area effects are included. The coefficient on the inverse hyperbolic sine of the rival count is 11.0 percentage points with year effects and 3.9 percentage points with firm-by-area effects. One more other area occupied last year raises occupancy by 1.2 to 1.9 percentage points. Column (4) drops the rival count, because within an area-year that count differs across firms only through the firm's own occupancy. With firm-by-area effects and a short panel, the coefficient on last year's occupancy is pulled toward zero, so the 9 percentage points that remain are a lower figure for the state dependence a sunk cost, or a serially correlated shock, would have to match.

Table 21: Persistence of area occupancy

A. Year-to-year transitions	
Occupied last year, occupied this year (share)	0.685
Not occupied last year, entered this year (share)	0.115
Occupied areas this year that are new this year (share)	0.305
Occupied areas last year that are vacated this year (share)	0.315
B. Occupancy spells	
Number of spells	20,607
Median spell length (years); mean spell length	1; 2.94
Spells of one year (share)	0.584
Entries that return to an area the firm occupied earlier in the sample (share)	0.610
Years away before a return: 1 year (share of returns)	0.495
2 years (share of returns)	0.209
3 or more (share of returns)	0.297
C. Three-year windows	
Occupied in all three years (count and share of windows)	31,328 (0.161)
Entered at t and still occupied at $t+1$	6,575 (0.034)
Entered at t and vacated at $t+1$	10,330 (0.053)
Vacated at t and re-entered at $t+1$	5,276 (0.027)
Vacated at t and still out at $t+1$	11,497 (0.059)
Out at $t-1$ and t, entered at $t+1$	10,607 (0.054)
Out in all three years	112,230 (0.576)
D. Portfolio size	
Firm-years with no occupied area (share)	0.399
Firm-years whose number of occupied areas equals last year's (share)	0.390
Correlation between this year's and last year's number of occupied areas	0.833

Notes: The panel covers listed U.S. electronics firms, 1990 to 2019, and twenty innovation areas. A firm occupies an area in a year when the average weight of its patents in that year on the area is at least one percent, the entry indicator used in the estimation. Transitions and spells use adjacent calendar years only. Panel C counts firm-area observations for which the firm is observed in three consecutive years.

Half of the returns come after one year away, 49.5 percent. Another 20.9 percent return after two years, and 29.7 percent after three or more. Among spells, 58.4 percent last one year, and the mean length is 2.94 years. In the three-year windows, 31,328 are occupied in all three years, 16.1 percent of windows, and 112,230 are out in all three, 57.6 percent. A firm that leaves is back the next year in 5,276 windows and still out in 11,497. Short spells, quick returns, and long stretches out of an area are all in the panel. A firm in any of those windows chooses again each year. The static comparisons use that year's bundle. Subsection B.5 asks how large a sunk cost would have to be to account for the one-year spells and the quick returns.

Table 22: Occupancy this year and occupancy last year

	(1)	(2)	(3)	(4)
Occupied last year	0.566	0.334	0.090	0.088
	(0.011)	(0.010)	(0.007)	(0.007)
Predicted valuation		0.076	0.103	0.097
		(0.005)	(0.010)	(0.010)
Rival count (inverse hyperbolic sine)		0.110	0.039	
		(0.004)	(0.012)	
Other areas occupied last year		0.012	0.018	0.019
		(0.001)	(0.001)	(0.001)
Firm-by-area effects	No	No	Yes	Yes
Area-by-year effects	No	No	No	Yes
Year effects	Yes	Yes	Yes	No
Observations	209,960	209,960	209,960	209,960
R-squared	0.328	0.421	0.524	0.525
Within R-squared			0.062	0.061

Notes: Linear probability models of occupancy on the firm-area-year panel, adjacent years only. Standard errors clustered by firm in parentheses. The coefficient on last year's occupancy is a descriptive measure of persistence and is not a structural parameter; with firm-by-area effects and a short panel it is biased toward zero. Predicted valuation is the first-stage fitted value used in the moment inequalities. Column (4) omits the rival count: within an area-year, the number of rivals varies across firms only through the firm's own occupancy.

B Model and empirical implementation

This appendix details the objects that Section 3 uses without derivation. Appendix B.1 restates the market-implied patent values of Kogan et al. (2017) in the notation of this paper and aggregates them to the firm-year-area level. Appendix B.2 consists of the characterization of the optimal bundle and then derives the three families of linear moment inequalities. Appendix B.3 writes about the subset sampling, the test statistic, and the static-game caveats. Appendix B.4 writes the add and drop profit gaps that recover the occupancy-to-scope ratio, and writes how swaps recover congestion. Appendix B.5 derives the one-period comparison that recovers a fixed-cost ratio and a sunk-cost ratio.

B.1 Market-implied patent values

Each patent p filed by firm i and granted in period t is assigned a dollar value A_{ipt} from the filtered stock-market reaction constructed by Kogan et al. (2017). I take their filtered series as given and only change notation so that the object lines up with the area-level valuation v_{ijt} in Section 3. What follows is their measurement model, written for a patent of firm i rather than for a generic grant j.

The USPTO publishes grants on a known announcement day. Absent other news, the change in the firm's market value on that day would recover only the unanticipated portion of the patent's private value. If ϖ_{ipt} is the market's ex-ante probability that the application is granted and ξ_{ipt} is

the present value of the associated monopoly rents, then

$$\Delta V_{ipt} \;=\; \left(1-\varpi_{ipt}\right)\xi_{ipt}. \tag{4}$$

The announcement therefore understates the full private value by the factor $1-\varpi_{ipt}$. The object is the unexpected increment associated with the grant, together with that scaling. The full present value of the underlying research program is a different quantity. LDA weights allocate that grant-level increment across innovation areas. Both steps introduce measurement error, which is why Section 4 projects realized valuations onto the firm's predetermined state. Following Kogan et al. (2017), I work with a three-day window $[t, t+2]$ around the grant date. Even inside that window the idiosyncratic return still mixes the patent with unrelated news. Write the market-adjusted return of firm i around the grant of patent p as

$$R_{ipt} \;=\; \nu_{ipt} + \varsigma_{ipt}, \tag{5}$$

where ν_{ipt} is the value of the patent as a fraction of the firm's market capitalization and ς_{ipt} is the residual return. Kogan et al. (2017) treat ν_{ipt} as a truncated normal, $\nu_{ipt} \sim N^{+}(0, \sigma^2_{\nu,it})$, and the residual as Gaussian, $\varsigma_{ipt} \sim N(0, \sigma^2_{\varsigma,it})$, independent of ν_{ipt}. The filtered fraction is then

$$\mathbb{E}[\nu_{ipt} \mid R_{ipt}] \;=\; \delta_{it}\, R_{ipt} + \sqrt{\delta_{it}}\,\sigma_{\varsigma,it}\,\frac{\varphi\left(-\sqrt{\delta_{it}}\, R_{ipt}/\sigma_{\varsigma,it}\right)}{1-\Phi\left(-\sqrt{\delta_{it}}\, R_{ipt}/\sigma_{\varsigma,it}\right)}, \tag{6}$$

where φ and Φ are the standard normal density and distribution and

$$\delta_{it} \;=\; \frac{\sigma^2_{\nu,it}}{\sigma^2_{\nu,it}+\sigma^2_{\varsigma,it}} \tag{7}$$

is the signal-to-noise ratio. Let M_{it} be the firm's market capitalization on the day before the announcement, and let N_{ipt} be the number of patents granted to the firm on the same day. The dollar value that I take from their filtered series is

$$A_{ipt} \;\equiv\; \xi_{ipt} \;=\; \left(1-\bar{\varpi}\right)^{-1}\frac{1}{N_{ipt}}\,\mathbb{E}[\nu_{ipt} \mid R_{ipt}]\, M_{it}. \tag{8}$$

The factor $(1-\bar{\varpi})^{-1}$ undoes the average grant probability in (4). The factor $1/N_{ipt}$ splits a day's filtered value equally across contemporaneous grants. I use their published filtered dollar series, in both nominal and 1982 dollars. The grant-day filter and the signal-to-noise ratio are those of Kogan et al. (2017).

The remaining step is the one that is specific to this paper. The grant-day filter delivers a single dollar value per patent. The entry model is defined on innovation areas. Using the LDA weights w_{pjt} of Appendix A.2, I allocate the patent across areas by

$$A_{ipjt} \;=\; w_{pjt}\, A_{ipt}. \tag{9}$$

Summing over the firm's patents in the period produces the realized area total

$$\sum_{p\in\mathcal{P}_{it}} A_{ipjt} \;=\; \sum_{p\in\mathcal{P}_{it}} w_{pjt}\, A_{ipt}. \tag{10}$$

Section 3 maps this total through a monotone transformation $b(\cdot)$ and a mean-zero measurement error,

$$v_{ijt} \;=\; b\left(\sum_{p\in\mathcal{P}_{it}} w_{pjt}\, A_{ipt}\right) + e_{ijt}. \tag{11}$$

In the empirical implementation, $b(\cdot)$ is the inverse hyperbolic sine. The valuation that enters the inequalities is the fitted value from projecting that transformed total on the firm's observable state. The residual e_{ijt} absorbs filter noise in A_{ipt}, sampling noise in the LDA weights, and any remaining aggregation error. Firms do not observe e_{ijt} at the time of entry, so $\mathbb{E}[e_{ijt} \mid \mathcal{F}_{ijt}] = 0$.

B.2 Derivation of the moment inequalities

I first take Result 1 to Result 2. I then derive Results 3–5 from those two results. Throughout, rivals' equilibrium counts $N_{jt,-i}$ are taken as given. The set of areas is finite. The unobservable ϵ_{ijt} is a composite ex-post forecast error with $\mathbb{E}[\epsilon_{ijt} \mid \mathcal{F}_{ijt}] = 0$. Instruments Z are a subset of $\mathcal{F}_{ijt}$, and $l(\cdot)$ maps them to a nonnegative scalar. All objects that appear in the deterministic indexes below are therefore measurable with respect to $\mathcal{F}_{ijt}$.

Write $S = \{j : d_{ijt} = 1\}$ for the observed bundle, $k_{it} = |S|$, and

$$\zeta^{e}_{ijt} \;=\; \check{v}_{ijt} + \alpha\, h(N_{jt,-i}) - \kappa$$

for the expected profitability index of area j. Result 1 gives

$$\mathbb{E}[\Pi_{it} \mid \mathcal{F}_{ijt}] \;=\; \sum_{j\in S} \zeta^{e}_{ijt} + \beta\, g(k_{it}). \tag{12}$$

Nash equilibrium requires that no unilateral deviation raise expected profit. For any alternative bundle S',

$$\mathbb{E}\left[\Pi_{it}(S') - \Pi_{it}(S) \;\middle|\; \mathcal{F}_{ijt}\right] \;\leq\; 0. \tag{13}$$

Characterization of the optimal bundle. The portfolio function g depends on the entered set only through how many areas it contains. Holding rivals' counts fixed, the firm can therefore rank areas by a single index and then choose how many of the highest-ranked areas to enter. I record that argument as three lemmas. They use only Result 1 and the restriction on g. The number of areas J is finite.

The first lemma is the swap calculation. The second says that, at a fixed size, the best bundles are those that never leave a better area out while keeping a worse one in. The third says that the firm then only has to compare those best-of-size values across the $J + 1$ possible sizes.

Lemma 1 (Swap identity) *Let S be a bundle of size k, and let $j \in S$ and $m \notin S$. The swapped bundle $S' = (S \setminus \{j\}) \cup \{m\}$ has the same size, and*

$$\mathbb{E}\left[\Pi_{it}(S') - \Pi_{it}(S) \;\middle|\; \mathcal{F}_{ijt}\right] \;=\; \zeta^{e}_{imt} - \zeta^{e}_{ijt}. \tag{14}$$

Proof. By (12), expected profit is the sum of the indexes of the entered areas plus βg of the number of entered areas. The swap drops j and adds m, so the two sums differ only by $\zeta^{e}_{imt} - \zeta^{e}_{ijt}$. The count is still k, so $\beta g(k)$ is the same on both bundles and cancels. ■

Lemma 2 (Fixed portfolio size) *Fix a size $k \in \{0, \ldots, J\}$. A bundle S of size k maximizes expected profit among all bundles of size k if and only if it satisfies the threshold property*

$$\min_{j \in S} \zeta^e_{ijt} \;\geq\; \max_{m \notin S} \zeta^e_{imt}, \tag{15}$$

where the minimum over the empty set is $+\infty$ and the maximum over the empty set is $-\infty$. Every such bundle attains

$$U^e_k \;=\; \sum_{r=1}^{k} \zeta^e_{(r)} + \beta\, g(k), \tag{16}$$

with $\zeta^e_{(1)} \geq \cdots \geq \zeta^e_{(J)}$ the decreasing rearrangement of the indexes. That k-set is unique if and only if $\zeta^e_{(k)} > \zeta^e_{(k+1)}$, reading $\zeta^e_{(0)} = +\infty$ and $\zeta^e_{(J+1)} = -\infty$.

Proof. The empty bundle ($k = 0$) and the full set of areas ($k = J$) satisfy (15) automatically: there is either nothing inside to compare, or nothing outside. I therefore treat a generic size k.

First, suppose S violates (15). Then the firm has entered some area j whose index is strictly below the index of some area m that it left out. Lemma 1 says that swapping m for j raises expected profit while keeping the same size. So a size-k maximizer cannot violate the threshold property.

Second, suppose S satisfies (15), and let T be any other bundle of the same size. The two sets differ by equally many areas: the extras in S and the extras in T can be lined up one for one. Every extra in S is an entered area, so by (15) its index is at least as large as the index of every area left out of S. Every extra in T was left out of S. Pairing them therefore gives

$$\sum_{j \in S \setminus T} \zeta^e_{ijt} \;\geq\; \sum_{m \in T \setminus S} \zeta^e_{imt}. \tag{17}$$

The areas that S and T share cancel. The portfolio term $\beta g(k)$ is the same on both bundles. Hence expected profit on S is at least as large as expected profit on T. Every threshold k-set is therefore optimal for that size.

Third, any threshold k-set must include every area whose index is strictly above the k-th largest index $\zeta^e_{(k)}$, and then enough areas tied at that cutoff to reach size k. All of those sets add the same k largest indexes, so they all deliver the same expected profit U^e_k in (16). If $\zeta^e_{(k)} > \zeta^e_{(k+1)}$, the cutoff is a single area and that set is unique. If several areas share the cutoff, any selection among those tied areas that fills out size k is optimal for that k. ■

Lemma 3 (Optimal size) *The $J+1$ numbers $U^e_0, \ldots, U^e_J$ attain a maximum. Write k^* for any maximizer. A bundle S maximizes expected profit over the power set if and only if $|S| = k^*$ for some such k^* and S satisfies* (15)*. In particular, every optimal bundle is a set of k^* areas with the largest indexes.*

Proof. Every bundle has some size k between 0 and J. Lemma 2 says that, within size k, the best expected profit is the single number U^e_k, and it is attained exactly on the threshold k-sets. The firm then compares the $J+1$ numbers $U^e_0, \ldots, U^e_J$. A finite list of real numbers has a largest value, so at least one best size k^* exists. The best expected profit in the whole problem is therefore $\max_k U^e_k$, and the bundles that attain it are exactly the threshold sets of those best sizes. ■

Lemmas 1–3 are Result 2. Several sizes can share that largest value of U^e_k. In that case any of those sizes is an optimal k^*.

Adding M areas. Fix an exogenous subset $\mathcal{M}$ of size M. Write $D^0_{\mathcal{M},it} \equiv \prod_{j\in\mathcal{M}}(1-d_{ijt})$ for the indicator that the realized bundle entered none of these areas. On the event $\{D^0_{\mathcal{M},it}=1\}$, one has $\mathcal{M}\cap S=\emptyset$, so the deviation bundle $S'=S\cup\mathcal{M}$ is well defined and has portfolio size $k_{it}+M$. Equation (12) yields

$$\mathbb{E}[\Pi_{it}(S') \mid \mathcal{F}_{ijt}] = \sum_{j\in S}\zeta^e_{ijt} + \sum_{j\in\mathcal{M}}\zeta^e_{ijt} + \beta\, g(k_{it}+M),$$
$$\mathbb{E}[\Pi_{it}(S) \mid \mathcal{F}_{ijt}] = \sum_{j\in S}\zeta^e_{ijt} + \beta\, g(k_{it}).$$

The expected profit difference is therefore

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] \;=\; \sum_{j\in\mathcal{M}}\zeta^e_{ijt} + \beta\big(g(k_{it}+M)-g(k_{it})\big) \;=\; \eta^+_{i,\mathcal{M},t}, \tag{18}$$

where $\Delta g^+_M(k_{it}) \equiv g(k_{it}+M)-g(k_{it})$ and

$$\eta^+_{i,\mathcal{M},t} \;=\; \sum_{j\in\mathcal{M}}\big(\check{v}_{ijt} + \alpha\, h(N_{jt,-i}) - \kappa\big) + \beta\,\Delta g^+_M(k_{it}).$$

Revealed preference (13) on the event $\{D^0_{\mathcal{M},it}=1\}$ implies $\eta^+_{i,\mathcal{M},t}\le 0$. Off that event the product $D^0_{\mathcal{M},it}\,\eta^+_{i,\mathcal{M},t}$ is zero. Hence

$$D^0_{\mathcal{M},it}\,\eta^+_{i,\mathcal{M},t} \;\le\; 0 \tag{19}$$

almost surely. Because $l(Z_{i,\mathcal{M},t})\ge 0$, the same product remains nonpositive after multiplication by the instrument. Taking expectations delivers the unconditional moment

$$\mathbf{M}_1(Z_{i,\mathcal{M},t};\theta) \;=\; \mathbb{E}\big[D^0_{\mathcal{M},it}\,\eta^+_{i,\mathcal{M},t}\,l(Z_{i,\mathcal{M},t})\big] \;\le\; 0, \tag{20}$$

which is Result 3.

Dropping M areas. For the same exogenous $\mathcal{M}$, write $D^1_{\mathcal{M},it} \equiv \prod_{j\in\mathcal{M}} d_{ijt}$ for the indicator that every area in the subset was entered. On the event $\{D^1_{\mathcal{M},it}=1\}$, one has $\mathcal{M}\subset S$ and therefore $k_{it}\ge M$. The deviation bundle is $S'=S\setminus\mathcal{M}$, with portfolio size $k_{it}-M$. The same expansion of (12) gives

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] \;=\; -\sum_{j\in\mathcal{M}}\zeta^e_{ijt} - \beta\big(g(k_{it})-g(k_{it}-M)\big) \;=\; -\eta^-_{i,\mathcal{M},t}, \tag{21}$$

where $\Delta g^-_M(k_{it}) \equiv g(k_{it})-g(k_{it}-M)$ and

$$\eta^-_{i,\mathcal{M},t} \;=\; \sum_{j\in\mathcal{M}}\big(\check{v}_{ijt} + \alpha\, h(N_{jt,-i}) - \kappa\big) + \beta\,\Delta g^-_M(k_{it}).$$

Revealed preference requires $-\eta^-_{i,\mathcal{M},t}\le 0$ on the event $\{D^1_{\mathcal{M},it}=1\}$. Therefore

$$D^1_{\mathcal{M},it}\,\big(-\eta^-_{i,\mathcal{M},t}\big) \;\le\; 0 \tag{22}$$

almost surely. Multiplying by $l(Z_{i,\mathcal{M},t})\ge 0$ and taking expectations yields

$$\mathbf{M}_2(Z_{i,\mathcal{M},t};\theta) \;=\; \mathbb{E}\big[D^1_{\mathcal{M},it}\,\big(-\eta^-_{i,\mathcal{M},t}\big)\,l(Z_{i,\mathcal{M},t})\big] \;\le\; 0, \tag{23}$$

which is Result 4.

Pairwise swaps. Let (j, m) be an ordered pair with $j \in S$ and $m \notin S$, so that $D^{\text{swap}}_{j,m,it} \equiv d_{ijt}(1 - d_{imt}) = 1$. The deviation replaces j with m. Portfolio size is unchanged, and therefore $\Delta g = 0$. The expected profit difference collapses to the difference of indexes,

$$\begin{aligned}\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] &= \zeta^e_{imt} - \zeta^e_{ijt} \\ &= \big(\check{v}_{imt} + \alpha\, h(N_{mt,-i}) - \kappa\big) - \big(\check{v}_{ijt} + \alpha\, h(N_{jt,-i}) - \kappa\big) \\ &= \big(\check{v}_{imt} + \alpha\, h(N_{mt,-i})\big) - \big(\check{v}_{ijt} + \alpha\, h(N_{jt,-i})\big) \;=\; \mu_{ijmt}(\theta).\end{aligned} \tag{24}$$

The per-area cost κ cancels. Revealed preference requires $\mu_{ijmt}(\theta) \le 0$ on the event $\{D^{\text{swap}}_{j,m,it} = 1\}$, so

$$D^{\text{swap}}_{j,m,it}\, \mu_{ijmt}(\theta) \;\le\; 0 \tag{25}$$

almost surely. The unconditional moment is

$$\mathbf{M}_3(Z_{ijmt}; \theta) \;=\; \mathbb{E}\big[D^{\text{swap}}_{j,m,it}\, \mu_{ijmt}(\theta)\, l(Z_{ijmt})\big] \;\le\; 0, \tag{26}$$

which is Result 5. Because the portfolio term drops out, $\mathbf{M}_3$ isolates the congestion parameter α from the scale of $g(\cdot)$.

From pathwise inequalities to unconditional moments. The three products (19), (22), and (25) are already nonpositive as random variables. Passing to an unconditional moment uses only that a nonnegative instrument cannot flip that sign.

Lemma 4 (Nonnegative measurable weights) *Let X be integrable and $\mathcal{F}_{ijt}$-measurable, with $X \le 0$ almost surely. Let W be nonnegative and $\mathcal{F}_{ijt}$-measurable, and suppose WX is integrable. Then $WX \le 0$ almost surely, and*

$$\mathbb{E}[WX] \;\le\; 0. \tag{27}$$

Proof. The event $\{W \ge 0\} \cap \{X \le 0\}$ has probability one, and on that event the product WX cannot be strictly positive. Hence $WX \le 0$ almost surely. For an integrable random variable Y,

$$\mathbb{E}[Y] \;=\; \mathbb{E}[Y\mathbf{1}_{\{Y \le 0\}}] + \mathbb{E}[Y\mathbf{1}_{\{Y > 0\}}].$$

The second term vanishes when $P(Y > 0) = 0$. The first term is the expectation of a nonpositive integrable function, so it is at most zero. Taking $Y = WX$ gives (27). ■

Because X is $\mathcal{F}_{ijt}$-measurable, $\mathbb{E}[X \mid \mathcal{F}_{ijt}] = X$, and the tower property writes the same object as

$$\mathbb{E}[WX] \;=\; \mathbb{E}\big[W\, \mathbb{E}[X \mid \mathcal{F}_{ijt}]\big]. \tag{28}$$

The right-hand side is the usual instrumenting identity that starts from a conditional moment $\mathbb{E}[X \mid \mathcal{F}_{ijt}] \le 0$. Here the complete-information assumption already gives the pathwise bound $X \le 0$, so Lemma 4 applies directly.

I apply the lemma three times, each time with $W = l(Z)$ for a nonnegative function of instruments contained in $\mathcal{F}_{ijt}$.

For adding M areas, set $X = D^0_{\mathcal{M},it}\, \eta^+_{i,\mathcal{M},t}$. The indicator $D^0_{\mathcal{M},it}$ is a function of the observed bundle.

Under the information structure of Section 3 the firm has no private type, so that bundle is a function of $\mathcal{F}_{ijt}$ once rivals' actions are taken as given. The index $\eta^{+}_{i,\mathcal{M},t}$ is a function of $\check{v}$, N, k_{it}, and θ. The first three objects lie in $\mathcal{F}_{ijt}$ or are conditioned on; θ is constant. Hence X is $\mathcal{F}_{ijt}$-measurable. Equation (19) is the pathwise bound $X \leq 0$. Lemma 4 applied to this pair (W, X) is Result 3.

For dropping M areas, set $X = D^{1}_{\mathcal{M},it}\,(-\eta^{-}_{i,\mathcal{M},t})$. The indicator is a function of the observed bundle. The index $\eta^{-}_{i,\mathcal{M},t}$ is a function of $\check{v}$, N, k_{it}, and θ, the same objects as in the add case. Hence X is $\mathcal{F}_{ijt}$-measurable. Equation (22) is the pathwise bound. Lemma 4 applied to this pair is Result 4.

For a pairwise swap, set $X = D^{\text{swap}}_{j,m,it}\,\mu_{ijmt}(\theta)$. The indicator is a function of (d_{ijt}, d_{imt}), and $\mu_{ijmt}(\theta)$ is a function of the two area indexes. Equation (25) is the pathwise bound. Lemma 4 applied to this pair is Result 5.

The only properties required of $l(\cdot)$ for these inequalities to be valid are nonnegativity, $\mathcal{F}_{ijt}$-measurability, and integrability of the product $l(Z)X$. In the reported estimates, $l(Z)$ is a collection of binary firm-year indicators that partition the sample by lagged R&D, assets, and patent flow. A bounded instrument makes $l(Z)X$ integrable whenever X is.

The baseline specification evaluates the inequalities on exogenous pairs and triplets ($M \in \{2, 3\}$) together with the swap pairs, as described in Section 3. The algebra above does not depend on the particular functional form of $g(\cdot)$. The estimates in Section 4 use $g(k) = \binom{k}{2}$.

B.3 Empirical implementation

This subsection details the search over a bounded parameter region, the profiled bounds, and the remaining static-game caveats. Section 3 states how subsets are chosen from $\check{v}_{ijt}$ and how the 99 percent sets are constructed.

Subset sampling. Evaluating every pair, triplet, and ordered swap among $J = 20$ areas is unnecessary for the inequalities. I therefore rank subsets from predicted values $\check{v}_{ijt}$ before looking at realized d_{ijt}. The inequalities keep the leading $|M|$ subsets of that ranking, with $|M| \in \{10, 20, 40\}$ on the instrument grid in Appendix C. I restrict joint deviations to $M \in \{2, 3\}$. I use pair and triplet add and drop deviations, which vary how many areas enter the portfolio term. Appendix B.4 writes the profit gaps. For add moments I take the subsets with the highest average predicted value, where the decision not to enter is most informative about the cost. For drop moments I take the subsets with the lowest average predicted value, where the decision to stay is most informative. For swaps I rank the 380 ordered pairs by divergence in $\check{v}$ and draw from the top, the bottom, and the middle of that ranking. The specification that uses all three instruments and the largest subset sample keeps $|M| = 40$ per deviation type.

Test statistic and profiling. The object is a confidence set for congestion and for the occupancy-to-scope ratio, not a point estimate. The test statistic $T(\theta)$ is the max-type studentized sample moment of Chernozhukov et al. (2019). The critical value $c(1 - a, \theta)$ is their two-step self-normalized procedure at $a = 0.01$. I search a bounded parameter region by simulated annealing (Grieco, 2014) and keep the best draws as starting values. If the test statistic still exceeds the critical value, I minimize that slack without constraints to obtain a feasible start. Profiled bounds for α and for

$\rho = \kappa/\beta$ then solve

$$\min \alpha, \quad \max \alpha \qquad \text{and} \qquad \min \rho, \quad \max \rho \qquad \text{subject to} \qquad T(\theta) \leq c(1-a,\theta).$$

Static game. The static game answers how a firm composes its contemporaneous portfolio and how rival counts in the same period shift expected profit. Swap deviations isolate congestion. Add and drop deviations recover the occupancy-to-scope ratio. A firm that leaves an area stops paying the yearly occupancy cost. The panel supplies valuations, lagged instruments, and the observed bundles. The design leaves out learning, option value, preemption, and irreversible research outlays. If firms occupy an area in part for later expansion, the occupancy-to-scope ratio is net of that option value and understates the purely contemporaneous occupancy cost. Under irreversible entry with a value of waiting it would overstate that cost. If entry is sticky, α mixes contemporaneous crowding with the shadow cost of a persistent rival stock. Firm and year-by-area effects absorb a time-invariant capability level and common area-time conditions. A sunk cost paid on entering an area can be bounded with the same revealed-preference logic, by comparing the observed path with a path that alters the portfolio in a single year and then returns to the observed path (Morales et al., 2019; Pakes et al., 2015). In that comparison the continuation values cancel except for the sunk cost saved or brought forward, so no equilibrium-selection rule and no transition for rival counts are needed. Preemption and learning would require a full dynamic game, which I do not impose. Appendix A.10 reports the persistence of occupancy in the panel, and Appendix B.5 carries out the one-period comparison.

B.4 Identification of congestion α and occupancy-to-scope ratio ρ

Section 3 derives the add, drop, and swap inequalities. This subsection writes the profit gaps that those comparisons recover, so that the objects reported in Section 4 are the ones the inequalities can recover.

A swap holds the number of occupied areas fixed. Consider a firm that occupies area j and does not occupy area m. Replacing j with m leaves the firm occupying the same number of areas, so the organizational portfolio term does not change and the firm still pays occupancy cost for the same number of areas. What can still differ is expected patent value and how crowded each area is. That comparison isolates congestion. That is why Table 9 reports a set for α.

An add or a drop of M areas changes how many areas the firm occupies. Write A for the sum of expected valuations and congestion on the areas that enter or leave the bundle. The β and κ piece of the profit index is

$$A + \beta\,\Delta g - M\kappa, \tag{29}$$

where Δg is the change in the portfolio function. Occupancy is treated as a cost, so κ and β share a sign when the ratio below is formed.

Pairwise interactions. The baseline portfolio function is $g(k) = \binom{k}{2}$. Adding M areas to a firm that already occupies k areas gives $\Delta g^{+} = Mk + \binom{M}{2}$, or $M\big(k + \frac{M-1}{2}\big)$. Substituting into (29) and writing $\rho = \kappa/\beta$ produces

$$A + M\beta\Big(k + \tfrac{M-1}{2} - \rho\Big). \tag{30}$$

Variation in k and in $M \in \{2,3\}$ traces ρ. Under pairwise interactions, ρ is the portfolio size at which one more area covers its own occupancy cost. To see that reading, set $M = 1$. Adding

one area creates k new pairs, so the extra scope benefit is βk and the extra occupancy cost is κ. Those two terms balance when $k = \rho$. The coefficient β remains a scale in front of the parenthesis, because expected value enters with coefficient one. I therefore report a profiled 99 percent set for ρ. On the row with all three instruments and the largest subset sample, a ratio of zero is kept and negative ratios are rejected. That set is bounded above. The lagged-patent row is bounded above and is not bounded away from zero.

Portfolio size times its inverse hyperbolic sine. When $g(k) = k\,\mathrm{asinh}(k)$, the increment is $\Delta g = (k+M)\,\mathrm{asinh}(k+M) - k\,\mathrm{asinh}(k)$, and (29) becomes $A + \beta(\Delta g - M\rho)$. The algebra is the same ratio. The units are the units of that increment, so the upper end of the set in Table 10 is not interchangeable with the pairwise upper end.

Linear portfolio. When $g(k) = k$, the increment is $\Delta g = M$, and (29) becomes $A + M\delta$ with $\delta = \beta - \kappa$. The last column on the linear row of Table 10 is a 99 percent set for that per-area gap. A larger δ means the scope benefit of occupying one more area exceeds the occupancy cost of that area.

The estimates in Section 4 use pair and triplet add and drop deviations together with the swap moments. Appendix B.3 shows how those subsets are chosen and how the 99 percent sets are constructed.

B.5 Fixed and sunk costs

To ask how large a sunk cost the panel can support, I extend the add, drop, and swap comparisons by one period, following Morales et al. (2019) and Pakes et al. (2015). The static game has one per-area cost κ. Here the firm pays a fixed cost κ_f in every year it occupies an area and a sunk cost κ_s in the first year of each spell. I keep the observed portfolios. The rest of this subsection details the comparison, then the ratios those inequalities recover.

Step 1: the observable flow. For firm i, area j, and year t, write the observable part of occupying the area as

$$\bar{f}_{ijt} = \check{v}_{ijt} + \alpha\, h(N_{jt,-i}) + \beta\, \Delta g,$$

the same value, congestion, and scope piece already in the static add and drop moments. The increment Δg is the change in the portfolio function for that comparison. Expected value enters this flow with coefficient one. The new object is the cost.

Step 2: two costs. Let $d_{ijt} \in \{0, 1\}$ be occupancy. The firm pays

$$C_{ijt} = \kappa_f\, d_{ijt} + \kappa_s\, \mathbf{1}\{d_{ij,t-1} = 0,\, d_{ijt} = 1\}.$$

The fixed cost is paid in every year of occupancy. The sunk cost is paid only when the firm enters an area it did not occupy the year before. The static game is the case $\kappa_s = 0$.

Write $\rho_f = \kappa_f/\beta$ and $\rho_s = \kappa_s/\beta$. Under pairwise interactions, adding one area to a firm that already occupies k areas creates k new pairs, so the extra scope term is βk and the extra fixed cost is κ_f. Those two terms balance when $k = \rho_f$. The sunk cost balances the same new pairs when

$k = \rho_s$. The net payoff of occupying the area is then

$$\check{v}_{ijt} + \alpha\, h(N_{jt,-i}) + \beta(\Delta g - \rho_f) - \beta\rho_s\, \mathbf{1}\{d_{ij,t-1} = 0,\, d_{ijt} = 1\}.$$

The coefficient β remains a scale in front of the parenthesis, because expected value enters with coefficient one. Variation in how many areas the firm already occupies traces ρ_f and ρ_s. I hold congestion at one value from the static set and report the pairs (ρ_f, ρ_s) the inequalities accept at that value. On the slice $\rho_s = 0$, the yearly ratio ρ_f is the occupancy-to-scope ratio of Appendix B.4.

Step 3: a one-period deviation. Compare the observed path with a path that flips occupancy in year t only, $d'_{ijt} = 1 - d_{ijt}$, and then follows the observed choice from year $t+1$ on. From year $t+2$ onward both paths have the same occupancy history, so those continuation values cancel. Year $t+1$ occupancy is also the same, so next year's flow $\bar{f}_{ij,t+1}$ cancels. The only term that can still differ at $t+1$ is the sunk cost, because the lag going into $t+1$ is d_{ijt} on the observed path and $1 - d_{ijt}$ on the deviation. Discount the $t+1$ correction at $\delta = 0.95$. Revealed preference requires that the observed path is at least as good as the deviation,

$$V^{\text{obs}} - V^{\text{dev}} = \big(d_{ijt}\bar{f}_{ijt} - C^{\text{obs}}_{ijt}\big) - \big(d'_{ijt}\bar{f}_{ijt} - C^{\text{dev}}_{ijt}\big) + \delta\big(C^{\text{dev}}_{ij,t+1} - C^{\text{obs}}_{ij,t+1}\big) \geq 0.$$

The observed status in year $t+1$ enters this difference in place of the firm's forecast, which is valid when forecast errors are uncorrelated with the instruments.

Step 4: the firm is out, and the deviation is to enter. On the observed path, $d_{ijt} = 0$, so year-t costs are zero. The deviation occupies the area and pays $\kappa_f + \kappa_s \mathbf{1}\{d_{ij,t-1} = 0\}$. At $t+1$, the observed path has lag 0 and therefore pays the sunk cost if $d_{ij,t+1} = 1$; the deviation has lag 1 and does not. The deviation therefore saves $\delta\kappa_s d_{ij,t+1}$ at $t+1$. Observed better than entering requires

$$\bar{f}_{ijt} \leq \kappa_f + \kappa_s\big(\mathbf{1}\{d_{ij,t-1} = 0\} - \delta d_{ij,t+1}\big).$$

A firm outside the area in both $t-1$ and t reveals $\bar{f}_{ijt} \leq \kappa_f + \kappa_s(1 - \delta d_{ij,t+1})$. A firm that has just left reveals $\bar{f}_{ijt} \leq \kappa_f - \delta\kappa_s d_{ij,t+1}$.

Step 5: the firm is in, and the deviation is to leave. On the observed path the firm pays $\kappa_f + \kappa_s \mathbf{1}\{d_{ij,t-1} = 0\}$ in year t. The deviation pays nothing in year t. At $t+1$, leaving today means that a return next year requires paying the sunk cost again, so the deviation costs an extra $\delta\kappa_s d_{ij,t+1}$. Staying better than leaving requires

$$\bar{f}_{ijt} \geq \kappa_f + \kappa_s\big(\mathbf{1}\{d_{ij,t-1} = 0\} - \delta d_{ij,t+1}\big).$$

A firm that has just entered reveals $\bar{f}_{ijt} \geq \kappa_f + \kappa_s(1 - \delta d_{ij,t+1})$. A firm that stayed reveals $\bar{f}_{ijt} \geq \kappa_f - \delta\kappa_s d_{ij,t+1}$.

The four groups are therefore

$$\begin{aligned}
(d_{ij,t-1}, d_{ijt}) = (0,0) \quad &\Rightarrow \quad \bar{f}_{ijt} \leq \kappa_f + \kappa_s(1 - \delta d_{ij,t+1}),\\
(d_{ij,t-1}, d_{ijt}) = (0,1) \quad &\Rightarrow \quad \bar{f}_{ijt} \geq \kappa_f + \kappa_s(1 - \delta d_{ij,t+1}),\\
(d_{ij,t-1}, d_{ijt}) = (1,1) \quad &\Rightarrow \quad \bar{f}_{ijt} \geq \kappa_f - \delta\kappa_s d_{ij,t+1},\\
(d_{ij,t-1}, d_{ijt}) = (1,0) \quad &\Rightarrow \quad \bar{f}_{ijt} \leq \kappa_f - \delta\kappa_s d_{ij,t+1}.
\end{aligned}$$

Next year's occupancy is in the formula because it shows whether the deviation prepays or postpones the sunk cost. A firm that is out this year and in next year would, by entering today, avoid paying the sunk cost next year. That saving is worth $\delta\kappa_s$. If next year's occupancy were omitted, the sunk cost would enter only through $1-\delta = 0.05$.

Write

$$w_{ijt} = \mathbf{1}\{d_{ij,t-1} = 0\} - \delta\, d_{ij,t+1}$$

and

$$L_{ijt} = \check{v}_{ijt} + \alpha\, h(N_{jt,-i}) + \beta\big(\Delta g - \rho_f - \rho_s w_{ijt}\big).$$

A firm that is out of the area in year t reveals $L_{ijt} \leq 0$. A firm that is in reveals $L_{ijt} \geq 0$. The four groups above are this pair of restrictions, with $w_{ijt} = 1 - \delta d_{ij,t+1}$ when the firm was out in $t-1$ and $w_{ijt} = -\delta d_{ij,t+1}$ when the firm was in.

Step 6: swaps. A swap leaves area j and enters area m in the same year, then returns to the observed path. Portfolio size is unchanged, so $\Delta g = 0$. The fixed cost is paid on m and ceases on j, so ρ_f cancels. Each area keeps the sunk-cost weight from Steps 4 and 5. The expected profit of the swap is

$$\big(\check{v}_{imt} - \check{v}_{ijt}\big) + \alpha\big(h(N_{mt,-i}) - h(N_{jt,-i})\big) - \beta\rho_s\big(w_{imt} - w_{ijt}\big).$$

When $\rho_s = 0$, the gap is the static swap and isolates congestion. When ρ_s is free, congestion and the sunk-cost ratio enter the same gap. Each row therefore holds one value of congestion from the static set, and the inequalities are asked which pairs (ρ_f, ρ_s) they accept at that value. I pool the swap comparisons across groups, each firm-year carrying its own sunk-cost weights. Because consecutive years of the same firm enter the same comparison, I cluster the inference by firm.

Step 7: the sets the inequalities recover. The test statistic and the self-normalized critical value are those of Section 3, with forty subsets per deviation family, all three instruments, and moments summed within firm before studentizing, at the one percent level. Each row holds one value of α from the static set in Table 9. At that value, and at a candidate pair (ρ_f, ρ_s), I minimize the criterion over the scale β. The pair is kept when the minimized criterion is non-positive. Table 23 reports the smallest and largest accepted value of each ratio. The scale stays inside that minimization. When $\rho_s = 0$, Steps 4 and 5 collapse to the static add and drop logic, still split by last year's status and still clustered by firm, and ρ_f is the occupancy-to-scope ratio of Appendix B.4. A sunk-cost ratio of zero is the cost structure of the static game. I compute the accepted pairs twice: once dropping a firm that is absent from the sample in the following year, and once treating that absence as having left.

The first-stage projection predicts realized value, which is zero in most firm-area-years outside an area, so predicted valuation rises with how often a firm has occupied an area. Firms that have just entered and firms that have just left therefore receive nearly the same prediction, which limits how far these comparisons separate ρ_s from ρ_f. The check is for the pairwise baseline of Table 9.

Table 23: Fixed-cost and sunk-cost ratios accepted by the one-period inequalities

Held fixed	α	Fixed-cost ratio ρ_f	Sunk-cost ratio ρ_s	Accepted pairs	Group that rejects the next larger sunk-cost ratio
Panel A: firms absent from the sample next year are dropped					
Static midpoint	-0.65035	$[0, 9.2]$	$[0, 7.4]$	4,644/12,221	firms that have just entered
Strong congestion	-1.116	$[0, 8.9]$	$[0, 7.6]$	4,540/12,221	firms that have just entered
Weak congestion	-0.1847	$[3.5, 9.4]$	$[0, 6.8]$	2,240/12,221	firms that have just entered
Static interior point	-1.03	$[0, 9]$	$[0, 7.5]$	4,548/12,221	firms that have just entered
Panel B: firms absent from the sample next year are treated as having left					
Static midpoint	-0.65035	$[0, 9.1]$	$[0, 7.7]$	4,765/12,221	firms that have just entered
Strong congestion	-1.116	$[0, 8.9]$	$[0, 7.9]$	4,705/12,221	firms that have just entered
Weak congestion	-0.1847	$[3.7, 9.3]$	$[0, 7.5]$	2,172/12,221	firms that have just entered
Static interior point	-1.03	$[0, 8.9]$	$[0, 7.9]$	4,779/12,221	firms that have just entered

Notes: Each row holds the congestion parameter at one value from the static set in Table 9 and reports the fixed-cost ratio ρ_f and the sunk-cost ratio ρ_s that the one-period inequalities of Appendix B.5 accept at the one percent level. The static midpoint is the midpoint of that congestion interval. Strong and weak congestion hold the congestion parameter at each end of that interval. The static interior point is a second point inside that interval. At each pair the criterion is minimized over the scale in front of the portfolio term, and the pair is kept when the minimized criterion is non-positive. The scale stays inside that minimization. The grid contains 121 values of the fixed-cost ratio, from 0 to 12, and 101 values of the sunk-cost ratio, from 0 to 10, hence $121 \times 101 = 12{,}221$ pairs per row. The reported intervals are the smallest and largest accepted values on that grid. Inference uses the self-normalized critical value with forty subsets per deviation family, all three instruments, and moments summed within firm before studentizing (763 firms). Comparison groups are defined by occupancy in the previous and current year; occupancy in the following year enters the payoff with discount factor 0.95. Swap comparisons are pooled over groups, each firm-year keeping its own sunk-cost terms. The last column names the comparison group whose inequality rejects the smallest sunk-cost ratio on the grid above the accepted range; for a row with no accepted pair it names the group that rejects at the pair closest to acceptance.

Every row of Table 23 keeps a sunk-cost ratio of zero, and every row is bounded above. At the static midpoint, Panel A keeps 4,644 of the 12,221 grid pairs, with ρ_f from 0 to 9.2 and ρ_s from 0 to 7.4. Strong congestion keeps 4,540 pairs, and the interior point keeps 4,548. Weak congestion, $\alpha = -0.1847$, keeps 2,240 pairs, and on that row the fixed-cost ratio starts at 3.5 and runs to 9.4, with the sunk-cost ratio from 0 to 6.8. When an extra rival reduces expected profit only a little, the inequalities ask for a larger fixed cost before they will accept the pair. Panel B treats a firm that is absent next year as having left. It keeps 4,765 pairs at the midpoint and 2,172 pairs under weak congestion. The upper ends stay within one ratio point of the Panel A ends.

On every row, the comparison that rejects the next larger sunk-cost ratio is the group of firms that have just entered. A firm in that group pays $\rho_f + \rho_s$ if it enters. If it stays out, it reveals that this sum was too large relative to predicted value and to the pairs the entry would have created. That comparison is what closes ρ_s from above. A sunk-cost ratio of zero stays inside the set on every row. That zero is the cost structure of the static game, and it is the cost structure the case study uses when it cuts ρ.

C Robustness of the identified sets

The baseline confidence sets in Section 4 use the pairwise-interaction portfolio function and add and drop deviations of size two and three. Table 24 reports the full instrument and subset-size grid. Table 25 repeats that grid after dropping size-three add and drop moments. Table 10 in the main text reports the portfolio-function comparison at the largest subset sample. This appendix

reports the pairwise-interaction grid.

Table 24: Estimated parameter bounds under different instrument setups

Setup	$\lvert M\rvert$	α	$\rho = \kappa/\beta$
No IV	20	$[-1.1084, -0.1111]$	—
RD_{it-1}	20	$[-1.1077, -0.2144]$	—
AT_{it-1}	20	$[-1.1088, -0.1444]$	—
PAT_{it-1}	20	$[-1.1106, -0.0966]$	$[0, 12.6250]$
RD_{it-1}, AT_{it-1}	20	$[-1.1109, -0.2052]$	—
RD_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.1992]$	$[0, 12.3750]$
AT_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.1453]$	$[0, 12.8750]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	10	$[-1.1112, 0.1287]^{*}$	$[0, 15.8750]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.1992]$	$[0, 12.8750]^{\dagger}$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	40	$[-1.1160, -0.1847]$	$[0, 12.3750]$

Notes: The table reports 99 percent confidence sets under pairwise interactions among entered areas, add and drop deviations of size two and three, and swap moments, on the full instrument and subset-size grid. Add and drop moments depend on the per-area occupancy cost and the pairwise scope benefit through their ratio $\rho = \kappa/\beta$. Swap moments identify congestion without that ratio. The table therefore reports α and a set for ρ. The congestion parameter α is the lowest and highest value among jointly feasible points. The set for ρ is obtained by freezing the ratio and searching the remaining parameters; a value is kept when the criterion is non-positive. A dash means that ratio is not identified. Where the ratio is identified, the set includes zero and is bounded above. On each row where the ratio is printed as an interval from zero, a ratio of zero is kept and ratios below zero are rejected, so that row is closed at zero. Congestion equal to zero is rejected on the instrumented rows other than the row with all three instruments and $|M| = 10$. Congestion above zero is rejected on those rows other than lagged R&D alone. The row with all three instruments and $|M| = 10$ keeps congestion equal to zero and congestion above zero. A dagger marks a confidence set that omits a single evaluated point in the interior (12.5 at RD_{it-1}, AT_{it-1}, PAT_{it-1} with $|M| = 20$). An asterisk marks an interval that contains zero. Table 9 reports the no-instrument, lagged-patent, and all-instrument largest-subset rows. Table 25 repeats this grid with add and drop deviations of size two only. Inference uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

C.1 Alternative portfolio functions

Under the pairwise-interaction technology, a positive β means that pairwise interactions scale with the number of unique pairs among entered areas, so the marginal value of the next area rises with portfolio size. Table 10 reports those alternatives at all three instruments and the largest subset sample. The last column is a 99% set for $\rho = \kappa/\beta$ on the pairwise and $k\,\text{asinh}(k)$ rows, and for $\delta = \beta - \kappa$ on the linear row. Appendix B.4 writes the profit gaps.

Three facts stand out.

First, applying the inverse hyperbolic sine, the square root, or the logarithm to portfolio size returns an empty 99% confidence set at all three instruments and the largest subset sample in Table 10. Those concave forms would make extra areas a diminishing synergy. No combination of congestion, scope, and occupancy cost satisfies the inequalities at the 99% critical value under those forms.

Second, a linear portfolio function identifies the per-area gap $\delta = \beta - \kappa$. At the largest subset sample that set is $[1.6250, 3.3750]$. One more area always adds the same scope benefit, so the last column on that row is that gap.

Third, multiplying portfolio size by its inverse hyperbolic sine remains feasible and keeps $\alpha < 0$, with $\rho \in (0, 4.1250]$. That upper end sits on a different scale from the pairwise upper end of

Table 25: Estimated parameter bounds, add and drop deviations of size two only

Setup	$\lvert M\rvert$	α	$\rho = \kappa/\beta$
No IV	20	$[-1.1050, -0.1106]$	—
RD_{it-1}	20	$[-1.1077, -0.2199]$	—
AT_{it-1}	20	$[-1.1059, -0.1541]$	—
PAT_{it-1}	20	$[-1.1091, -0.1016]$	$[0] \cup [8.1250, 12.6250]$
RD_{it-1}, AT_{it-1}	20	$[-1.1083, -0.2177]$	—
RD_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.2054]$	$[0, 12.3750]$
AT_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.1449]$	$[0, 12.8750]^{\dagger}$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	10	$[-1.1081, 0.1209]^{*}$	$[0, 15.3750]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.2054]$	$[0, 12.3750]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	40	$[-1.1128, -0.1964]$	$[0, 12.3750]$

Notes: The table uses the same pairwise-interaction portfolio function and the same instrument and subset-size grid as Table 24, with add and drop deviations of size two only. Add and drop moments depend on the per-area occupancy cost and the pairwise scope benefit through their ratio $\rho = \kappa/\beta$. Swap moments identify congestion without that ratio. The table therefore reports α and a set for ρ. The congestion parameter α is the lowest and highest value among jointly feasible points. The set for ρ is obtained by freezing the ratio and searching the remaining parameters; a value is kept when the criterion is non-positive. A dash means that ratio is not identified. Where the ratio is identified, the set includes zero and is bounded above. On each row where the ratio is printed as an interval from zero, a ratio of zero is kept and ratios below zero are rejected, so that row is closed at zero. On the lagged-patent row, a ratio of zero is kept and ratios below zero are rejected. The positive ratios kept on that row are the interval printed beside zero. Congestion equal to zero is rejected, and congestion above zero is rejected, on the instrumented rows other than the row with all three instruments and $|M| = 10$. That row keeps congestion equal to zero and congestion above zero. The row with no instrument rejects congestion equal to zero and congestion above zero. A dagger marks a confidence set that omits a single evaluated point in the interior (12.5 at AT_{it-1}, PAT_{it-1} with $|M| = 20$). An asterisk marks an interval that contains zero. Table 9 reports the no-instrument, lagged-patent, and all-instrument largest-subset rows under add and drop deviations of size two and three. Inference uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

12.3750, because one more area changes $g(k)$ by a different amount. I keep the pairwise-interaction function as the baseline because it is the technology written in Section 3, and because its set for ρ at that sample is bounded above.

Taken together, at all three instruments and the largest subset sample the feasible forms preserve crowding out. Under a concave portfolio function the identified set empties.

The case study in Section 5 uses the pairwise row of that comparison. Under pairwise interactions the marginal value of the next area rises with the number of areas already occupied, the set for ρ includes zero and stops at 12.3750, and congestion stays negative. The linear technology is feasible and answers a different question. There the per-area gap $\delta = \beta - \kappa$ lies in $[1.6250, 3.3750]$, so throughout that set the scope benefit of one more area exceeds its occupancy cost. The numerical exercises scale a ratio ρ and a pair value b. That scaling is the pairwise technology, in which the extra scope from one more area is β times the number of areas already occupied.

The technology that multiplies portfolio size by its inverse hyperbolic sine is feasible as well, and its set for ρ stops at 4.1250. For a firm that already occupies five areas, the extra scope from a sixth area is about 3.39β, against five new pairs under $g(k) = \binom{k}{2}$. A cut of 25 percent in ρ therefore removes a different quantity of scope under the two functions. The portfolios in Appendix D are built from the pairwise increment, $g(k+1) - g(k) = k$. The concave technologies stay out of those exercises, because at this sample the inequalities return an empty set.

C.2 Alternative entry thresholds

As explained in the main text and previous Appendix, I define entry when the firm's text exceeds the threshold $\bar{w} \approx 1\%$ (see Appendix A.3). For the sensitivity analysis, I re-estimate the specification in the last row of Table 9 at $\tilde{w} = 0.5\%$ and at $\tilde{w} = 2\%$. Then, I rebuild rival counts from the new entry thresholds and keep the predicted valuations, the instruments, and the firm-year sample of the main specification. Table 26 reports the 99% sets and the average number of areas a firm occupies in a year.

Table 26: Estimated parameter bounds by entry threshold

Threshold $\bar{w}$	Mean k_{it}	α	$\rho = \kappa/\beta$
1%	5.37	$[-1.1160, -0.1847]$	$[0, 12.3750]$
0.5%	6.19	$[-1.3366, -0.3148]$	—
2%	4.34	—	—

Notes: The table reports 99% confidence sets for congestion α and occupancy-to-scope ratio ρ in the specification of the last row of Table 9: pairwise interactions among entered areas, add and drop deviations of size two and three, swap moments, all three instruments, and the largest subset sample ($|M| = 40$). Each row changes the weight threshold $\bar{w}$ in both places where it enters the entry indicator. Patent weights below $\bar{w}$ are set to zero, and a firm enters an area when the average of those weights over its patents in that year is at least $\bar{w}$. Rival counts are rebuilt from the new entry indicators. The firm-year sample, the predicted valuations, and the instruments are those of the main specification. The first row repeats Table 9, and the other two thresholds are the lines at 0.5 and 2 percent in Figure 1. Mean k_{it} is the average number of areas a firm occupies in a year in the estimation sample. On the rows at 1 percent, I further profile each parameter and find that I reject the possibility of $\alpha \geq 0$ and $\rho < 0$. At the 0.5 percent threshold, a negative ratio is kept. A dash in the last column means that ratio is not identified. At the 2 percent threshold, the inequalities found no jointly feasible point, so both columns are empty. Inference uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

The first row repeats the last row of Table 9 which uses $\bar{w} = 1\%$. The second row uses $\tilde{w} = 0.5\%$, and in this case the average firm occupies 6.19 areas in a year. This cutoff $\tilde{w} = 0.5\%$ lies on the left hill of Figure 1, among the leftover shares, so leftover shares now count toward entry. I still find that an extra rival still lowers expected profit. Congestion α lies in $[-1.3366, -0.3148]$, and from further profiling I reject the possibility that $\alpha \geq 0$. However, the occupancy-to-scope ratio ρ is not identified. With those leftover shares now counted, the 99% set also contains values at which the gain from occupying more areas at once, β, and the cost of one more area, κ, are both close to zero. Near such values, a small change in either one moves their ratio far in either direction. The set therefore keeps negative ratios as well as positive ones, and the negative ratios I find pair a cost just above zero with a gain just below zero.

The third row uses $\tilde{w} = 2\%$, and the average firm now only occupies 4.34 areas in a year. This threshold $\tilde{w} = 2\%$ line lies above the fourth-largest share of the median patent, about 0.9 percent, and below its third-largest share, about 8 percent (see Figure 2). At this threshold, notice that the fourth area of the median patent does not count toward entry. As for the result, I find that there is no combination of congestion, scope, and occupancy cost that satisfies the inequalities at the 99% critical value, so both columns of the row are empty.

D Details on Numerical Simulation

This appendix provides notes on how I build the portfolios in Section 5 and how I read the cells in the tables. It then details the area score under each exercise. In that section the rival counts are

those implied by the portfolios. The same exercises with the observed counts are at the end of this appendix. Let $\mathcal{B}$ be the four semiconductor areas named in that section: memory, lithography and fabrication, packaging, and the transistor.

D.1 Parameters

The exercises use the 99 percent confidence set in the third row of Table 9. That row is the specification with all three instruments and the largest subset sample. The comparisons are the add, drop, and swap moments, and the portfolio term counts pairwise interactions among entered areas. Throughout, $h(N) = \operatorname{asinh}(N)$ and $g(k) = \binom{k}{2}$, as in Section 3. The portfolio coefficient is a pair value $b = \beta$. The per-area cost in an ordinary area is then $\kappa = b\rho$. I evaluate $b \in \{0.25, 1, 4\}$.

At each pair value I keep pairs (α, ρ) that the inequalities accept at $\theta = (\alpha, b, b\rho)$. I evaluate that criterion on a grid. Congestion runs from -1.50 up to just below zero in steps of 0.05, and the grid includes the published endpoints -1.1160 and -0.1847. The occupancy-to-scope ratio runs from 0 to 12.375 in steps of 0.25, and the grid includes the endpoint 12.375. A pair is kept when the criterion at that θ is zero or negative, at the same 99 percent level as Table 9. The grid keeps 9 pairs at $b = 0.25$, 9 pairs at $b = 1$, and 29 pairs at $b = 4$. That collection is smaller than the rectangle formed by the printed intervals for α and ρ.

The fixed-size columns of Tables 29 and 11 walk α from -1.1160 to -0.1847 in steps of 0.02. Those two columns depend only on congestion. The extra-area columns, and both margins of Table 12, use the kept pairs at the column's pair value. Halving congestion maps each accepted α to $\alpha/2$, including values whose half lies outside the confidence set. Each reported range is the minimum and maximum of the object over the parameters used in that column. Tables 27 and 28 use the same pairs and hold rival counts at their observed values.

D.2 Building the portfolio

For a bundle S with $k = |S|$ areas, expected profit is

$$\mathbb{E}[\Pi(S)] = \sum_{j \in S} \big(\check{v}_j + \alpha\, h(N_j) - \kappa_j\big) + b\, g(k).$$

I build the bundle in two ways. First, I hold k at the firm's observed number of areas. The portfolio term $b\, g(k)$ is then the same for every bundle of that size, and a cost that is the same in every area is paid the same number of times. The firm occupies the k areas with the largest area score $\zeta_j = \check{v}_j + \alpha\, h(N_j) - \kappa_j$. Second, I start from the bundle the firm actually occupies and allow k to rise. Adding one empty area m changes expected profit by

$$\Delta_m = \big(\check{v}_m + \alpha\, h(N_m) - \kappa_m\big) + b\, k,$$

because $g(k+1) - g(k) = k$. In an ordinary area, $\kappa_m = b\rho$, so $\Delta_m = \check{v}_m + \alpha\, h(N_m) + b(k - \rho)$. The firm adds the empty area with the largest positive increment, raises k by one, and repeats. Occupied areas stay occupied. When two empty areas tie, I add the one with the smaller topic number.

D.3 Rival counts

The tables in Section 5 use rival counts that match the portfolios. I take each year on its own. Let $d_{ijt} = 1$ when firm i occupies area j in the current assignment, and let F_{jt} be the number of entrants who appear in the patent counts for that area and year and are outside the sample. Those entrants stay where the data put them. The rival count in the profit function is the sum of the other firms' entry,

$$N_{jt,-i} = F_{jt} + \sum_{i' \neq i} d_{i'jt}.$$

When a firm moves, the indicators in that sum change, so the rival count changes for every firm scored after that move.

I start from the portfolios in the data. I then take the firms in the sample one at a time, in the order of the sample. For the firm at hand I recompute $N_{jt,-i}$ from the current assignment and rescore the areas with the area score in the previous subsection. Under the fixed-size rule I replace the portfolio with the top k areas when the sum of area scores rises, and k is the observed number. Under the extra-area rule I rebuild the portfolio from the observed bundle by the one-area increments above, and I replace the current portfolio when the rebuild differs. Areas the firm occupies in the data stay occupied. If the firm moves, I update the assignment before I score the next firm. I repeat until a full pass leaves every portfolio unchanged.

I make that calculation twice. The first uses the scores before the policy and starts from the observed portfolios. The second uses the policy scores and starts from the first assignment. Each cell is the change from the first assignment to the second.

As an illustration, consider the following: three firms in one year show the arithmetic. Each firm keeps one area. In the tables the penalty on rivals is $\alpha\, h(N_{jt,-i})$. In this illustration I set that penalty at 3 per other firm, and I set $F_{jt} = 0$.

	Value in memory	Value in display	Starting area
Firm 1	10	6	Memory
Firm 2	10	4	Memory
Firm 3	15	8	Display

Firm 1 faces one other firm in memory and one in display. Memory scores $10 - 3 = 7$ and display scores $6 - 3 = 3$, so firm 1 stays in memory. Firm 2 faces the same counts. Memory scores $10 - 3 = 7$ and display scores $4 - 3 = 1$, so firm 2 stays in memory. Firm 3 faces two other firms in memory and none in display. Memory scores $15 - 6 = 9$ and display scores 8, so firm 3 moves to memory.

On the next pass, memory contains all three firms. Firm 1 faces two other firms in memory and none in display. Memory scores $10 - 6 = 4$ and display scores 6, so firm 1 moves to display. Firm 2 then faces one other firm in each area. Memory scores $10 - 3 = 7$ and display scores $4 - 3 = 1$, so firm 2 stays in memory. Firm 3 faces one other firm in each area. Memory scores $15 - 3 = 12$ and display scores $8 - 3 = 5$, so firm 3 stays in memory. A further pass leaves every portfolio unchanged. Memory contains firms 2 and 3, and each of those two was scored against one other firm. Display contains firm 1, who was scored against no other firm.

Scored once against the counts in the starting assignment, firm 3 moves to memory and firms 1 and 2 stay, so memory contains three firms. Once the sum is rebuilt, firm 1 moves to display, and

memory contains firms 2 and 3.

D.4 Reading the cells

Each cell is the change, in percentage points, in an area's share of occupied slots, from the same parameters before the policy to the policy. The last row is the change, in percentage points, in the share of firm-years that add at least one area. With the number of areas held fixed, that last row is zero. In the extra-area columns, occupied areas stay occupied, so a lower share means that other areas account for more of a larger total.

D.5 A subsidy based on predicted value

The scores in this subsection and the next are the scores in Section 5, where the rival count is the count from the other firms' current portfolios, and in Tables 27 and 28, where the rival count is the observed count.

I add a quarter of predicted value inside $\mathcal{B}$:

$$\mathbb{E}[\Pi'] = \sum_{j \in S} (\check{v}_j + \alpha\, h(N_j) - \kappa_j) + \sum_{j \in S \cap \mathcal{B}} 0.25\, \check{v}_j + b\, g(k).$$

The fixed-size column uses the first construction above and walks α on the published interval. The pair value cancels. The firm ranks by

$$\zeta_j = \begin{cases} 1.25\, \check{v}_j + \alpha\, h(N_j) & j \in \mathcal{B}, \\ \check{v}_j + \alpha\, h(N_j) & j \notin \mathcal{B}, \end{cases}$$

and keeps the top k.

The extra-area columns use the second construction, at the kept pairs for that b. The firm adds area m when

$$\Delta_m = \begin{cases} 1.25\, \check{v}_m + \alpha\, h(N_m) + b(k - \rho) & m \in \mathcal{B}, \\ \check{v}_m + \alpha\, h(N_m) + b(k - \rho) & m \notin \mathcal{B}. \end{cases}$$

Before the policy, every area uses the second line.

D.6 A lower occupancy cost in semiconductor areas

I replace ρ with 0.75ρ only inside $\mathcal{B}$. A semiconductor area's cost falls from $b\rho$ to $0.75\, b\rho$, which raises its score by $0.25\, b\rho$.

The fixed-size columns use the first construction above. The firm ranks by

$$\zeta_j = \begin{cases} \check{v}_j + \alpha\, h(N_j) + 0.25\, b\rho & j \in \mathcal{B}, \\ \check{v}_j + \alpha\, h(N_j) & j \notin \mathcal{B}, \end{cases}$$

and keeps the top k. The extra-area columns use the second construction. The firm adds area m when

$$\Delta_m = \begin{cases} \check{v}_m + \alpha\, h(N_m) + b(k - \rho) + 0.25\, b\rho & m \in \mathcal{B}, \\ \check{v}_m + \alpha\, h(N_m) + b(k - \rho) & m \notin \mathcal{B}. \end{cases}$$

Before the policy, every area uses the second line. Both constructions use the kept pairs at that column's pair value.

D.7 Halving rival congestion

I replace α with $\alpha/2$. Costs are unchanged. Each firm is scored once, against the observed rival counts.

The fixed-size column uses the first construction above. The firm ranks the twenty areas by

$$\zeta_j = \check{v}_j + \frac{\alpha}{2}\, h(N_j)$$

and keeps the top k, where k is the observed number of areas. Before the policy the same ranking uses α. The pair value cancels: once k is held fixed, $b(k-\rho)$ is the same for every area.

The extra-area columns use the second construction, at the kept pairs for that b. The firm adds empty area m when

$$\Delta_m = \check{v}_m + \frac{\alpha}{2}\, h(N_m) + b(k-\rho) > 0,$$

raises k by one, and repeats. Before the policy the same line uses α. Occupied areas stay occupied.

Holding the number of areas fixed, firms move toward more crowded areas, because those areas gain the most when the congestion penalty is cut in half. The moves are about one to two percentage points. "Display Panel Tech (LCD/OLED)" falls by 1.25 to 1.86 percentage points. "Integrated Circuit Design, Verification, and Testing" gains the most, between 0.56 and 1.37. Memory, video, and imaging also lose share. Audio and board-level hardware gain share.

The fixed-size column is a reshuffling of the observed number of areas. Cutting the penalty in half adds $(\alpha/2 - \alpha)\, h(N) = (-\alpha/2)\, h(N)$ to each area's score, and that addition is larger where $h(N)$ is larger. Display is the thinnest area in Table 18, and it loses the most. Memory loses 0.41 to 1.31 percentage points, imaging 0.41 to 1.37, video 0.24 to 1.10, and the transistor 0.32 to 1.01. Optics loses 0.17 to 0.75. On the other side of the same ranking, audio gains 0.36 to 1.10, board-level hardware 0.37 to 1.01, clock circuits 0.37 to 0.78, and power-supply electronics 0.30 to 0.90. Lithography and packaging barely move. The firm is reordering areas by $\check{v}_j + (\alpha/2)\, h(N_j)$, with k held at the observed count, so one area's gain is another area's loss.

Allowing extra areas, the same halving induces expansion when a pair is cheap, and little expansion when a pair is expensive. The share of firm-years that add at least one area rises by 16.13 to 24.56 percentage points at $b = 0.25$, by 6.08 to 9.40 at $b = 1$, and by 0.18 to 2.39 at $b = 4$. The new slots raise the share of display, memory, video, and imaging. At $b = 0.25$, display gains 1.10 to 1.48 percentage points, memory 0.66 to 0.88, video 0.56 to 0.79, and imaging 0.56 to 0.73. Integrated circuit design, audio, and board-level hardware account for a smaller share of the larger total: at the same pair value those three fall by 0.90 to 1.28, 0.77 to 1.06, and 0.58 to 0.83. A cheap pair makes the one-area increment positive for many empty areas. The areas that were held back by a steep congestion penalty are the ones the firm adds.

D.8 A lower occupancy cost in every area

I replace ρ with 0.75ρ in every area, so $\kappa_j = 0.75\, b\rho$ everywhere. Congestion stays as it is. Each firm is scored once, against the observed rival counts.

For two bundles S and S' of the same size, the difference in expected profit after the cut is

$$\mathbb{E}[\Pi'(S)] - \mathbb{E}[\Pi'(S')] = \sum_{j \in S} \big(\check{v}_j + \alpha\, h(N_j)\big) - \sum_{j \in S'} \big(\check{v}_j + \alpha\, h(N_j)\big).$$

The new cost is the same in every area, so the top-k bundle is unchanged and the share change at a fixed size is zero. Table 30 therefore has extra-area columns only.

At the kept pairs for that b, the firm adds area m when

$$\Delta_m = \check{v}_m + \alpha\, h(N_m) + b\big(k - 0.75\rho\big) > 0.$$

Before the policy the same line uses ρ.

Expansion is small when a pair is cheap and larger when a pair is expensive. The share of firm-years that add at least one area rises by 0.00 to 1.33 percentage points at $b = 0.25$, by 6.72 to 8.31 at $b = 1$, and by 10.00 to 12.80 at $b = 4$. Area shares move by less than one percentage point. Display gains the most. Integrated circuit design, audio, and clock circuits account for a smaller share of the larger total.

The relief in every area is $0.25\, b\rho$. A high pair value makes that relief large, which is why expansion rises with b. In the halving exercise a cheap pair produces the large expansion, because halving α does not scale with b. Here the cost cut does scale with b. At $b = 4$, display gains 0.65 to 0.83 percentage points, memory 0.39 to 0.49, video 0.34 to 0.42, imaging 0.34 to 0.41, and the transistor 0.31 to 0.39. Integrated circuit design loses 0.66 to 0.78, audio 0.52 to 0.63, and clock circuits 0.42 to 0.52. The new areas are the ones whose score $\check{v} + \alpha\, h(N)$ clears the lower threshold $b(k - 0.75\rho)$. Because the cut is the same in every area, two bundles of equal size keep the same ranking, and every change in Table 30 is expansion from the observed portfolio.

At $b = 0.25$ the relief is small. The share of firm-years that add an area rises by at most 1.33 percentage points, and every area share moves by at most 0.09. At $b = 1$ that share rises by 6.72 to 8.31 percentage points, and display gains 0.48 to 0.61. The areas that gain are the ones with room under the lower threshold: display, memory, video, imaging, and the transistor. Integrated circuit design, audio, and clock circuits account for a smaller share of the larger total. Occupied areas stay occupied, so those three lose share because the new slots are added elsewhere.

D.9 Holding rival counts fixed

Tables 27 and 28 repeat the semiconductor exercises with each firm scored once, against the rival counts in the data. The halving exercise and the uniform occupancy-cost cut use that same construction. In those tables every firm faces the observed counts.

Table 27: Case I: a 25% subsidy based on predicted value, with rival counts held fixed

Area	Fixed size	Extra areas		
		$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.84, 0.96]	[0.05, 0.07]	[0.02, 0.02]	[0.00, 0.01]
Audio and Speech-Signal Processing	[−0.16, −0.07]	[−0.03, −0.03]	[−0.03, −0.02]	[−0.02, −0.00]
Semiconductor Lithography and Fabrication Processes	[0.70, 0.86]	[0.03, 0.05]	[0.01, 0.01]	[−0.00, 0.01]
Integrated Circuit Design, Verification, and Testing	[−0.23, −0.18]	[−0.04, −0.03]	[−0.03, −0.02]	[−0.02, −0.00]
PCB and Electronic Hardware with Connectors	[−0.27, −0.15]	[−0.03, −0.02]	[−0.02, −0.02]	[−0.01, 0.00]
Video Recording, Encoding, and Media Processing	[−0.28, −0.19]	[−0.00, 0.01]	[0.01, 0.01]	[−0.00, 0.01]
Semiconductor Packaging and Interconnect Bonding	[0.75, 0.89]	[0.04, 0.05]	[0.01, 0.01]	[0.00, 0.01]
Thin-Film Materials, Deposition and Chemistry	[−0.24, −0.15]	[−0.01, −0.01]	[−0.01, −0.00]	[−0.00, 0.00]
Wireless Base-Station and Mobile Communications	[−0.31, −0.23]	[−0.00, 0.00]	[0.01, 0.02]	[0.00, 0.01]
Clock Generation, Timing Circuits, and Signal Processing	[−0.15, −0.10]	[−0.03, −0.03]	[−0.02, −0.01]	[−0.02, −0.00]
Power-Supply and Power-Management Electronics	[−0.24, −0.18]	[−0.02, −0.02]	[−0.02, −0.02]	[−0.01, −0.00]
Networking Protocols and Data Infrastructure	[−0.20, −0.16]	[−0.01, 0.01]	[0.01, 0.02]	[−0.00, 0.01]
Transistor Gate Structures and Semiconductor Device Physics	[0.70, 0.80]	[0.06, 0.06]	[0.02, 0.02]	[−0.00, 0.01]
Imaging Sensors and Camera Pipelines	[−0.36, −0.24]	[0.01, 0.02]	[0.01, 0.02]	[0.00, 0.01]
Processor Architecture, Caches, Instruction Execution	[−0.15, −0.10]	[−0.02, −0.01]	[−0.00, 0.00]	[−0.01, 0.00]
Display Panel Tech (LCD/OLED)	[−0.22, −0.16]	[0.01, 0.02]	[0.03, 0.03]	[0.00, 0.02]
Rotational Electromechanics and Magnetic Actuation	[−0.24, −0.20]	[−0.02, −0.01]	[−0.02, −0.02]	[−0.00, 0.00]
Optics, Lasers, and Photonics	[−0.33, −0.19]	[−0.01, −0.00]	[−0.00, 0.01]	[0.00, 0.01]
Analog and Differential Amplifier Circuits	[−0.32, −0.23]	[−0.03, −0.02]	[−0.02, −0.01]	[−0.01, −0.00]
Mobile UI and Content Services	[−0.19, −0.12]	[−0.02, −0.02]	[−0.01, −0.01]	[−0.01, −0.00]
Firm-years adding an area	0	[0.83, 0.96]	[0.33, 0.38]	[0.07, 0.21]

Notes: Each area row is the change, in percentage points, in that area's share of occupied slots. The fixed-size column holds the number of areas at the observed count. The extra-area columns start from the observed portfolio and keep occupied areas occupied. The last row is the change in the share of firm-years that add at least one area. Rival counts stay at their observed values.

Table 28: Case II: a 25% occupancy-to-scope ratio reduction, with rival counts held fixed

Area	Fixed size			Extra areas		
	$b = 0.25$	$b = 1$	$b = 4$	$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.00, 2.22]	[4.48, 5.28]	[4.57, 7.10]	[0.00, 0.05]	[0.29, 0.36]	[0.47, 0.60]
Audio and Speech-Signal Processing	[−0.24, 0.00]	[−0.92, −0.72]	[−1.56, −0.75]	[−0.02, 0.00]	[−0.31, −0.26]	[−0.55, −0.47]
Semiconductor Lithography and Fabrication Processes	[0.00, 2.18]	[6.00, 6.16]	[6.15, 6.37]	[−0.00, 0.02]	[0.15, 0.20]	[0.31, 0.41]
Integrated Circuit Design, Verification, and Testing	[−0.33, 0.00]	[−1.18, −0.89]	[−2.01, −0.91]	[−0.02, 0.00]	[−0.33, −0.28]	[−0.61, −0.53]
PCB and Electronic Hardware with Connectors	[−0.25, 0.00]	[−1.11, −0.95]	[−1.69, −0.97]	[−0.01, 0.00]	[−0.23, −0.19]	[−0.35, −0.30]
Video Recording, Encoding, and Media Processing	[−1.32, 0.00]	[−2.11, −2.06]	[−2.13, −1.60]	[−0.00, 0.01]	[0.13, 0.14]	[0.19, 0.22]
Semiconductor Packaging and Interconnect Bonding	[0.00, 2.39]	[6.00, 6.12]	[6.10, 6.31]	[−0.00, 0.03]	[0.11, 0.16]	[0.22, 0.31]
Thin-Film Materials, Deposition and Chemistry	[−0.34, 0.00]	[−0.90, −0.74]	[−1.41, −0.78]	[−0.00, 0.00]	[−0.02, −0.01]	[−0.02, 0.00]
Wireless Base-Station and Mobile Communications	[−0.97, 0.00]	[−1.74, −1.71]	[−1.81, −1.73]	[−0.00, 0.00]	[0.05, 0.06]	[0.05, 0.08]
Clock Generation, Timing Circuits, and Signal Processing	[−0.26, 0.00]	[−1.00, −0.80]	[−1.45, −0.82]	[−0.02, 0.00]	[−0.26, −0.21]	[−0.45, −0.39]
Power-Supply and Power-Management Electronics	[−0.33, 0.00]	[−1.31, −1.11]	[−1.99, −1.13]	[−0.01, 0.00]	[−0.20, −0.17]	[−0.34, −0.31]
Networking Protocols and Data Infrastructure	[−0.91, 0.00]	[−1.60, −1.56]	[−1.64, −1.59]	[0.00, 0.01]	[0.08, 0.09]	[0.10, 0.12]
Transistor Gate Structures and Semiconductor Device Physics	[0.00, 2.42]	[4.95, 5.63]	[5.04, 6.88]	[0.00, 0.03]	[0.23, 0.29]	[0.40, 0.51]
Imaging Sensors and Camera Pipelines	[−1.72, 0.00]	[−2.83, −2.62]	[−2.85, −1.79]	[0.00, 0.01]	[0.14, 0.15]	[0.20, 0.24]
Processor Architecture, Caches, Instruction Execution	[−0.33, 0.00]	[−1.08, −0.88]	[−1.47, −0.91]	[−0.01, 0.00]	[−0.07, −0.05]	[−0.16, −0.11]
Display Panel Tech (LCD/OLED)	[−0.26, 0.00]	[−2.85, −2.70]	[−2.91, −2.20]	[−0.00, 0.01]	[0.29, 0.35]	[0.53, 0.62]
Rotational Electromechanics and Magnetic Actuation	[−0.30, 0.00]	[−1.14, −0.95]	[−1.72, −0.99]	[−0.01, 0.00]	[−0.16, −0.13]	[−0.25, −0.21]
Optics, Lasers, and Photonics	[−1.09, 0.00]	[−1.82, −1.67]	[−1.84, −1.20]	[−0.01, 0.00]	[0.06, 0.08]	[0.18, 0.21]
Analog and Differential Amplifier Circuits	[−0.26, 0.00]	[−1.06, −0.86]	[−1.66, −0.90]	[−0.02, 0.00]	[−0.17, −0.14]	[−0.27, −0.24]
Mobile UI and Content Services	[−0.28, 0.00]	[−0.96, −0.80]	[−1.51, −0.83]	[−0.01, 0.00]	[−0.13, −0.10]	[−0.25, −0.19]
Firm-years adding an area	0	0	0	[0.00, 0.50]	[4.54, 5.69]	[8.86, 10.67]

Notes: Each area row is the change, in percentage points, in that area's share of occupied slots. The fixed-size column holds the number of areas at the observed count. The extra-area columns start from the observed portfolio and keep occupied areas occupied. The last row is the change in the share of firm-years that add at least one area. Rival counts stay at their observed values.

Table 29: Halving rival congestion

Area	Fixed size	Extra areas		
		$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[−1.31, −0.41]	[0.66, 0.88]	[0.26, 0.38]	[0.01, 0.10]
Audio and Speech-Signal Processing	[0.36, 1.10]	[−1.06, −0.77]	[−0.51, −0.34]	[−0.14, −0.01]
Semiconductor Lithography and Fabrication Processes	[−0.08, 0.03]	[0.20, 0.33]	[0.10, 0.17]	[−0.00, 0.05]
Integrated Circuit Design, Verification, and Testing	[0.56, 1.37]	[−1.28, −0.90]	[−0.62, −0.40]	[−0.16, −0.02]
PCB and Electronic Hardware with Connectors	[0.37, 1.01]	[−0.83, −0.58]	[−0.39, −0.26]	[−0.10, −0.01]
Video Recording, Encoding, and Media Processing	[−1.10, −0.24]	[0.56, 0.79]	[0.25, 0.38]	[0.01, 0.10]
Semiconductor Packaging and Interconnect Bonding	[−0.17, 0.04]	[0.07, 0.20]	[0.05, 0.08]	[0.00, 0.03]
Thin-Film Materials, Deposition and Chemistry	[0.22, 0.44]	[−0.18, −0.08]	[−0.07, −0.05]	[−0.03, 0.01]
Wireless Base-Station and Mobile Communications	[−0.28, −0.09]	[0.31, 0.43]	[0.15, 0.21]	[0.00, 0.07]
Clock Generation, Timing Circuits, and Signal Processing	[0.37, 0.78]	[−0.90, −0.66]	[−0.40, −0.29]	[−0.10, −0.01]
Power-Supply and Power-Management Electronics	[0.30, 0.90]	[−0.65, −0.47]	[−0.30, −0.21]	[−0.08, −0.00]
Networking Protocols and Data Infrastructure	[−0.30, −0.16]	[0.41, 0.56]	[0.21, 0.27]	[0.00, 0.08]
Transistor Gate Structures and Semiconductor Device Physics	[−1.01, −0.32]	[0.40, 0.61]	[0.19, 0.30]	[0.01, 0.06]
Imaging Sensors and Camera Pipelines	[−1.37, −0.41]	[0.56, 0.73]	[0.24, 0.36]	[0.01, 0.10]
Processor Architecture, Caches, Instruction Execution	[0.16, 0.47]	[−0.11, −0.09]	[−0.05, −0.03]	[−0.03, −0.00]
Display Panel Tech (LCD/OLED)	[−1.86, −1.25]	[1.10, 1.48]	[0.45, 0.67]	[0.02, 0.17]
Rotational Electromechanics and Magnetic Actuation	[0.32, 0.78]	[−0.60, −0.40]	[−0.27, −0.19]	[−0.07, −0.00]
Optics, Lasers, and Photonics	[−0.75, −0.17]	[0.33, 0.52]	[0.13, 0.22]	[0.01, 0.06]
Analog and Differential Amplifier Circuits	[0.26, 0.74]	[−0.66, −0.52]	[−0.32, −0.22]	[−0.07, −0.01]
Mobile UI and Content Services	[0.24, 0.63]	[−0.24, −0.15]	[−0.11, −0.06]	[−0.02, −0.00]
Firm-years adding an area	0	[16.13, 24.56]	[6.08, 9.40]	[0.18, 2.39]

Table 30: A lower occupancy cost in every area

Area	Extra areas		
	$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.00, 0.06]	[0.29, 0.34]	[0.39, 0.49]
Audio and Speech-Signal Processing	[−0.05, 0.00]	[−0.45, −0.37]	[−0.63, −0.52]
Semiconductor Lithography and Fabrication Processes	[−0.00, 0.01]	[0.12, 0.15]	[0.20, 0.24]
Integrated Circuit Design, Verification, and Testing	[−0.06, 0.00]	[−0.55, −0.44]	[−0.78, −0.66]
PCB and Electronic Hardware with Connectors	[−0.04, 0.00]	[−0.34, −0.28]	[−0.45, −0.35]
Video Recording, Encoding, and Media Processing	[0.00, 0.05]	[0.28, 0.33]	[0.34, 0.42]
Semiconductor Packaging and Interconnect Bonding	[−0.00, 0.00]	[0.06, 0.07]	[0.11, 0.16]
Thin-Film Materials, Deposition and Chemistry	[−0.00, 0.00]	[−0.07, −0.03]	[−0.08, −0.05]
Wireless Base-Station and Mobile Communications	[−0.00, 0.01]	[0.16, 0.20]	[0.19, 0.24]
Clock Generation, Timing Circuits, and Signal Processing	[−0.05, 0.00]	[−0.37, −0.31]	[−0.52, −0.42]
Power-Supply and Power-Management Electronics	[−0.04, 0.00]	[−0.28, −0.23]	[−0.38, −0.32]
Networking Protocols and Data Infrastructure	[−0.00, 0.03]	[0.21, 0.25]	[0.25, 0.31]
Transistor Gate Structures and Semiconductor Device Physics	[0.00, 0.03]	[0.21, 0.26]	[0.31, 0.39]
Imaging Sensors and Camera Pipelines	[0.00, 0.05]	[0.26, 0.33]	[0.34, 0.41]
Processor Architecture, Caches, Instruction Execution	[−0.03, 0.00]	[−0.05, −0.03]	[−0.09, −0.06]
Display Panel Tech (LCD/OLED)	[0.00, 0.09]	[0.48, 0.61]	[0.65, 0.83]
Rotational Electromechanics and Magnetic Actuation	[−0.03, 0.00]	[−0.25, −0.21]	[−0.33, −0.25]
Optics, Lasers, and Photonics	[0.00, 0.02]	[0.14, 0.20]	[0.28, 0.33]
Analog and Differential Amplifier Circuits	[−0.04, 0.00]	[−0.29, −0.23]	[−0.35, −0.31]
Mobile UI and Content Services	[−0.01, 0.00]	[−0.09, −0.07]	[−0.17, −0.13]
Firm-years adding an area	[0.00, 1.33]	[6.72, 8.31]	[10.00, 12.80]